Informatica

Advanced Topic

CDI Ingestion for MDM SaaS

Lab Guide

CONTENTS

Lab Environment 3
Image Passwords 4
Lab Files 5
LAB: Overview 6
Purpose: 6
LAB: Prepare Local Environment 7
Purpose: 7
Part 1: Install Cloud Secure Agent 7
Part 2: Assign privileges in IDMC 14
Part 3: Setup the application 14
Part 4: Copy files 15
LAB Part 1: Simple Customer file 16
Purpose: 16
Instructions/Steps – Create Connection: 16
Instructions/Steps – Create Project and Source System: 19
Instructions/Steps – Create mapping: 21
Instructions/Steps – Create Mapping Task and Taskflow: 35
Instructions/Steps – Create Business 360 job and test: 40
LAB Part 2: Multiple child elements 44
Purpose: 44
Instructions/Steps – Create Mapping, Task, and Taskflow: 44
Instructions/Steps – Business360 job and test: 53
LAB Part 3: Custom error handling 55
Purpose: 55
Instructions/Steps – Create Crosswalk: 55
Instructions/Steps – Create CDQ validation: 60
Instructions/Steps – Create CDI mapping with validation: 67
LAB Part 4: Customer relationships 75
Purpose: 75
Instructions/Steps – Create CDI mapping 75
Instructions/Steps – Create job definition and run 78

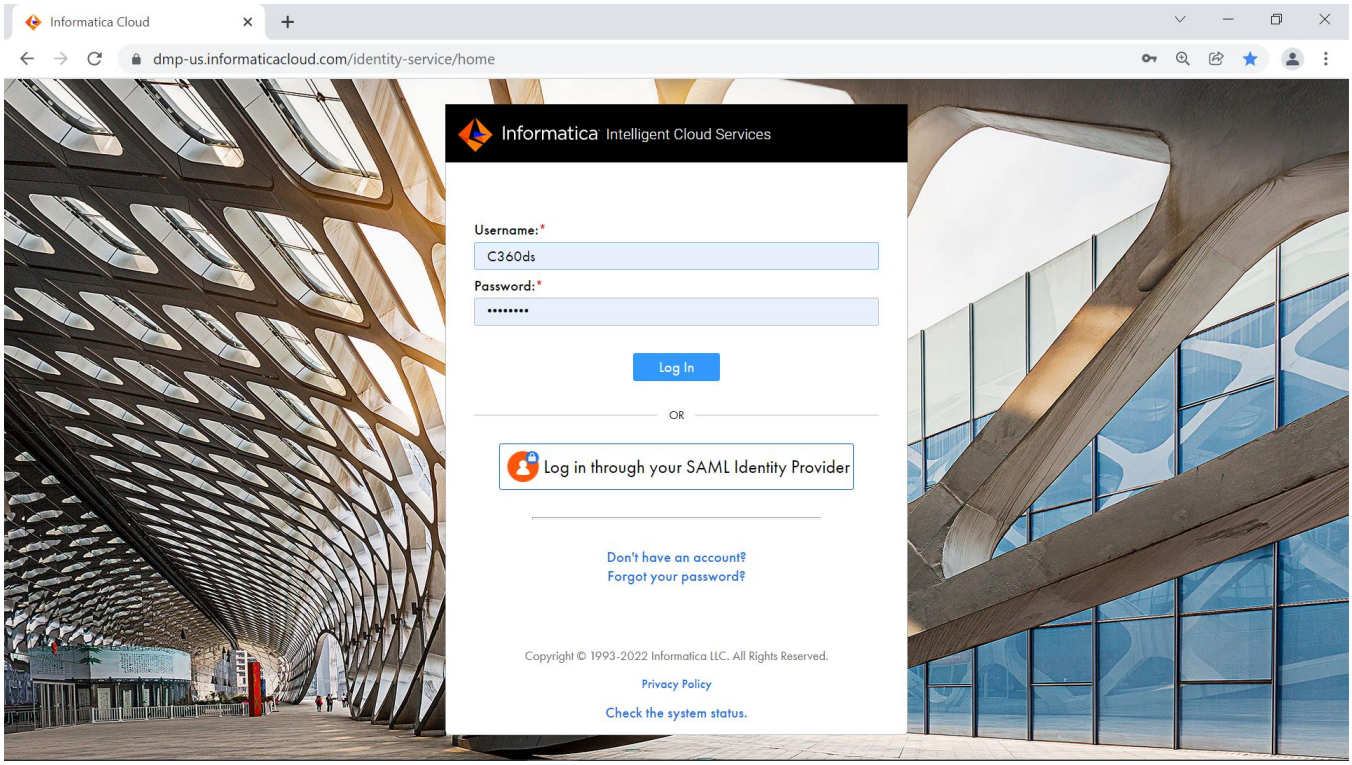
© Informatica.

Lab Environment

This class uses a MDM SaaS cloud image for the lab. You can access the environment via a browser; Google Chrome is recommended.

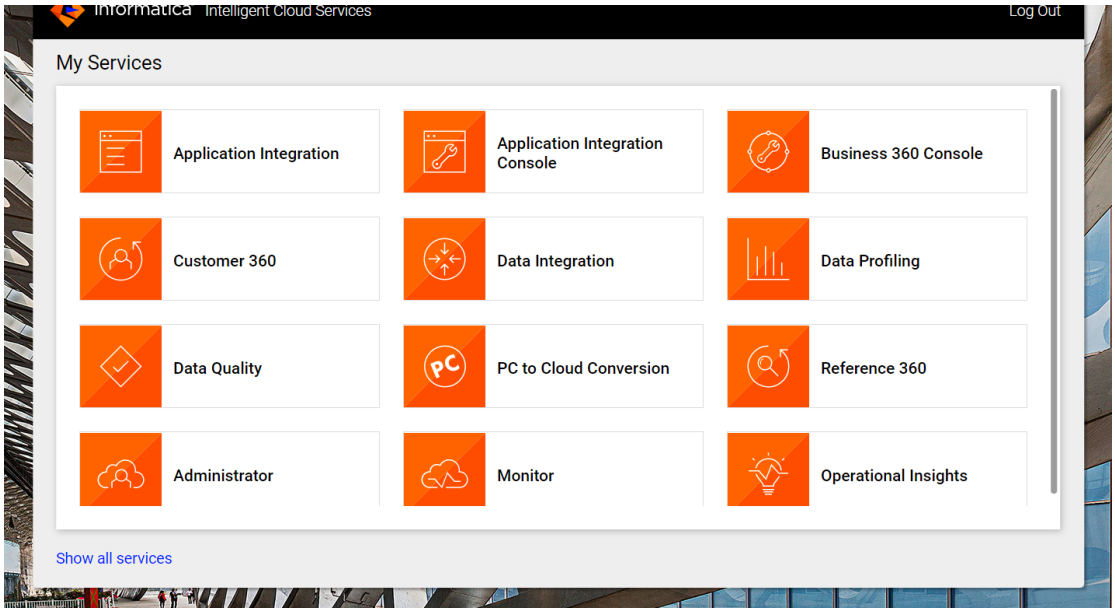
Log in using the credentials you have received.

The url to log in is https://dmp-us.informaticacloud.com .



© Informatica.

You see the list of available Services on the main page. Which ones appear depends on the permissions of your user ID.



For this lab, you will use the following services:

- Business 360 Console
- Customer 360
- Data Integration
- Data Quality

Image Passwords

Here are the passwords needed for this lab:

Box/Application	User Name	Password
IDMC	Provided in email	Provided in email

© Informatica.

Lab Files

The following files are in the LAB_FILES_Reference360.zip file.

Filename	Use
Customer_Part_1.csv	Customer file with one customer address
Customer_Part_2.csv	Customer file with multiple addresses and phone numbers per customer
Customer_Part_3.csv	Customer file with various phone types that need to be validated.
Customer_Relationships.csv	File containing relationships between customers.

****

THE LAB IMAGE HAS A LIMITED DURATION AND CANNOT BE EXTENDED, so please schedule accordingly.

When you order your image, schedule it for a time when you are available to work on the lab.

****

Send an email to DE_On_Demand_Labs@informatica.com with the following information at least 24 hours before you want your image:

Your name

Course name

Company Name

Date you want image available

Your timezone

© Informatica.

LAB: Overview

DURATION: 3 hours

Purpose:

This lab will guide you through developing Cloud Data Ingestion processes for MDM SaaS entities and relationships. The mappings will increase in design complexity to show the various scenarios you can encounter in a customer's project.

You will start with the preparation, consisting of installing the secure agent, setting authorizations, and copying the files to your Windows environment.

The first section will have you create a mapping for a customer with one address and phone number. The target entity is Organization. The root object, as well as different child elements, will be loaded.

In the second section, you will create a more complex mapping for a customer with multiple addresses and phone numbers. This ingestion process will use out-of-the-box error handling to catch faults.

The third section will show custom error handling in a mapping with different inputs. You will build a custom data quality cleanse process to validate the input data against the reference data in MDM SaaS. Validation errors will be caught in the mapping and written to an error file.

The final section shows how to load relationships between the entities you have loaded.

© Informatica.

LAB: Prepare Local Environment

DURATION: 20 minutes

Purpose:

This lab will guide you through the preparation steps for this workshop:

1. Install the Cloud Secure Agent
2. Setup authorizations
3. Setup the application
4. Copy the source files

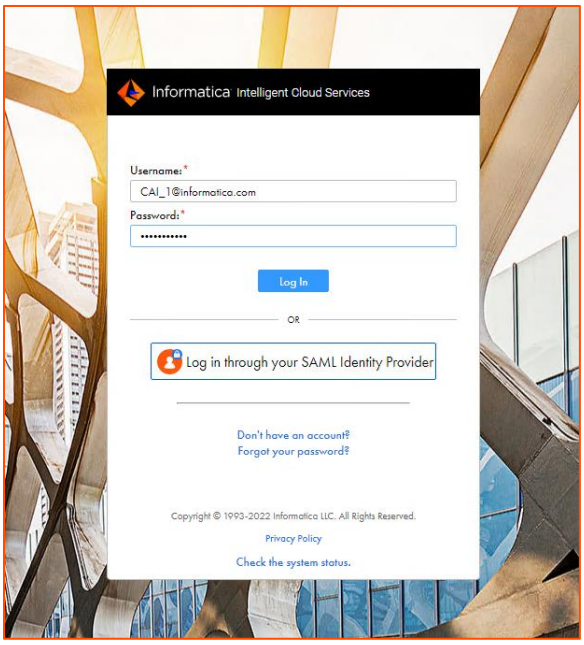
Part 1: Install Cloud Secure Agent

The computer you are working on has no Secure Agent installed.

Please follow the following instructions. Starting a Secure Agent for the first time may take 15-30 minutes to be accessible and usable. You don't have to wait for this to complete; you can continue with the remainder of this lab. Part 3, setting up the Customer360 application, also takes about 15 minutes, so ensure you do these two steps in parallel to save time.

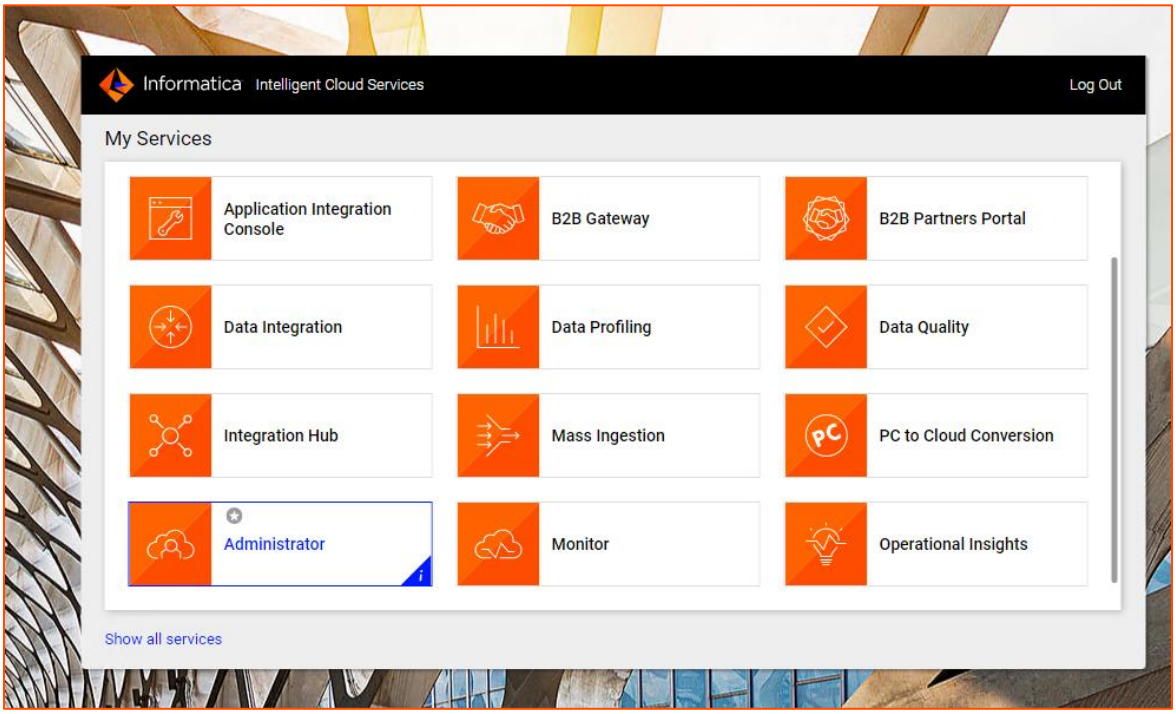
1. Login to the IICS instance using

   https://dmp-us.informaticacloud.com/identity-service/home

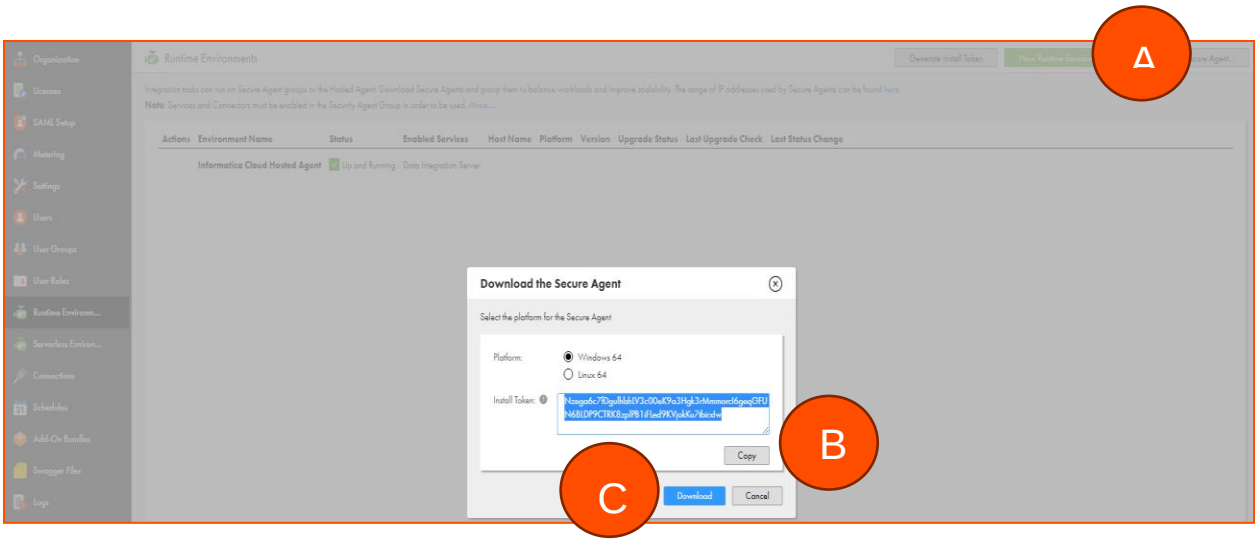


© Informatica.

2. Open up the Administrator Service

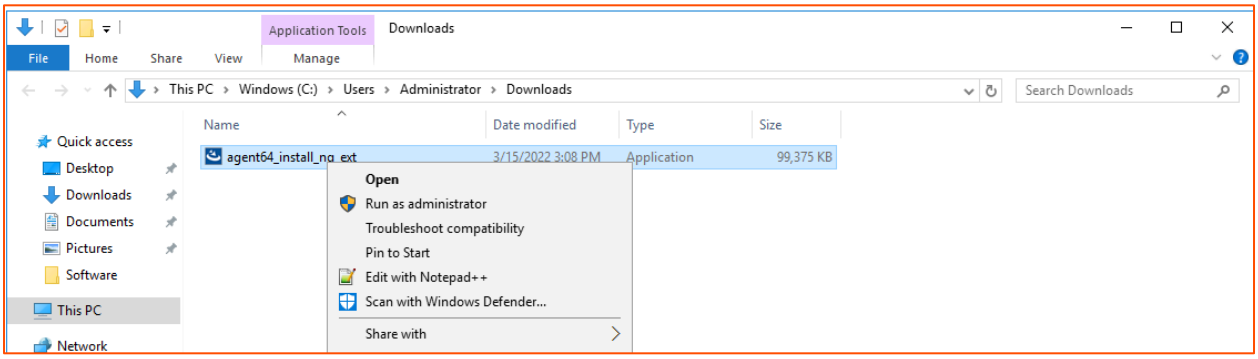


3. Go to Run Time Environments
   a. Click on Download Secure Agent
   b. Copy the Install Token and save it to a local file for future reference
   c. Click Download button to download the executable

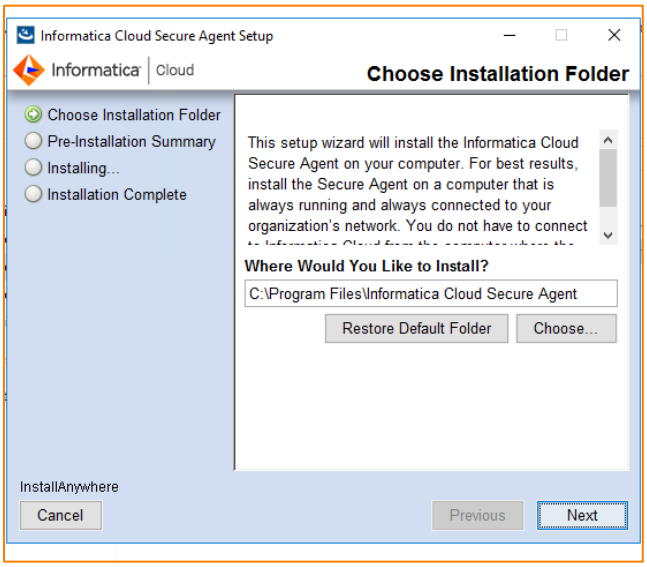


© Informatica.

4. Go to the location where the executable was downloaded (normally, the Downloads folder), right-click the file, and Run as Administrator

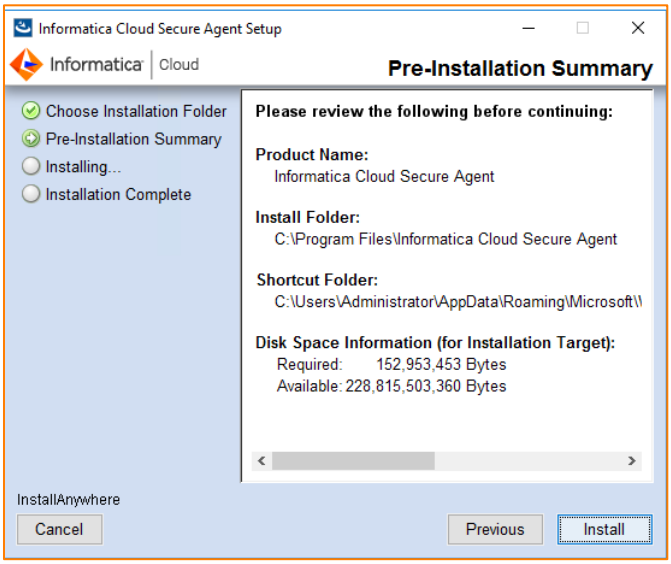


5. Leave the installation directory as default and click Next

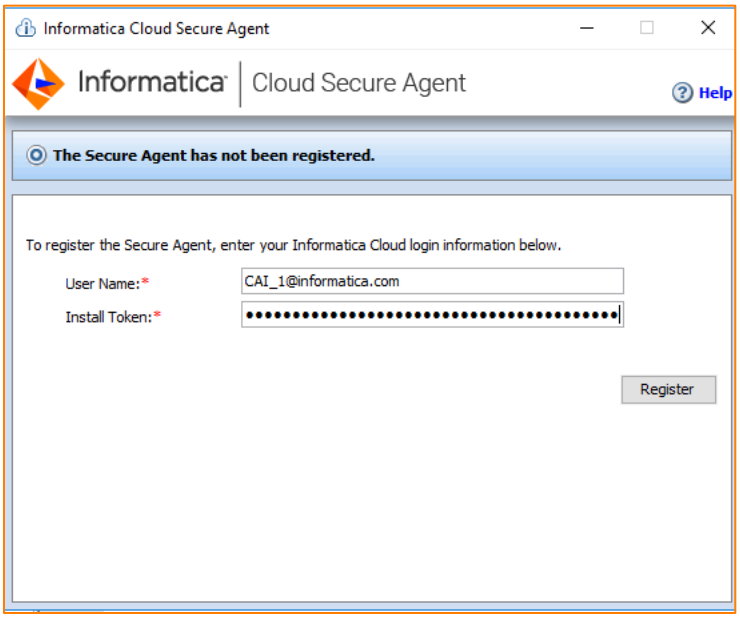


© Informatica.

6. Click Install to start the installation



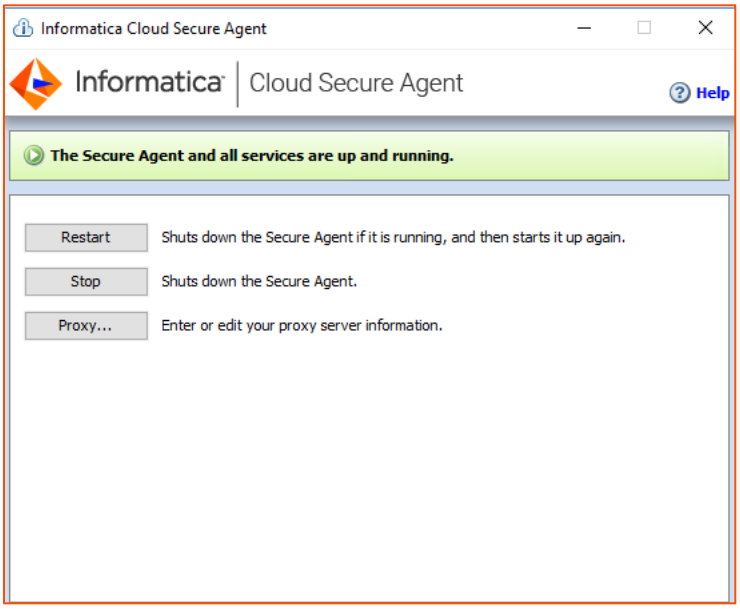
7. Enter the IDMC user name (or email) you received and used to log in. Paste the Install Token you saved.



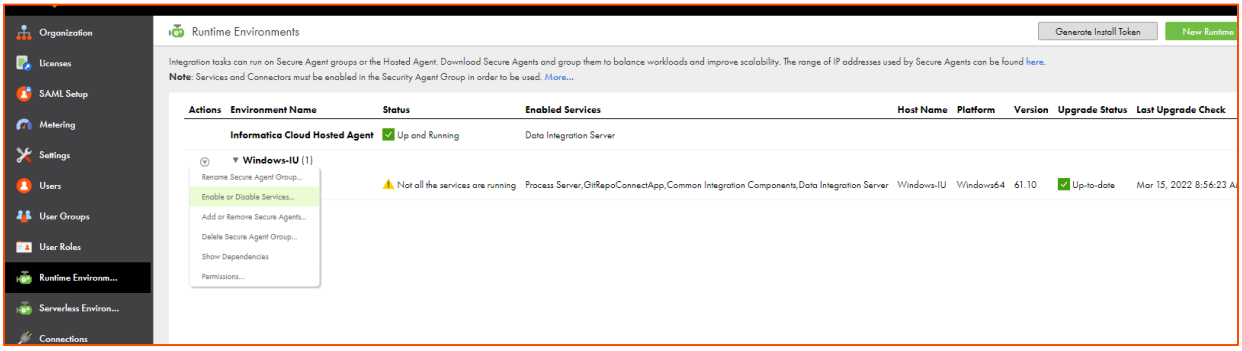
**Note, if you forgot the Install token, go back to Run Time Environments and Generate a new one.

© Informatica.

8. An application window will pop up like below. However, even if it states that all services are up and running, that is not entirely accurate. The next steps will show how to get current validation and/or reduce the number of services to accelerate startup.

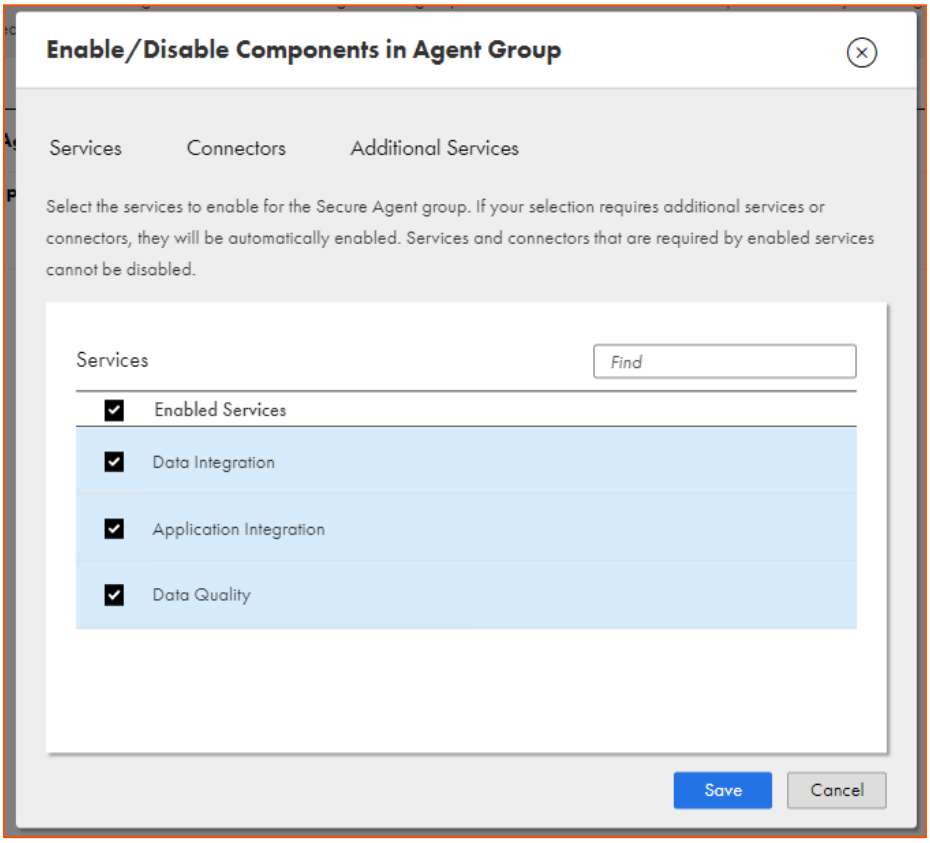


9. Navigate back to Run Time Environments in the Administrator service
10. Click on the arrow to the left of your newly created Secure Agent (group) and choose Enable or Disable Services

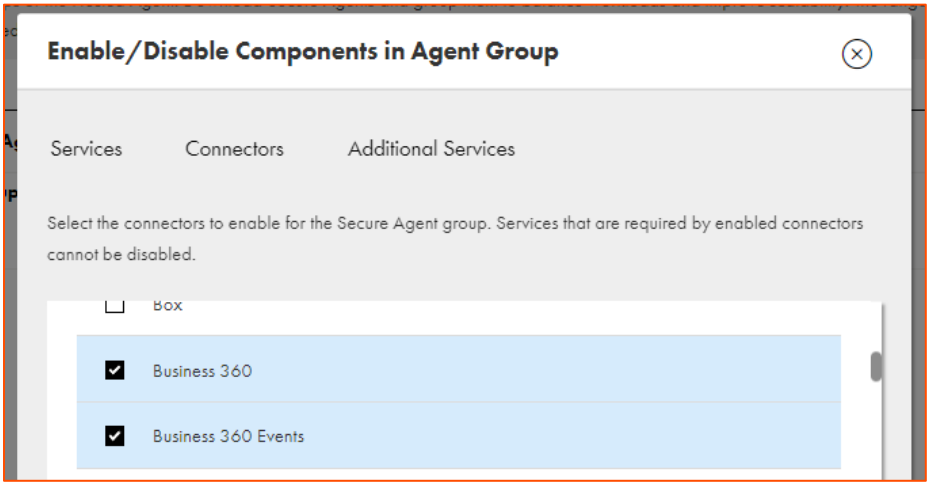


© Informatica.

11. Check and uncheck the following services to accelerate the startup and click OK

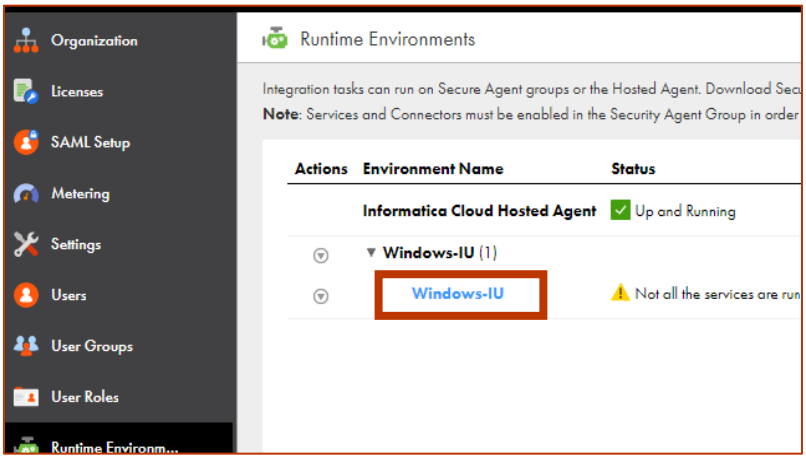


12. Click on Connectors in the same panel and enable the Business 360 and Business 360 Events connectors:

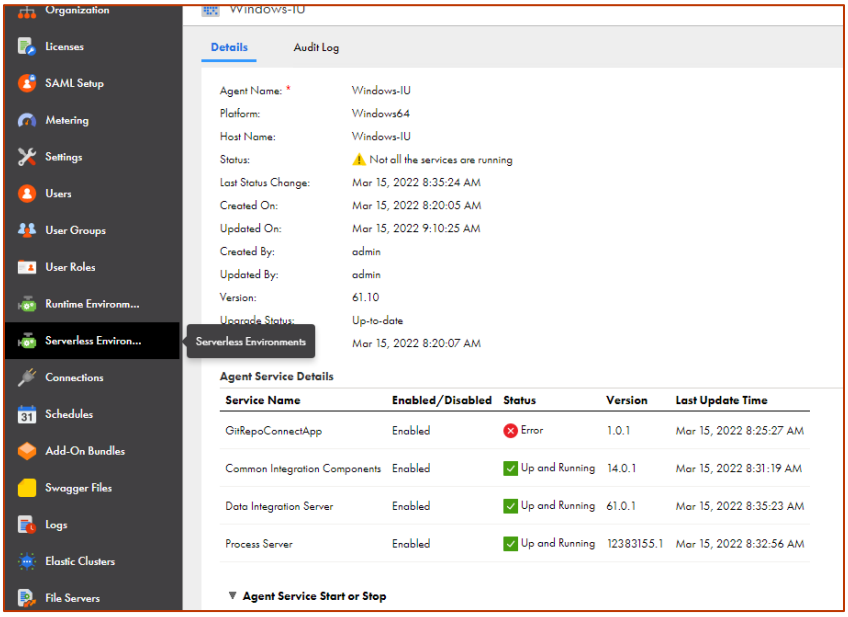


© Informatica.

13. Within Runtime Environments, click the hyperlink for your Secure Agent



14. After 10-15 minutes, you should see that the services are up and running. It is acceptable that GitHub has an error if it is enabled.



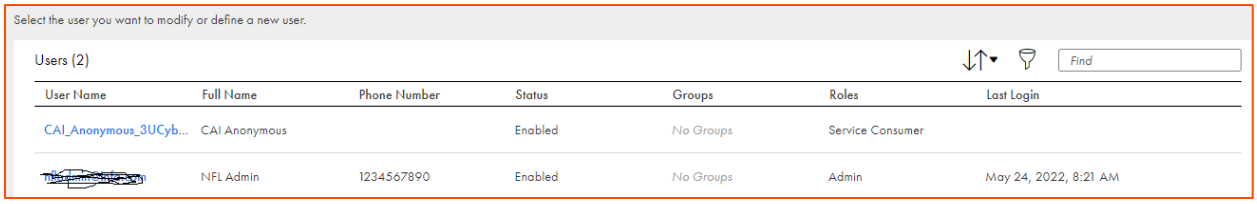
© Informatica.

Part 2: Assign privileges in IDMC

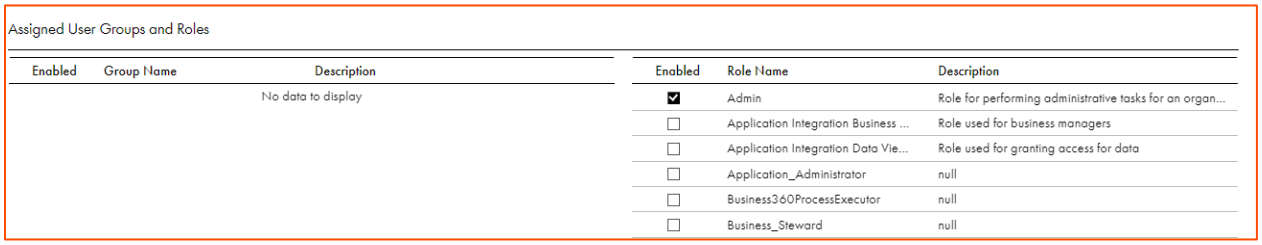
Although your userid has Admin privileges in the IDMC org, this is insufficient to perform all the necessary tasks.

You will have to assign a few more.

1. In IDMC, go to Administrator
2. Click on Users. You will see your user here:



3. Open the user to see the assigned roles:



4. Scroll down and assign yourself the following roles:
    a. Data Integration Data Previewer
    b. Reference 360 Administrator
    c. Reference 360 Business Steward
5. Log out and back in to activate the roles.

Part 3: Setup the application

At the creation of the IDMC org, the Customer360 application is not set up automatically. You will have to do this manually.

1. Login to IDMC.
2. Go to the Customer360 Service.
3. You will see a screen to set up the application; this is importing and deploying the assets.
4. Click Setup. This will take 10-15 minutes to complete.

© Informatica.

Part 4: Copy files

You can download the zip file with this workshop's lab files from the course's resource section.All files are in .csv format.

It is recommended to keep the files on your laptop for analysis purposes because the image does not have excel installed.

Follow the below steps:

1. On the image, go to Windows explorer and navigate to the C: drive
2. Right-click to create a new folder
3. Give the folder a meaningful name, like Files
4. From your laptop, copy the zip file to this location
5. Right-click on the file and click Extract All
6. Change the target location to C:\Files
7. Extract the files.

The following files are in the LAB_FILES.zip file.

Filename	Use
Customer_Part_1.csv	Customer file with one customer address.
Customer_Part_2.csv	Customer file with multiple addresses and phone numbers per customer.
Customer_Part_3.csv	Customer file with various phone types that need to be validated.
Customer_Relationships.csv	File containing relationships between customers.

THIS CONCLUDES THE PREPARATION

© Informatica.

LAB Part 1: Simple Customer file

DURATION: 30 minutes

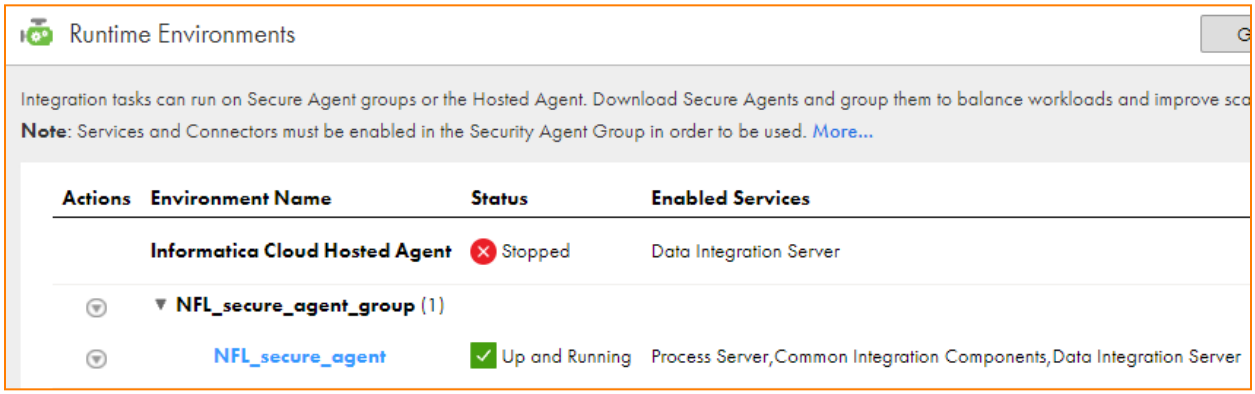
Purpose:

The purpose of this lab is threefold:

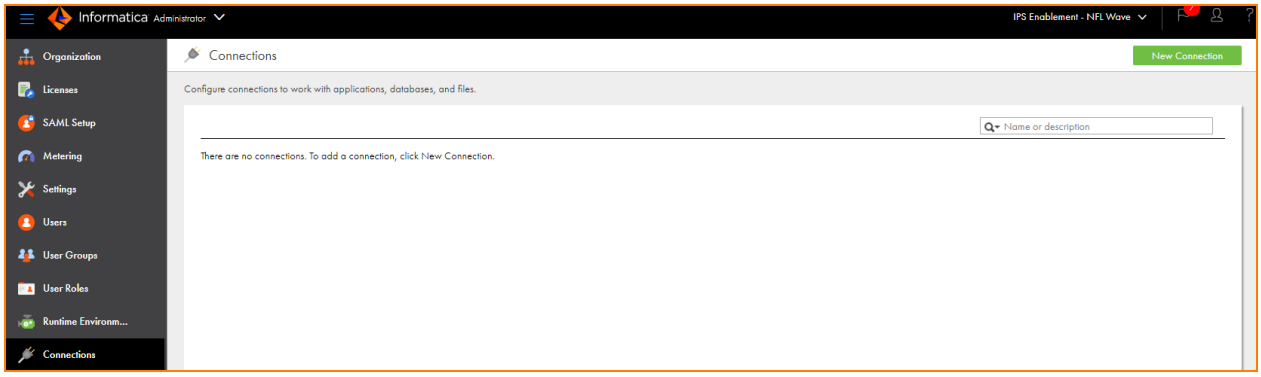
- Create connections
- Create a source system
- Create a mapping for Customer_Part_1.csv

Instructions/Steps – Create Connection:

1. Log into IDMC and go to Administrator.
2. First, check whether your cloud secure agent is running. Go to Runtime Environments and check the status:

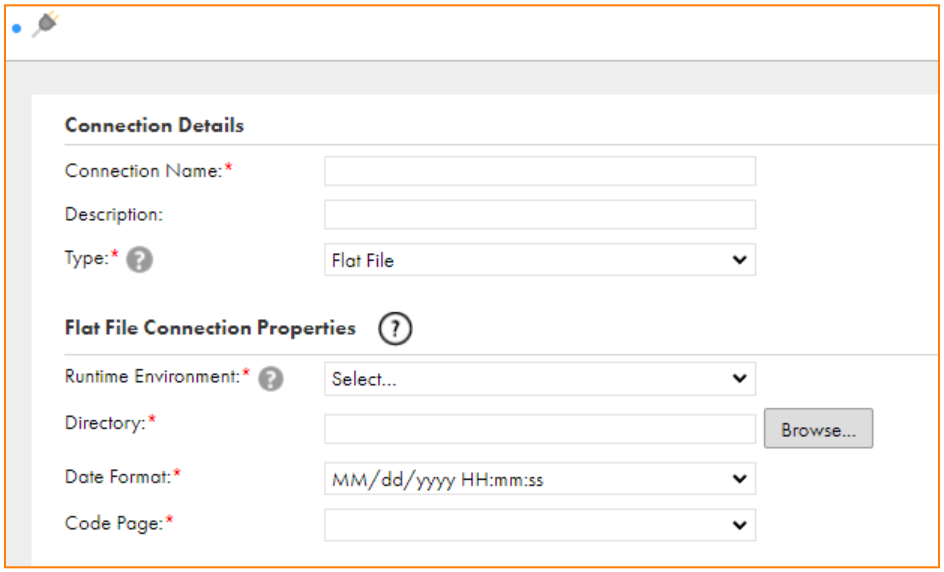


3. Click on the secure agent to see the detailed status if something is not running.
4. Go to Connections and click on New Connections at the top right of the screen:

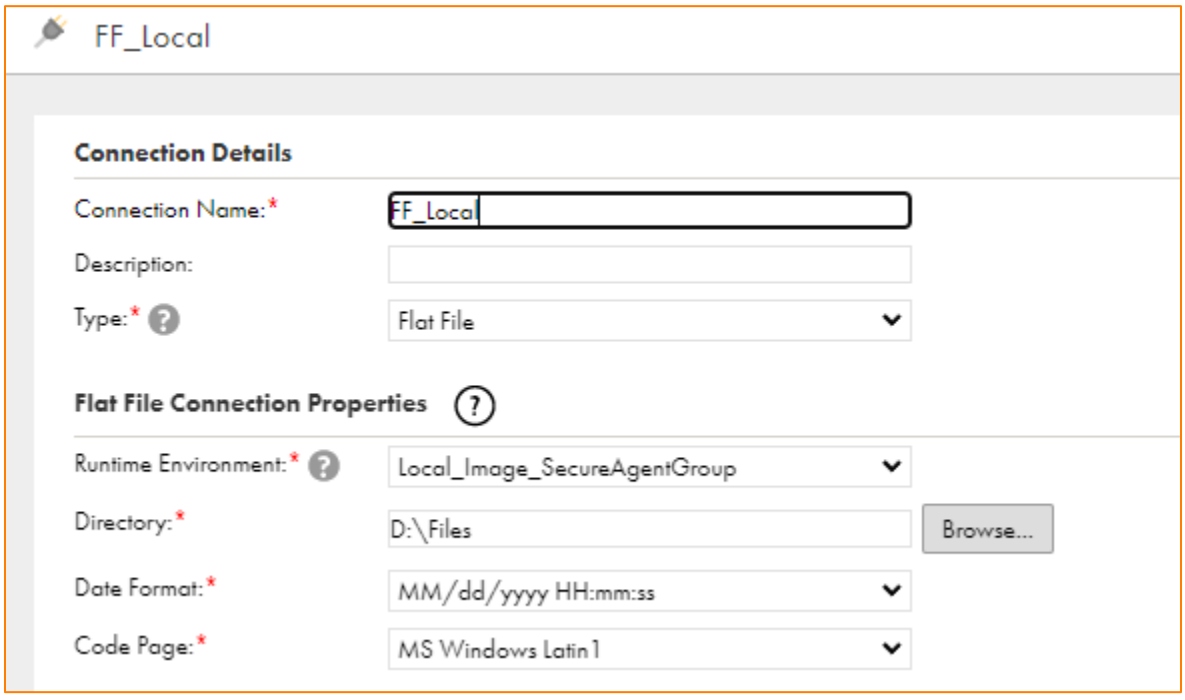


© Informatica.

5. Select the type as Flat File from the dropdown list. The properties screen will appear:



6. Give the connection a name, e.g., FF_Local.
7. Select the Runtime Environment and point it to your secure agent group
8. Click Browse to find the directory where you stored the data files, or enter the directory manually. Browse does not work when the secure agent runs as Local System.
9. Leave the data format as is
10. Select the Code Page MS Windows Latin 1.
11. Save
12. The connection will now look like this:

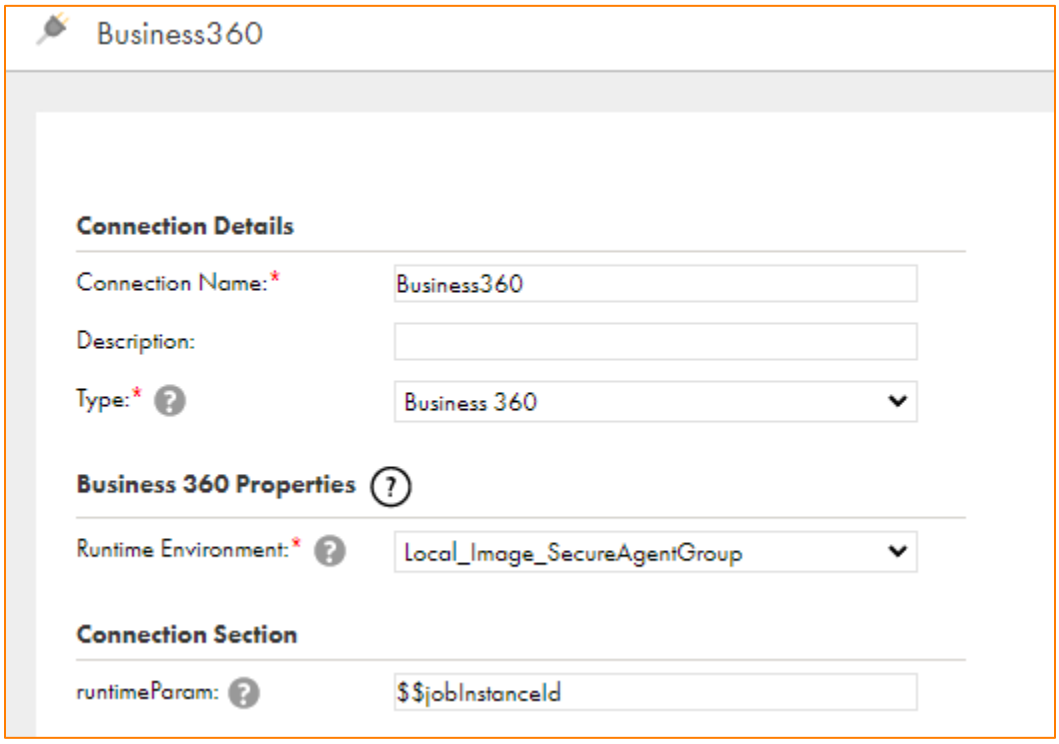


© Informatica.

13. Test the connection. It should succeed, Typical errors are the secure agent not running, or the directory does not exist.

The second connection is the Business 360 connector.

14. Click on New Connection
15. Select the type Business 360 from the dropdown list.
16. Give the connection a name
17. Select the Runtime Environment as your local secure agent group
18. The connection should look like this:



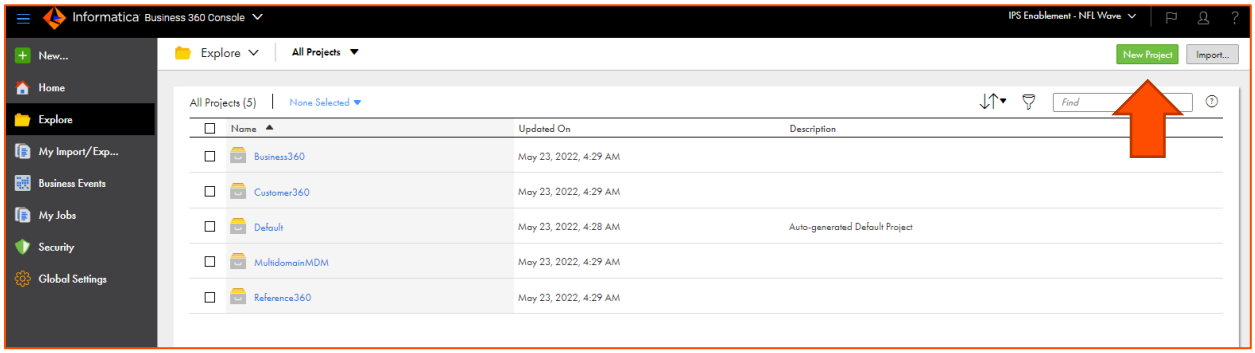
19. Test the connection. It should succeed. If not, check whether you have enabled the connection type in the Secure Agent Group.

© Informatica.

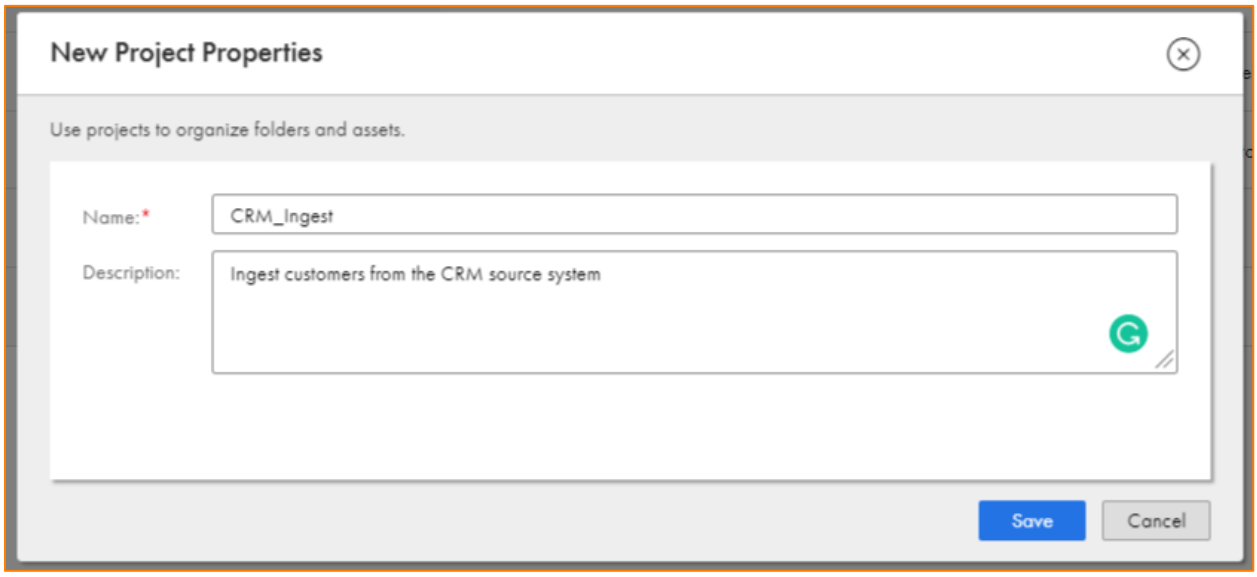
Instructions/Steps – Create Project and Source System:

First, you must create a project and a source system in the Business360 Console. You can modify the out-of-the-box projects, however, no new assets can be added. It is a best practice to create a project per source system to group all assets created.

1. Log in to Business 360.
2. Create a project to store the items you will be creating. Go to Explore and Select New Project.

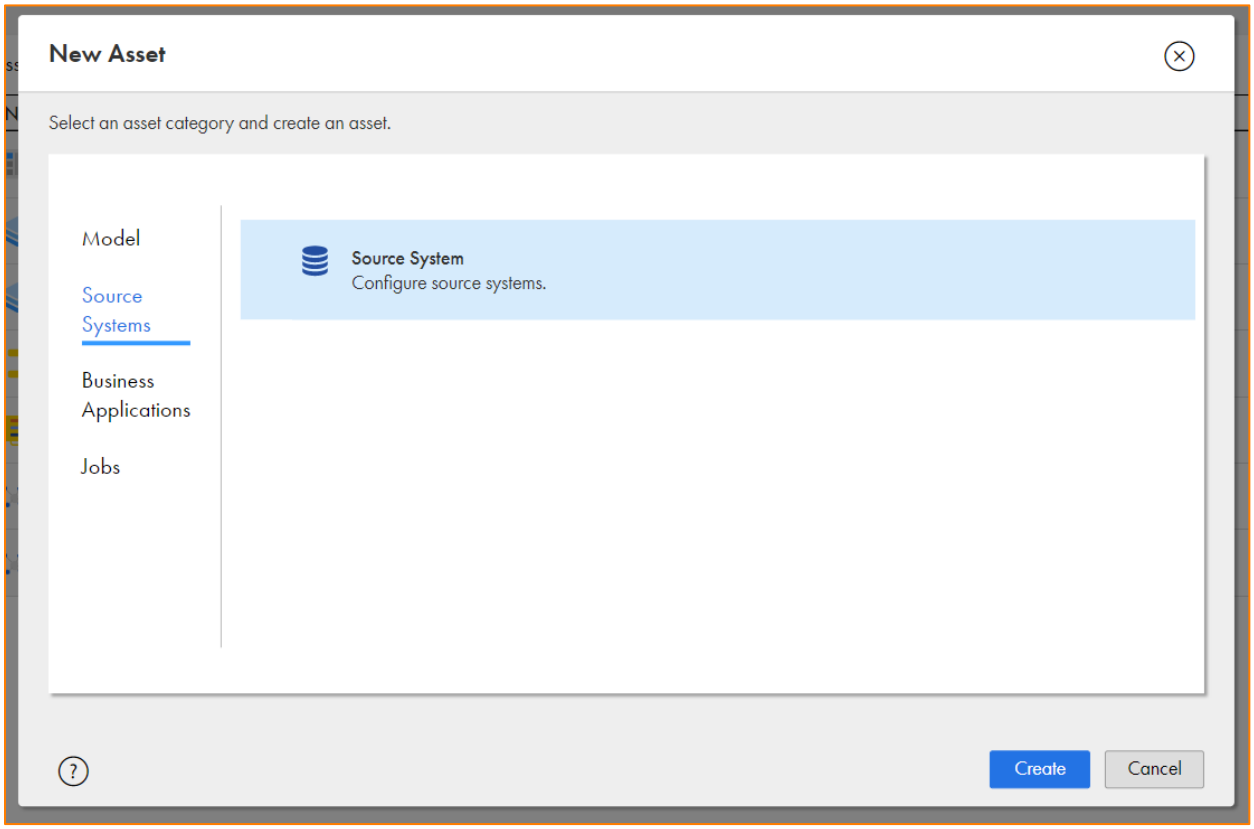


3. Provide the Name for the new project. This is the project you will use when storing every new asset you create.

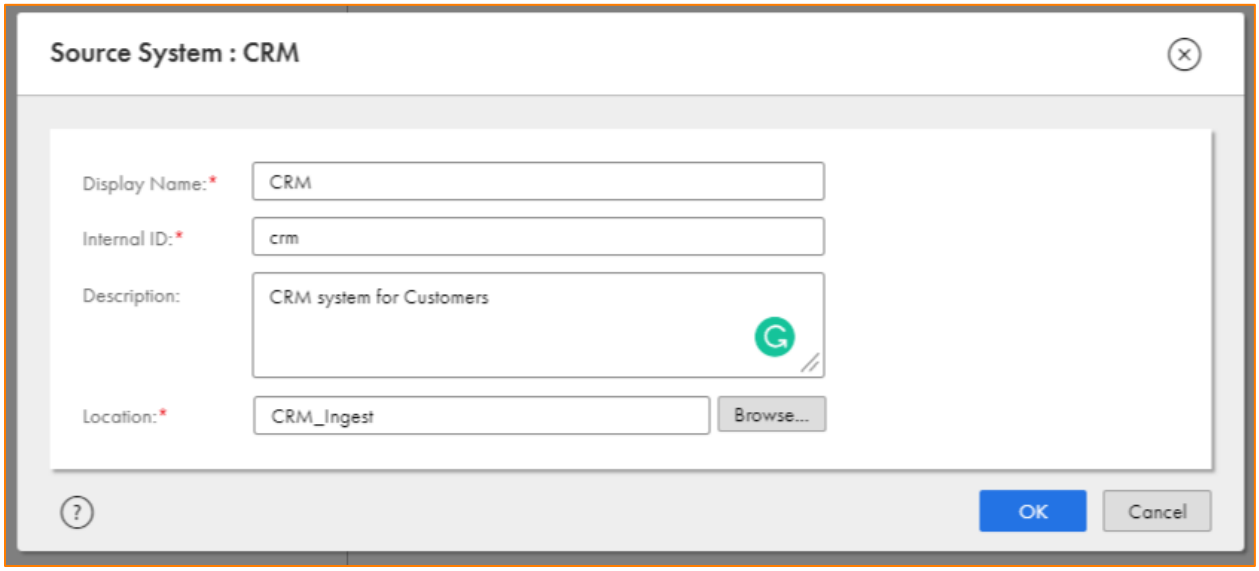


© Informatica.

4. Create a new source system called Customer. Go to New – Source System and click Create.



5. Give it the name CRM and description.



6. Select the new project as the location and click OK.
7. On the next screen, click Save.

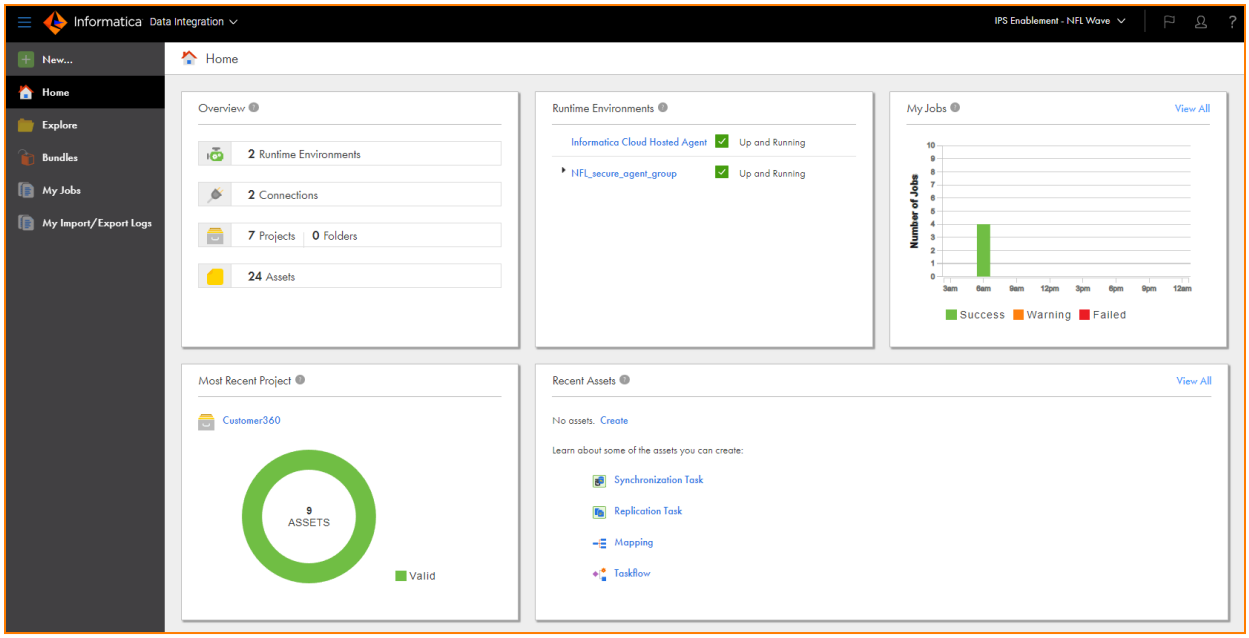
© Informatica.

Instructions/Steps – Create mapping:

The first mapping you will create is relatively simple. It reads the customers from the source file, sets default values needed to load the data, and writes the customers to the Organization Business entity.

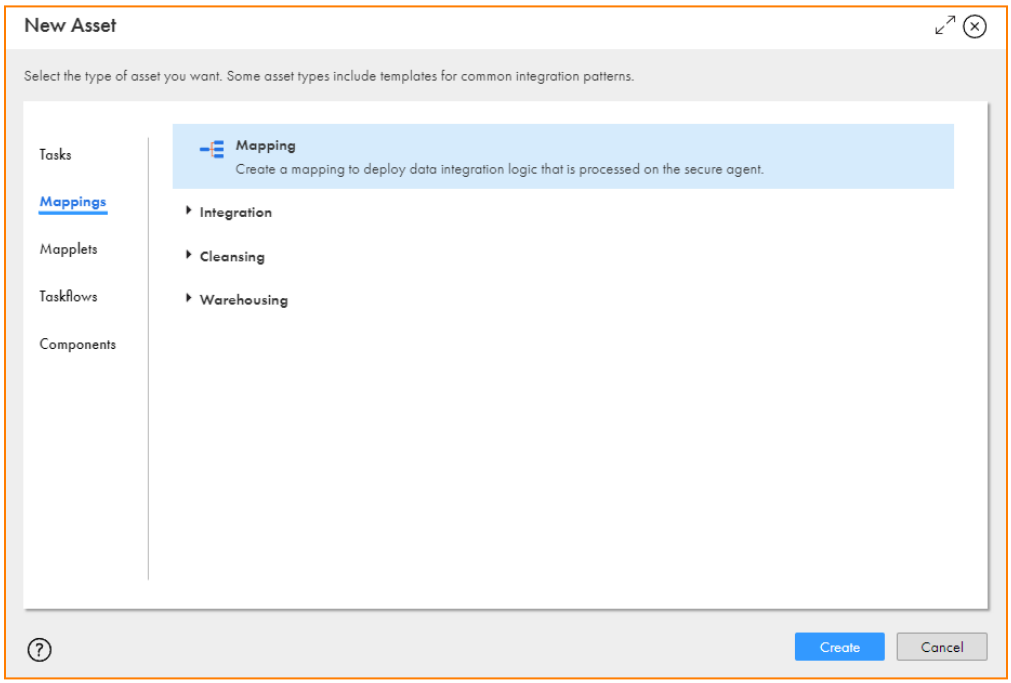
To create the CDI mapping, follow these steps.

1. Login in IDMC and open Data Integration. You will see a screen like this:

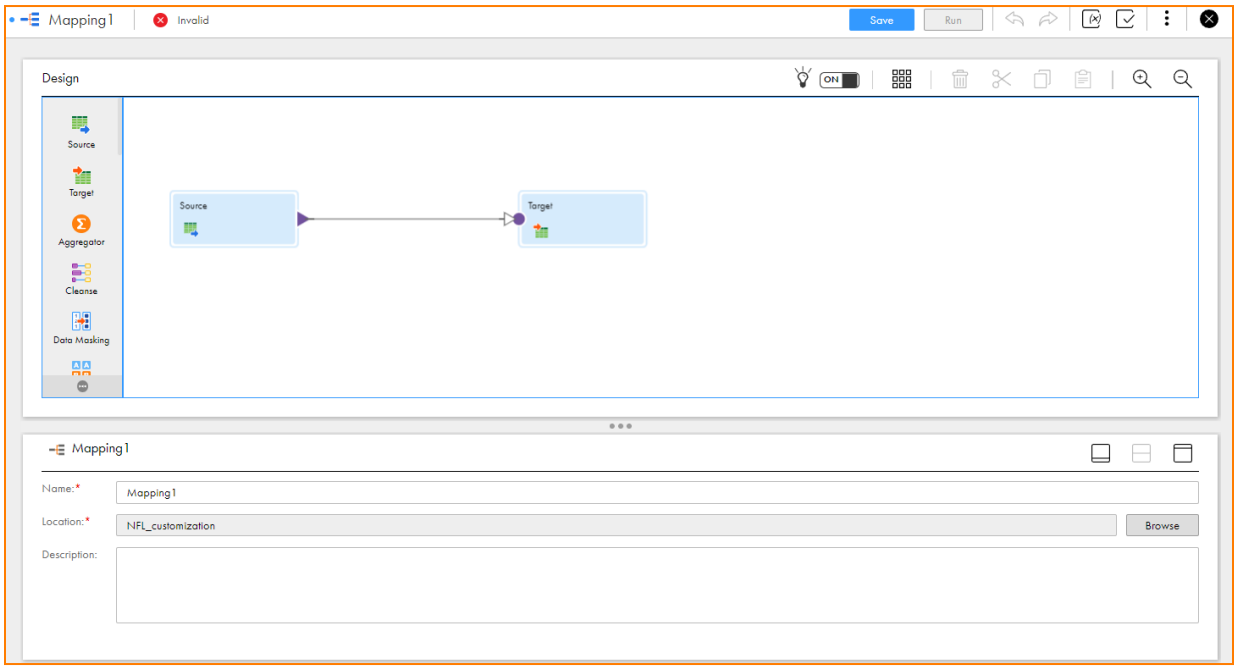


© Informatica.

2. Click New and Click on the Mappings tab on the left:

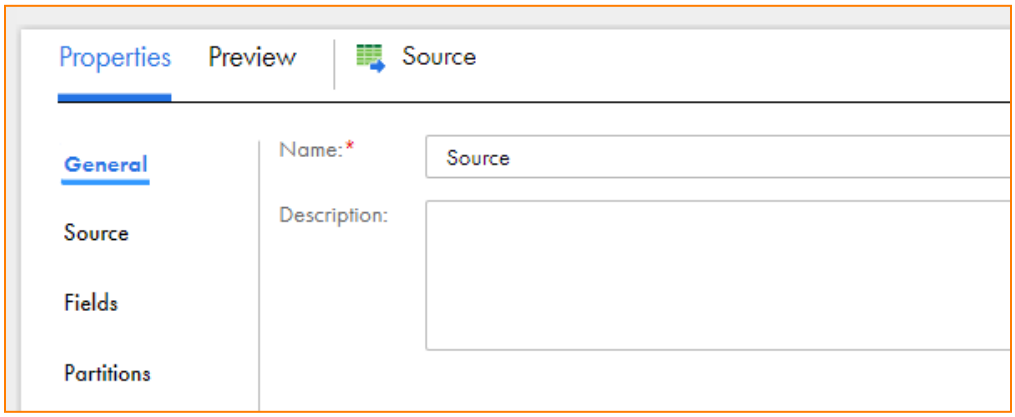


3. Highlight mapping and click Create. The mapping canvas will open with the basic mapping structure:

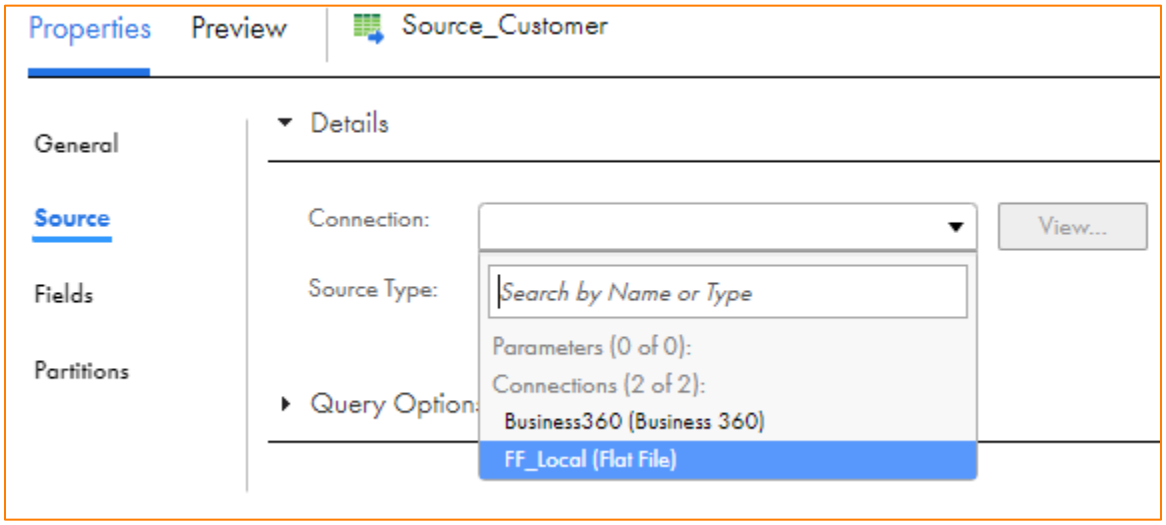


© Informatica.

4. Name the mapping m_Customer_Part_1.
5. Set the location to the CRM_Ingest project.
6. Click on the Source object in the canvas. The properties panel will open at the bottom of the page:



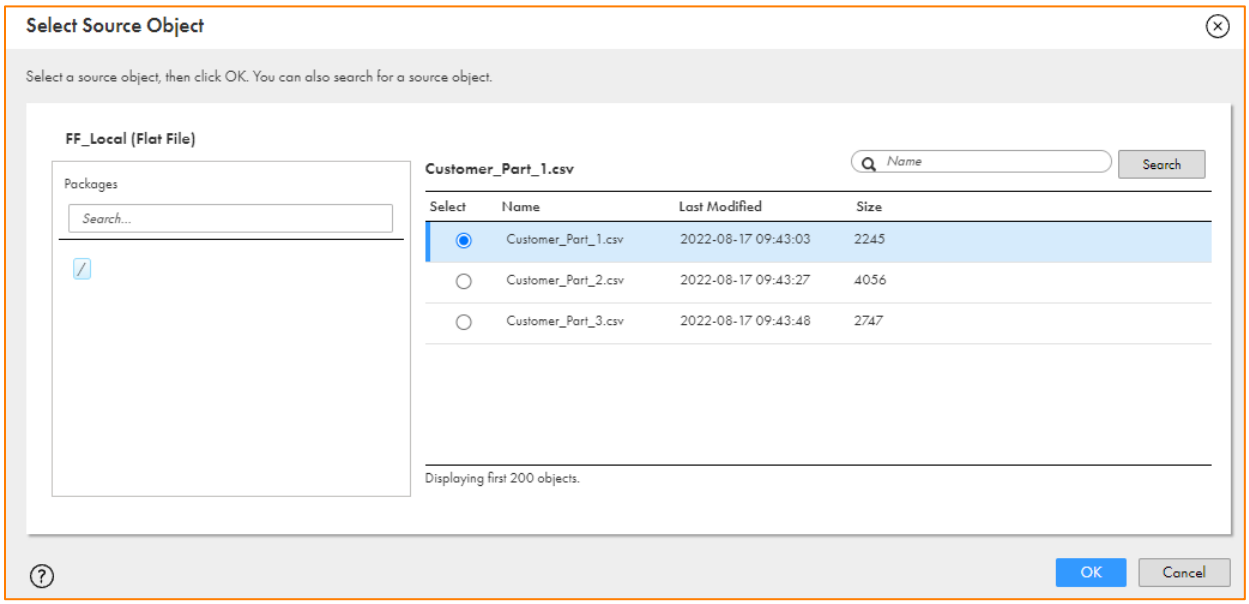
7. Give the source object a name, like Source_Customer
8. Click on the Source tab on the left to specify the source to use
9. From the connections pull-down menu, select your flat file connection:



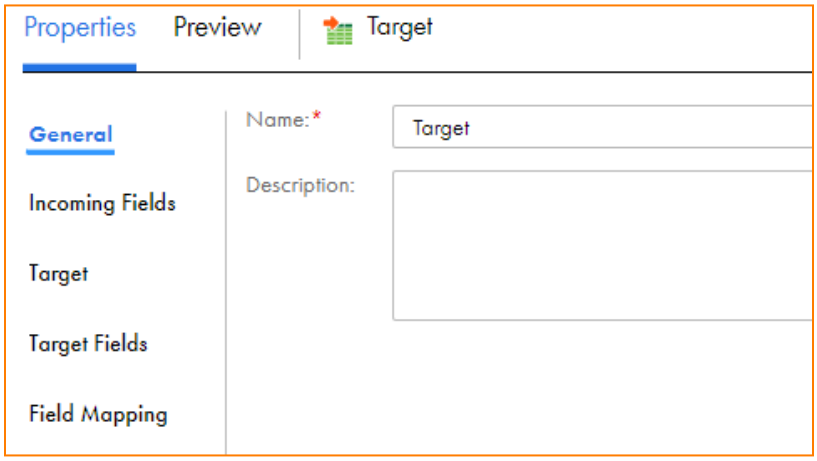
10. Click on Select next to the Object field to select the source file

© Informatica.

11. Select the file "Customer_Part_1.csv" you have profiled before and click OK:



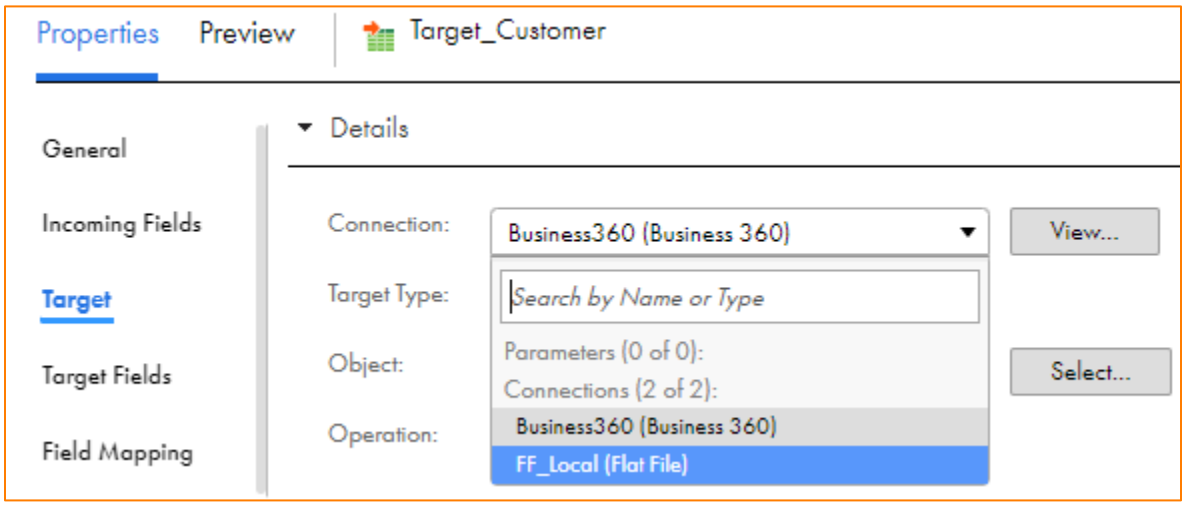
12. Click on Fields to see the metadata imported.
13. Save the mapping. It is always good to save regularly.
14. Now, click on the Target object in the canvas. The properties will again appear at the bottom of the screen:



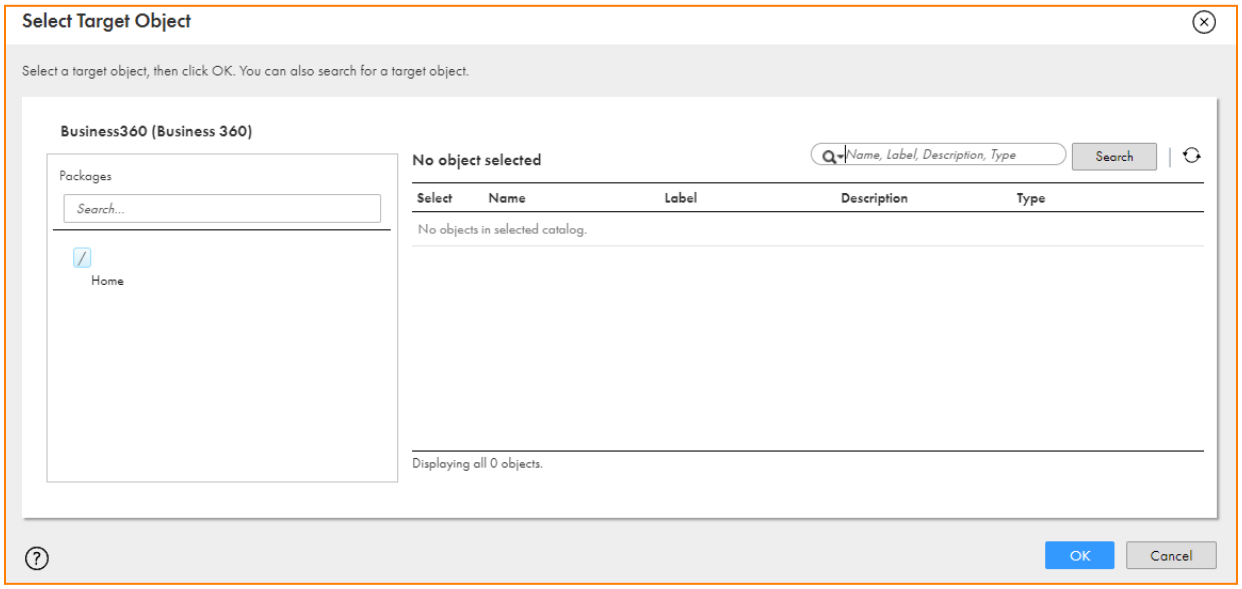
15. Give it a name, like Target_Customer

© Informatica.

16. Click on Target and select the Business360 connection:

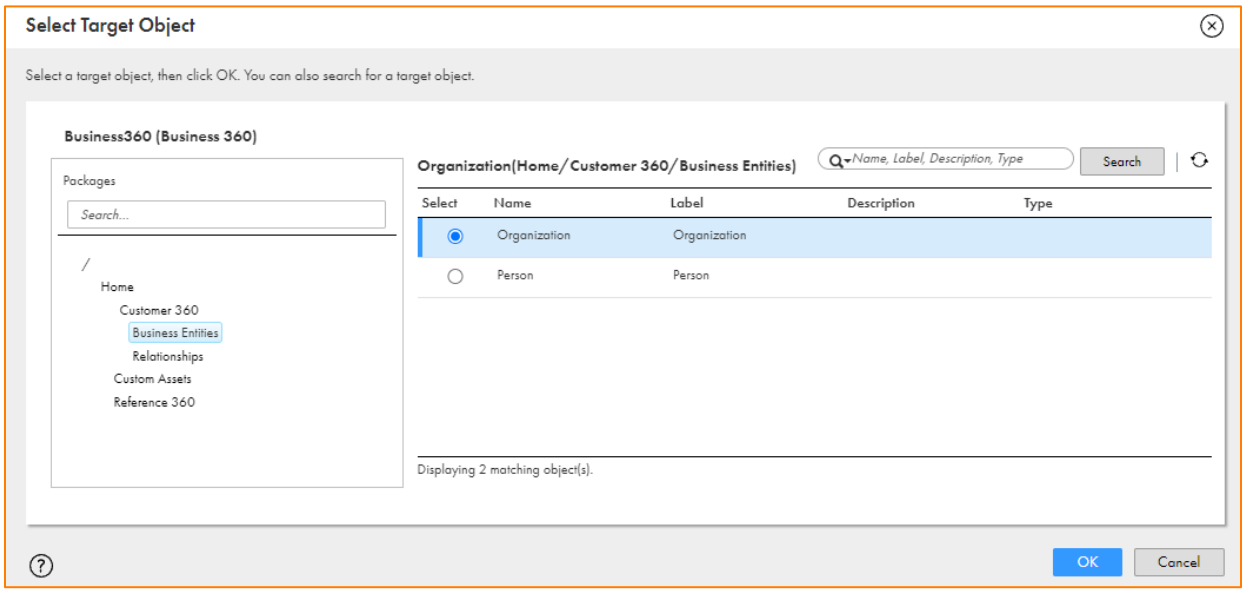


17. Click Select next to the Object field. The following screen appears:

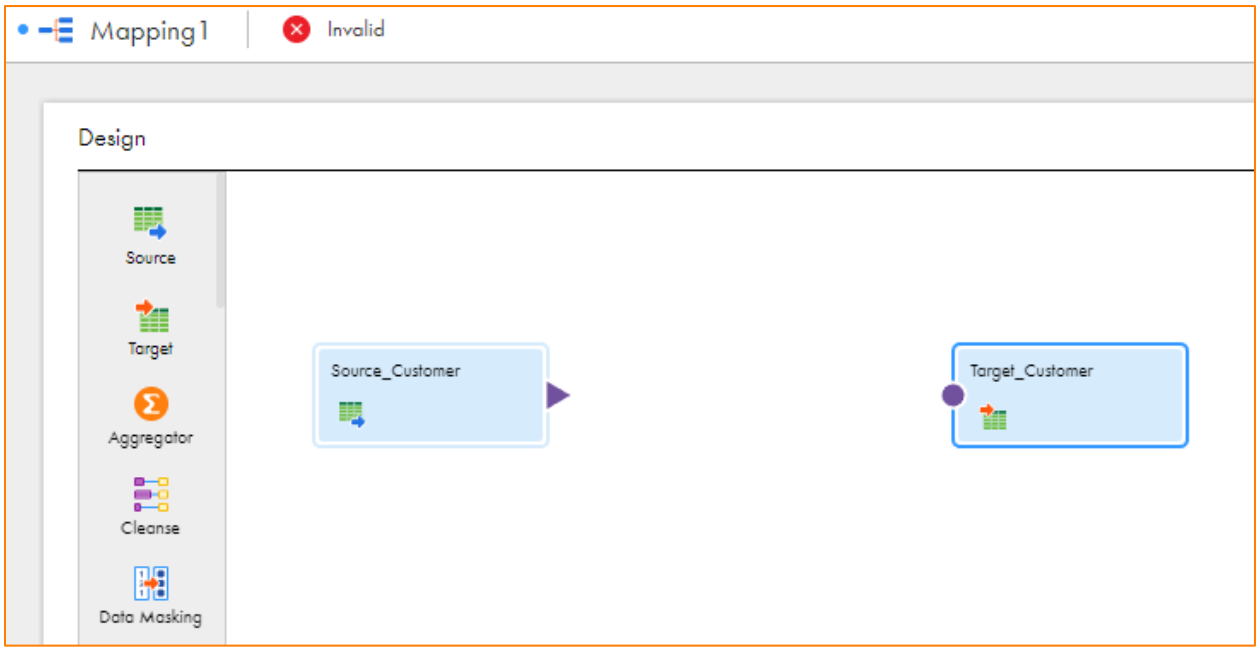


© Informatica.

18. Click Home, Customer 360, Business Entities. You will see the two out-of-the-box entities here.



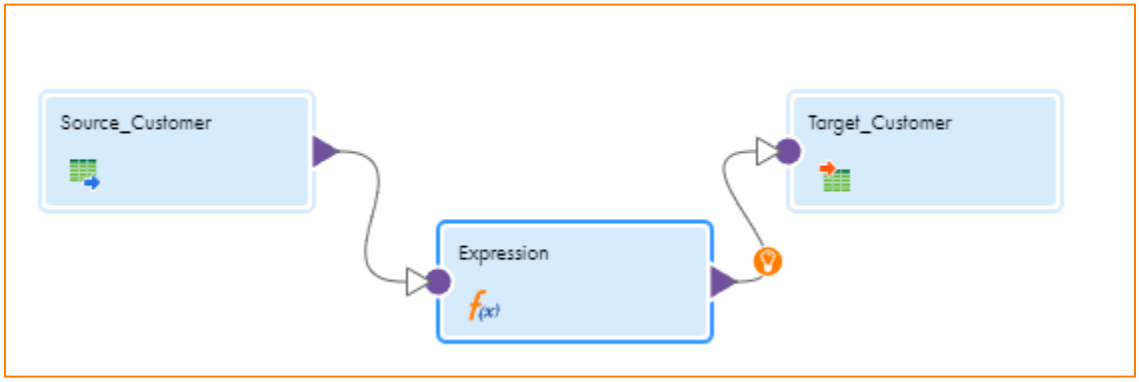
19. Click Organization and click OK
20. You see the link between the source and target disappear. This is normal behavior:



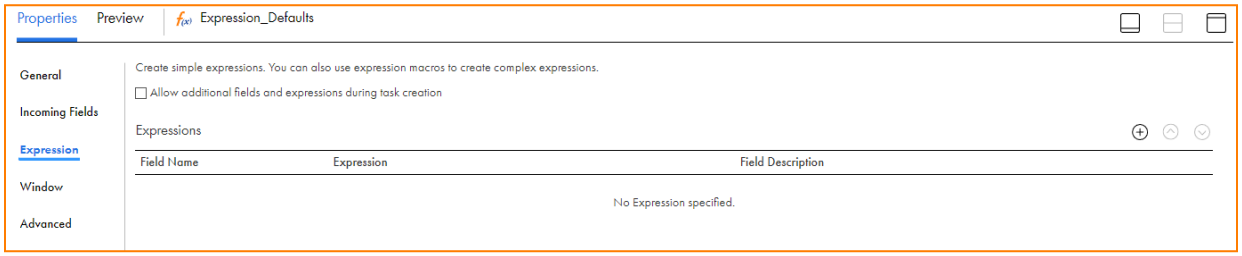
21. Now, you will have to set some default values. You do this by adding an Expression. On the toolbar on the left, scroll down until you see it and drag it into the canvas.

© Informatica.

22. Connect the source to the expression by click-and-drag from the output triangle on the source to the input circle of the expression. Repeat this from the expression to the target object. The mapping will now look like this:

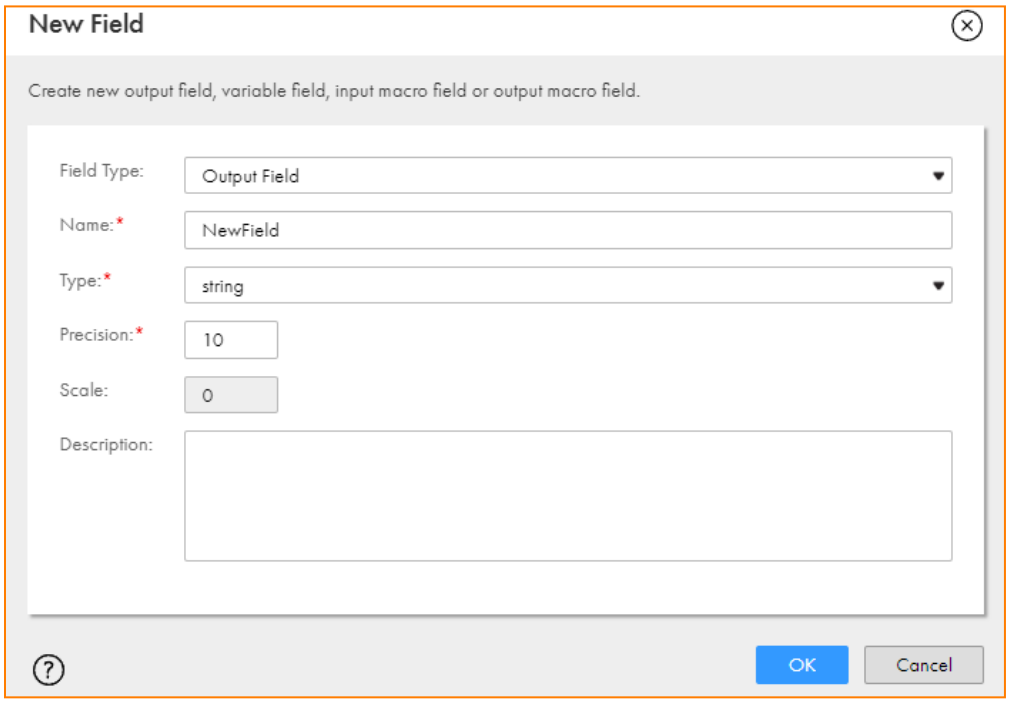


23. Click on the expression and name it Expression_Defaults.
24. Click on the expression tab. The following panel appears:

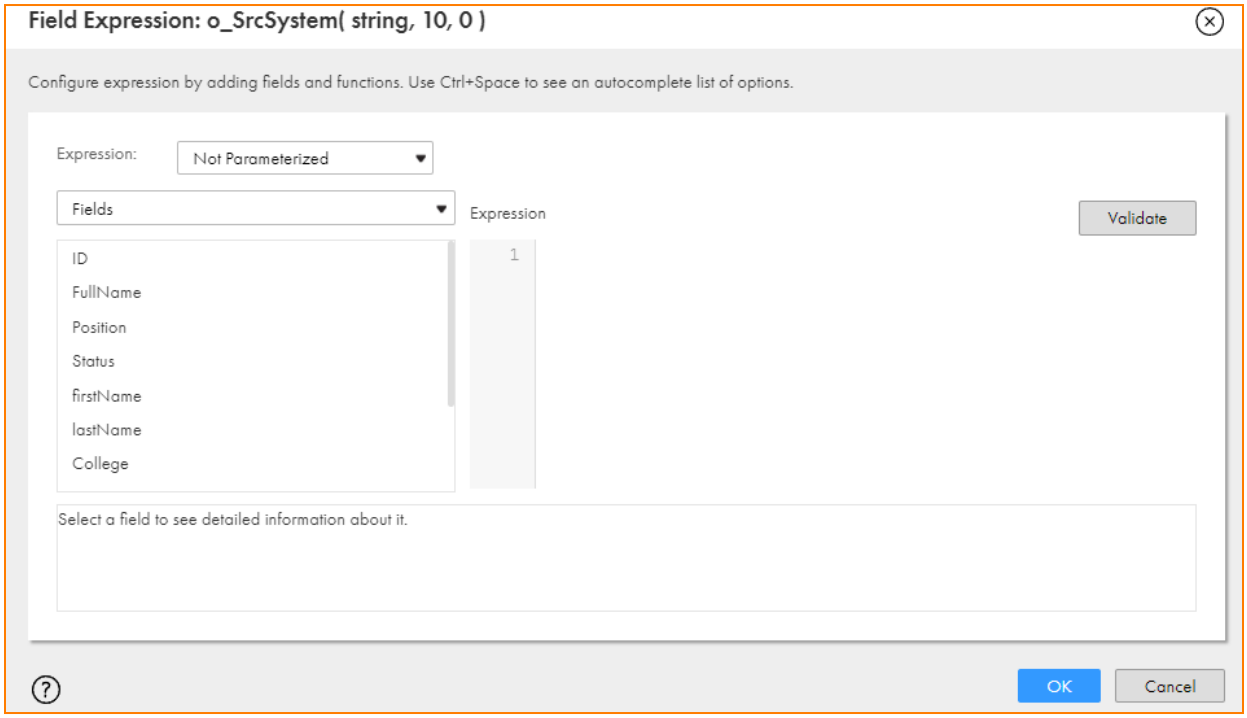


© Informatica.

25. Click on the highlighted "+" button to add an expression. The New Field panel appears:



26. Create a new output field called o_SrcSystem, with type and precision string (10). You will see it appear in the expression tab. Click Configure. The expression editor opens:



© Informatica.

27. In the Expression box, type 'CRM', including the quotes. Click Validate to check whether you entered the string correctly. Click Ok.
28. Add more output fields according to the below table:

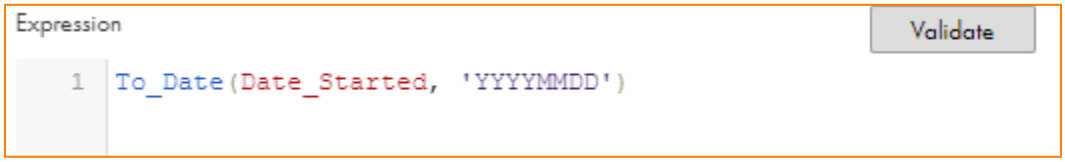
Field Name	Expression
o_sourcePKey_Address	DUNS_Id || '-Ship'
o_sourcePKey_Phone	DUNS_Id || '-Phone'
o_UsageType	'Business'
o_AddressType	'Shipping'
o_PhoneType	'Telephone'
o_AltIdentifierType	'DUNS'

29. Add one more field, called o_DateOfIncorporation of type date/time.
30. Click Configure to add the expression
31. Click on the Fields pull-down menu and select Built-in functions
32. Expand the Conversion group and click To_Date. The syntax appears in the expression editor, and you see the help at the bottom of the pane:

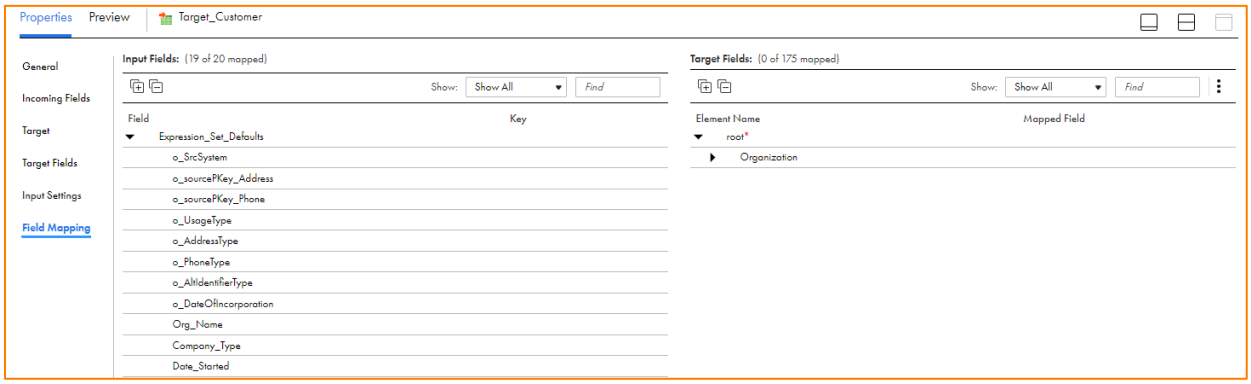


© Informatica.

33. Highlight the text 'string', go back to Fields, and click on Date_Started. You will see the text 'Add>' appear. Click on it. This will replace the highlighted string with the Date_Started field.
34. Complete the function by specifying the Birthdate format in the input file, which you can find in the profile output. The completed expression should be: "To_Date(Date_Started, 'YYYYMMDD')"



35. Click Validate to validate the expression and click OK.
36. The expression is complete; now it is time to configure the target fields. Click on the Target object and click on the Field Mapping tab. The easiest way to do the file mapping is to maximize the properties tab by using the maximize button on the top right. The screen will look like this:

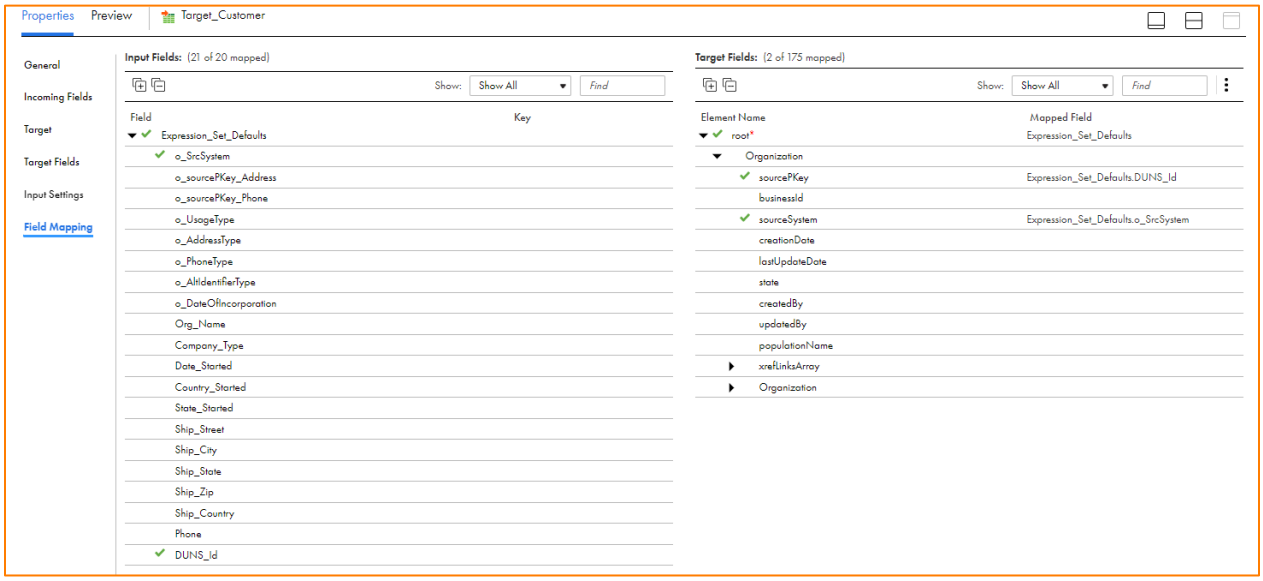


37. Expand the Organization container on the right. You see two required (starred) fields here:

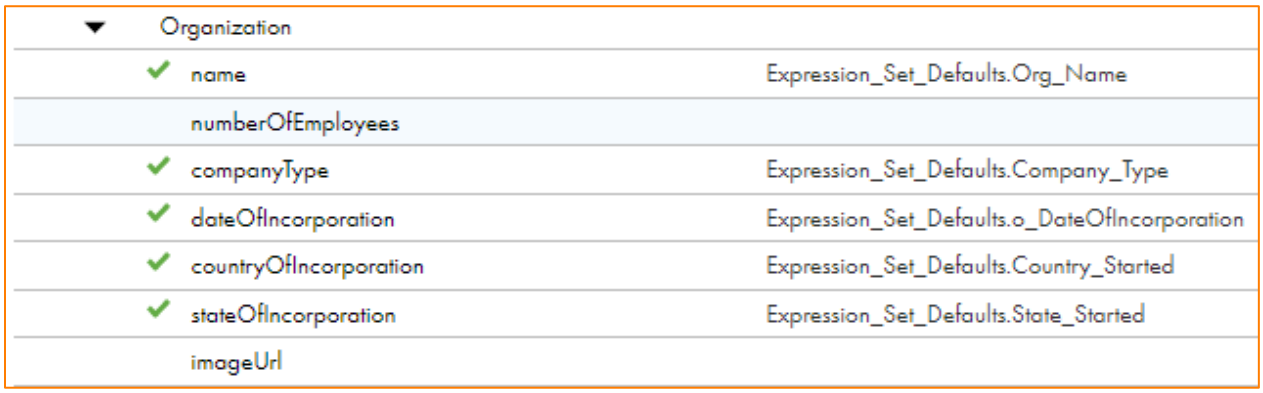


© Informatica.

38. Map the DUNS_Id from the source to sourcePKey and o_SrcSystem to sourceSystem. You can drag and drop the fields. It will now look like this:



39. Expand the Organization entity at the bottom and map the fields as shown here. Ensure you use the converted date field o_DateOfIncorporation.



© Informatica.

40. Expand the PostalAddressArray, and expand PostalAddress. Map the fields as shown here:

PostalAddressArray	
PostalAddress	Expression_Set_Defaults
sourcePKey	Expression_Set_Defaults.o_sourcePKey_Address
parentId	Expression_Set_Defaults.DUNS_Id
fieldGroup_State	
defaultIndicator	
addressType	Expression_Set_Defaults.o_AddressType
usageType	Expression_Set_Defaults.o_UsageType
addressStatus	
addressLine1	Expression_Set_Defaults.Ship_Street
addressLine2	
addressLine3	
country	Expression_Set_Defaults.Ship_Country
state	Expression_Set_Defaults.Ship_State
city	Expression_Set_Defaults.Ship_City
county	
postalCode	Expression_Set_Defaults.Ship_Zip

41. Expand the PhoneCommunicationsArray, expand PhoneCommunication, and map the fields as shown here:

PhoneCommunicationArray	
PhoneCommunication	Expression_Set_Defaults
sourcePKey	Expression_Set_Defaults.o_sourcePKey_Phone
parentId	Expression_Set_Defaults.DUNS_Id
fieldGroup_State	
defaultIndicator	
doNotCallIndicator	
phoneUsageType	Expression_Set_Defaults.o_UsageType
phoneType	Expression_Set_Defaults.o_PhoneType
phoneNumber	Expression_Set_Defaults.Phone
phoneNumberExtension	

© Informatica.

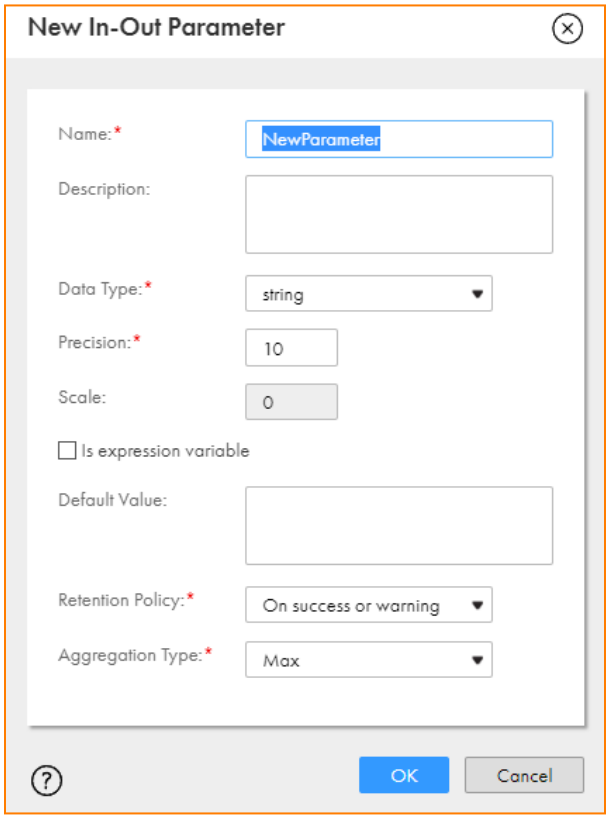
42. Expand AlternateIdentifierArray, expand AlternateIdentifier, and map the fields as shown:

AlternateIdentifierArray	
AlternateIdentifier	Expression_Set_Defaults
sourcePKey	Expression_Set_Defaults.DUNS_Id
parentId	Expression_Set_Defaults.DUNS_Id
fieldGroup_State	
idType	Expression_Set_Defaults.o_AltIdentifierType
altIdValue	Expression_Set_Defaults.DUNS_Id

43. Save the mapping. The mapping should now be valid.
44. One more thing to do; the mapping needs a runtime parameter. Click on the Parameters icon at the top of the screen:

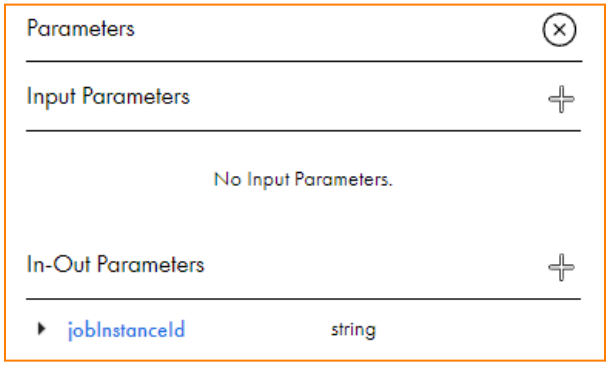


45. Click on the "+" sign for In-Out Parameters. The following screen comes up:



© Informatica.

46. Give the parameter the name "jobInstanceId". The name is case-sensitive
47. Change the Precision to 100
48. Click Ok. The parameter is now created:



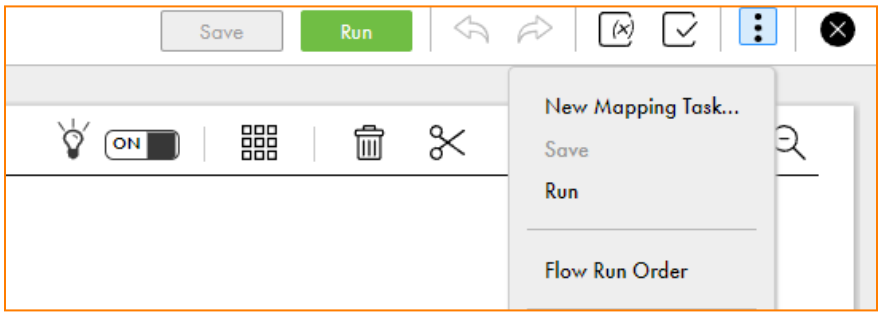
49. Save the mapping

© Informatica.

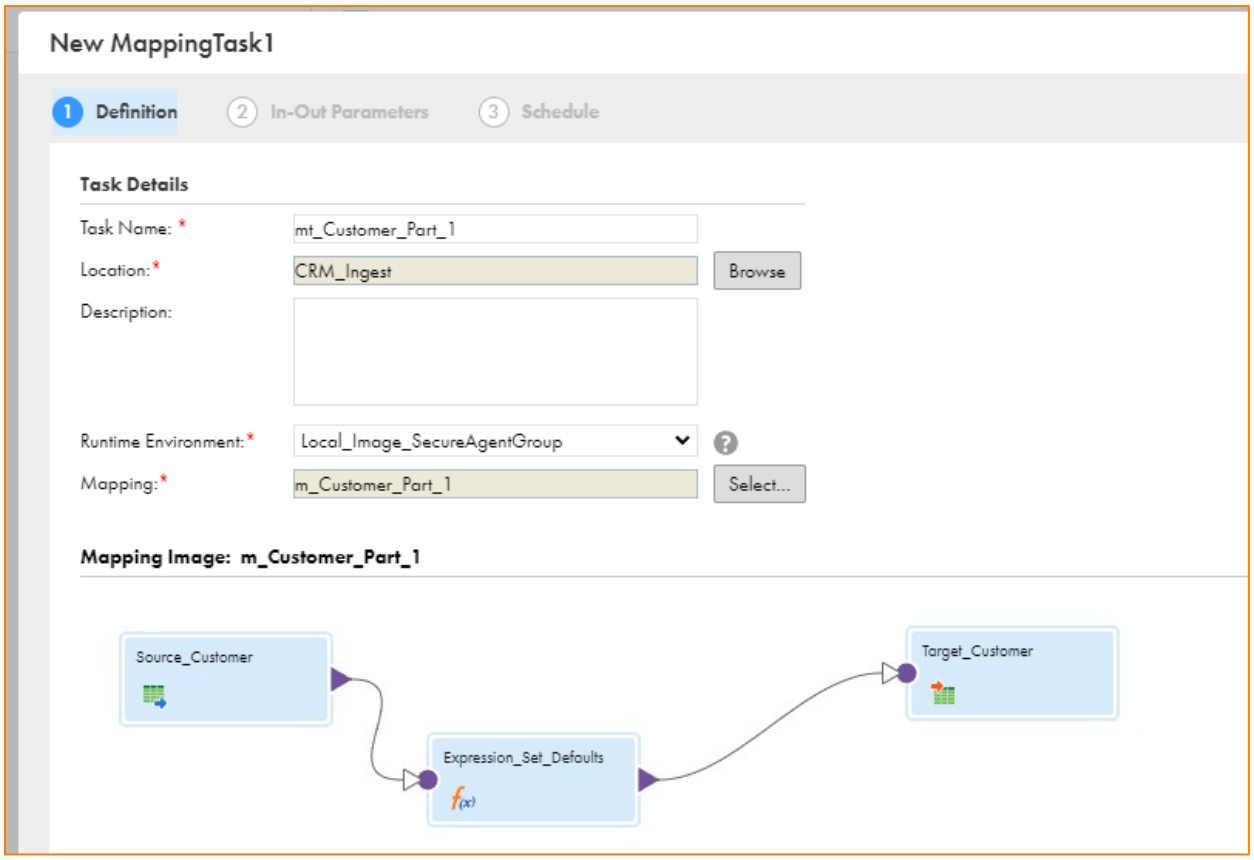
Instructions/Steps – Create Mapping Task and Taskflow:

The following steps are creating the mapping task and taskflow in CDI. The taskflow will then be used in a job definition in Business 360.

1. Click on the ellipsis at the top right of the screen and select New Mapping Task:



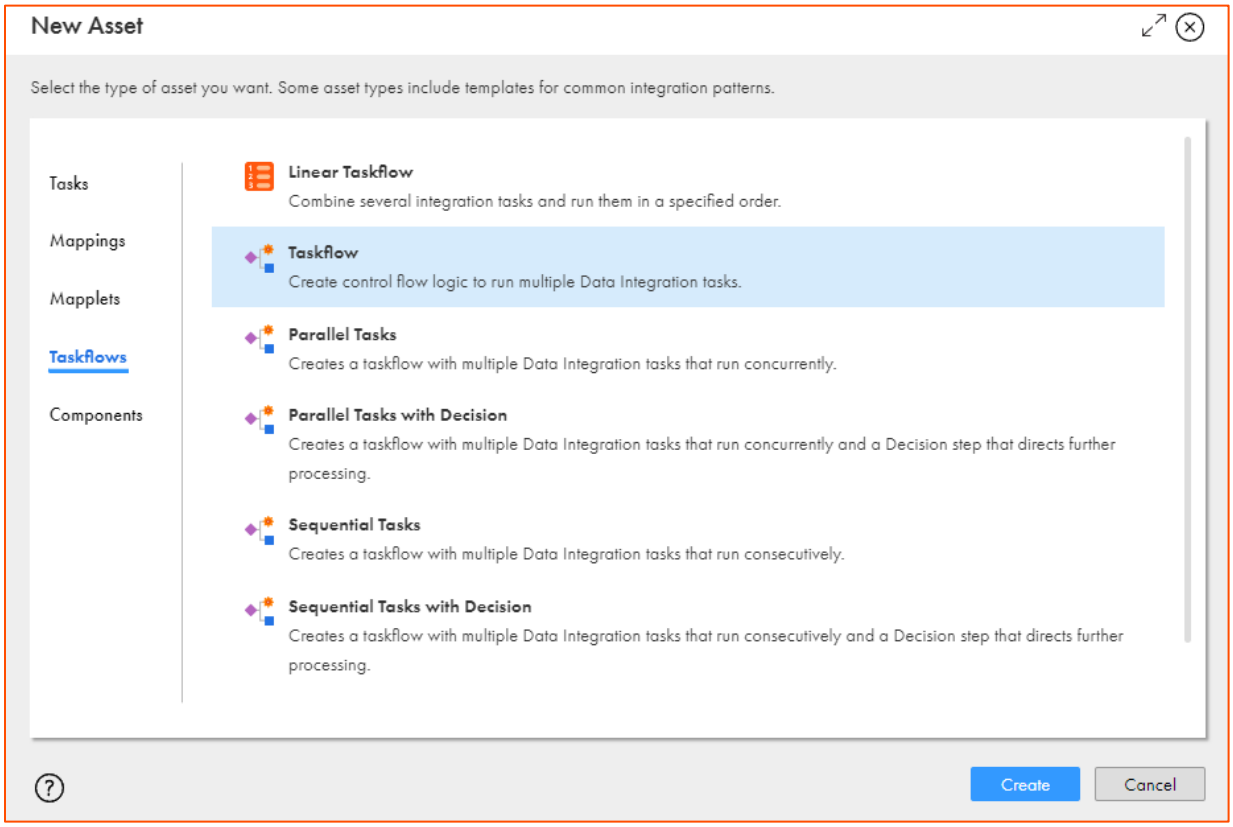
2. Give the mapping task a name, like mt_Customer_Part_1, and check whether the location is correct.



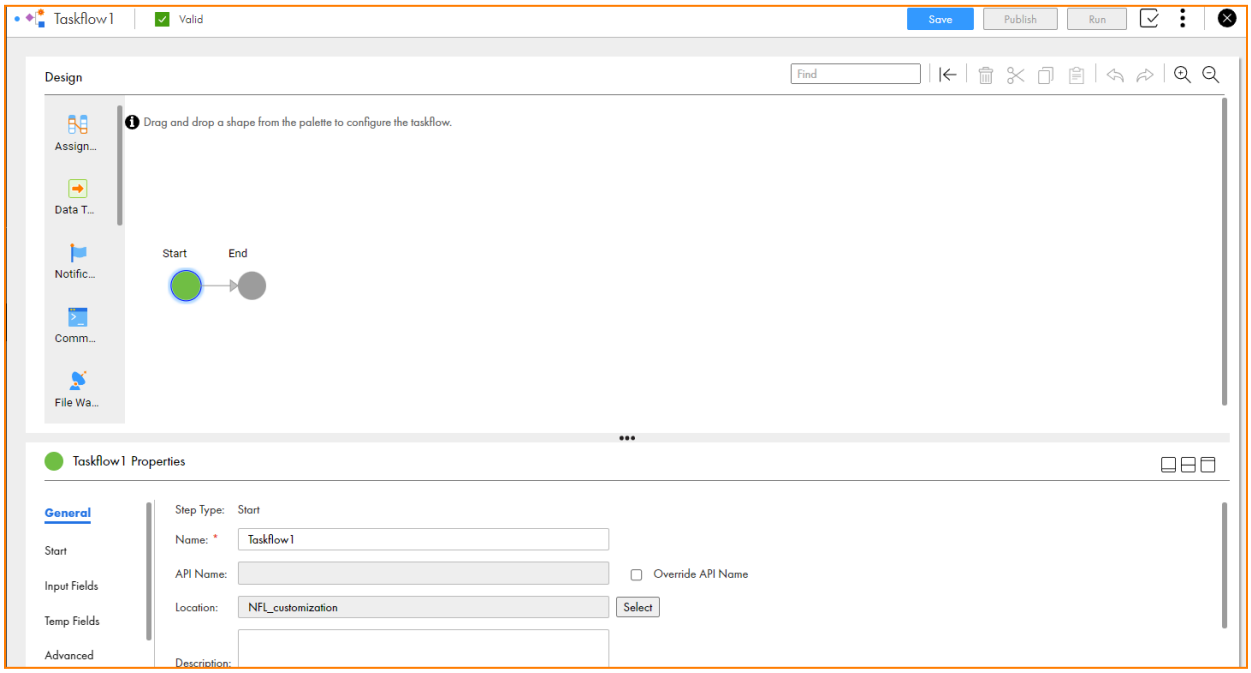
3. Click Next. You will see the In-Out parameter jobInstanceId.

© Informatica.

4. Click Next and click Finish
5. Next is to create the taskflow. Click New at the top left of the screen and click the Taskflow panel:

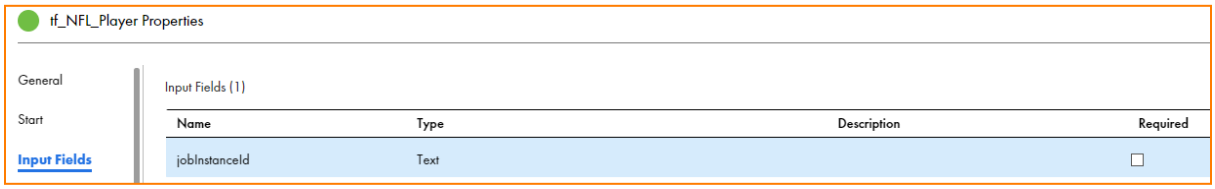


6. Click Taskflow and click Create. The design canvas will open:

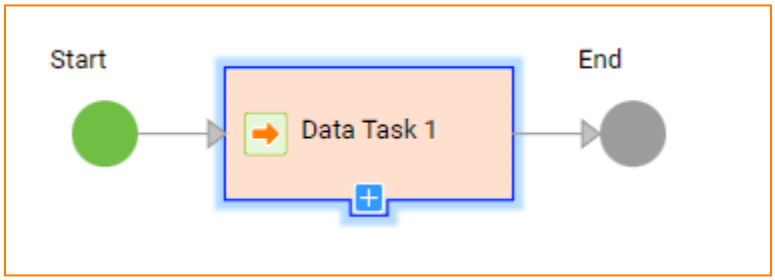


© Informatica.

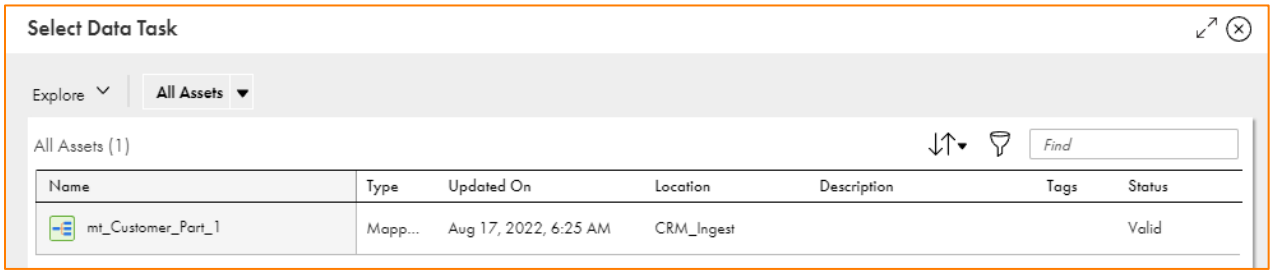
7. Name the taskflow tf_Customer_Part_1 and check whether the location is your project.
8. Click on Input fields.
9. Click on the "+" sign to add a field. Name it (case sensitive) jobInstanceId.



10. Click on Data Task in the canvas and drag it between the start and end steps. It will now look like this:



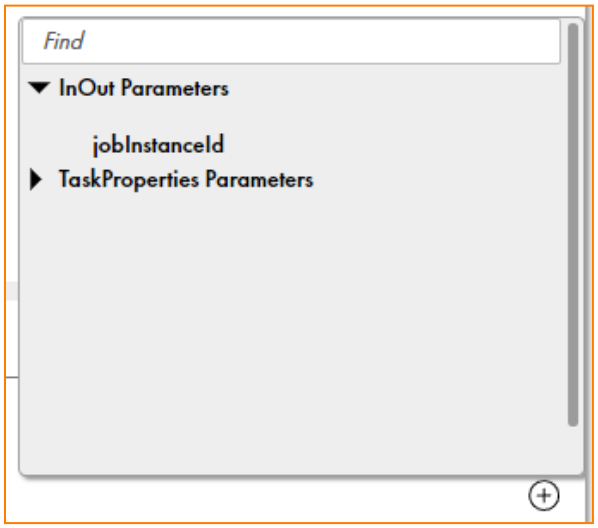
11. In the properties, give the task a name; the best practice is to use the mapping task name, so in this case, mt_Customer_Part_1.
12. Go to the Data Task tab and configure the task to execute. Click on Select, and the following screen pops up:



13. Highlight the task and click Select. The task metadata will be imported.

© Informatica.

14. Click on Input Fields and click on the "+" sign at the right to add an input field. Expand the InOut Parameters container. You will see the parameter jobInstanceId:

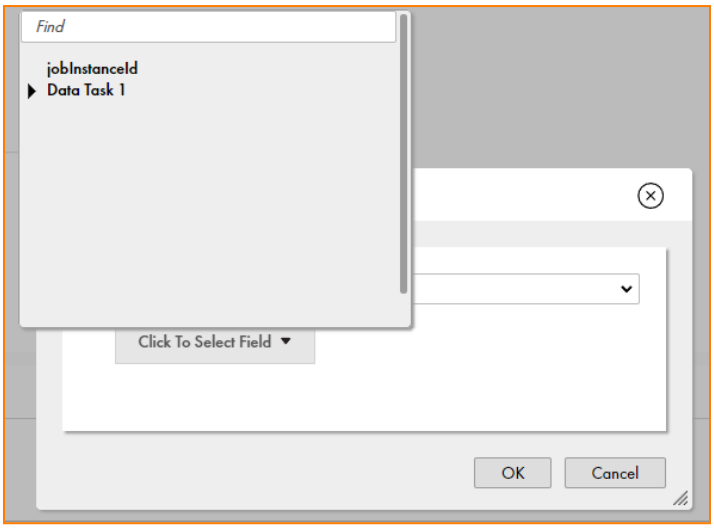


15. Select the parameter to add it as an input field:

Input Fields (1)

Name	Type	Value	Path
jobInstanceId	Text		mt-NFL-Player → Inout → jobInstanceId

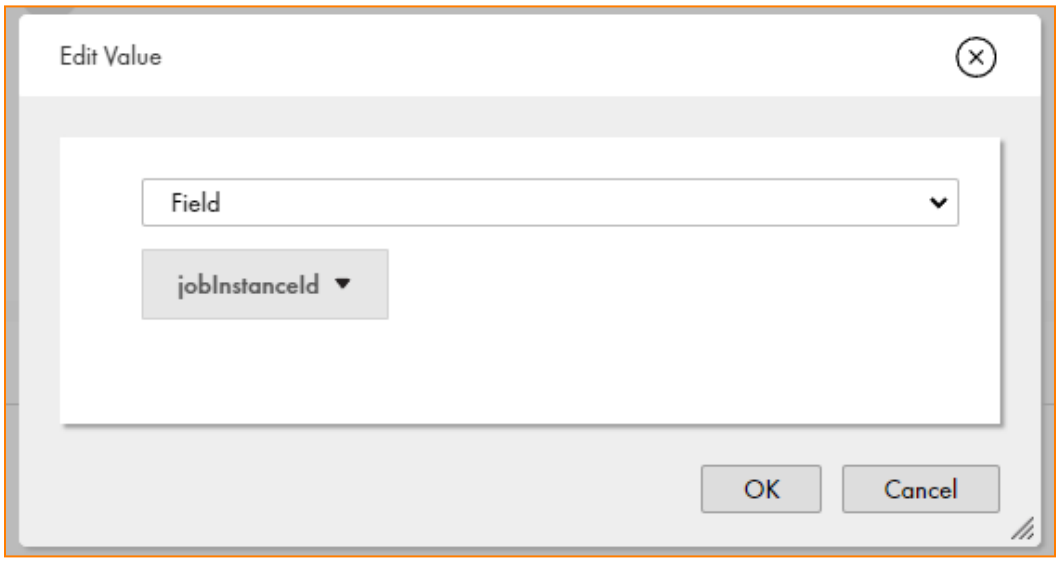
16. Hover over the field, and a Pencil icon will appear on the right. Click on it. The Edit Value popup appears.
17. In the pull-down menu, change Content to Field and click on "Click to Select Field". A list of available fields pops up:



.

© Informatica.

18. Select the field jobInstanceId. The Edit Value popup will now look like this:



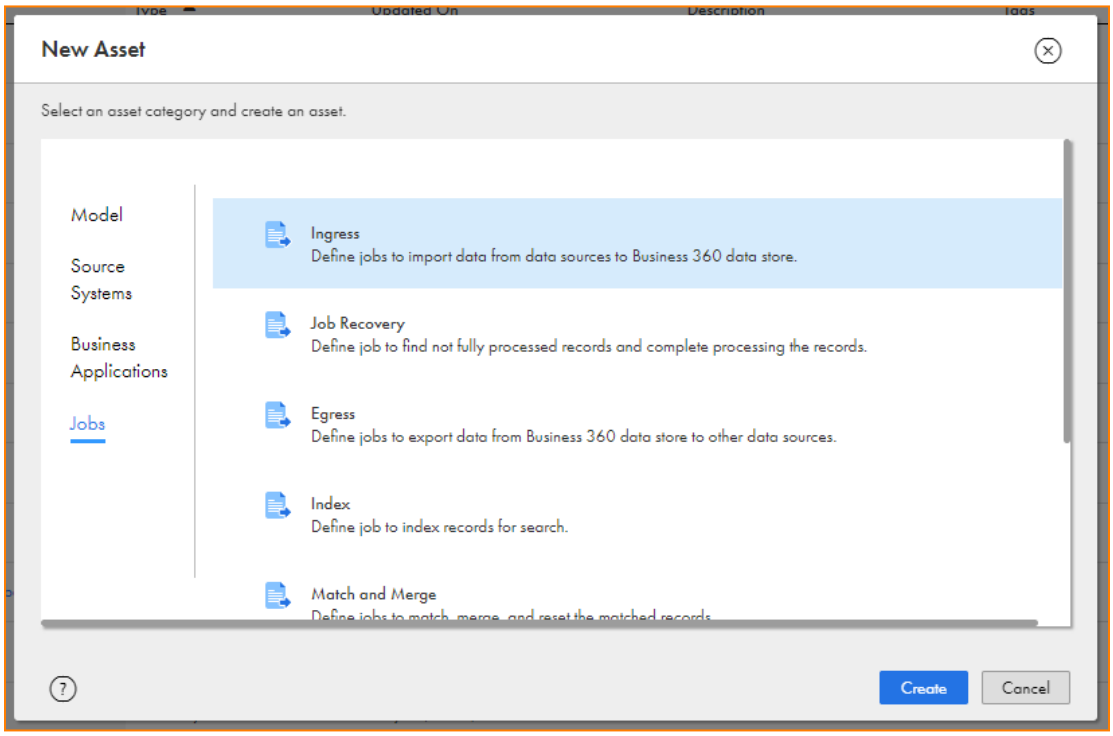
19. Click OK and Save the taskflow.
20. Click Publish to publish the taskflow for use in Business 360.

This completes the actions in CDI. The following steps are in the Business 360 Console to create the Ingress job.

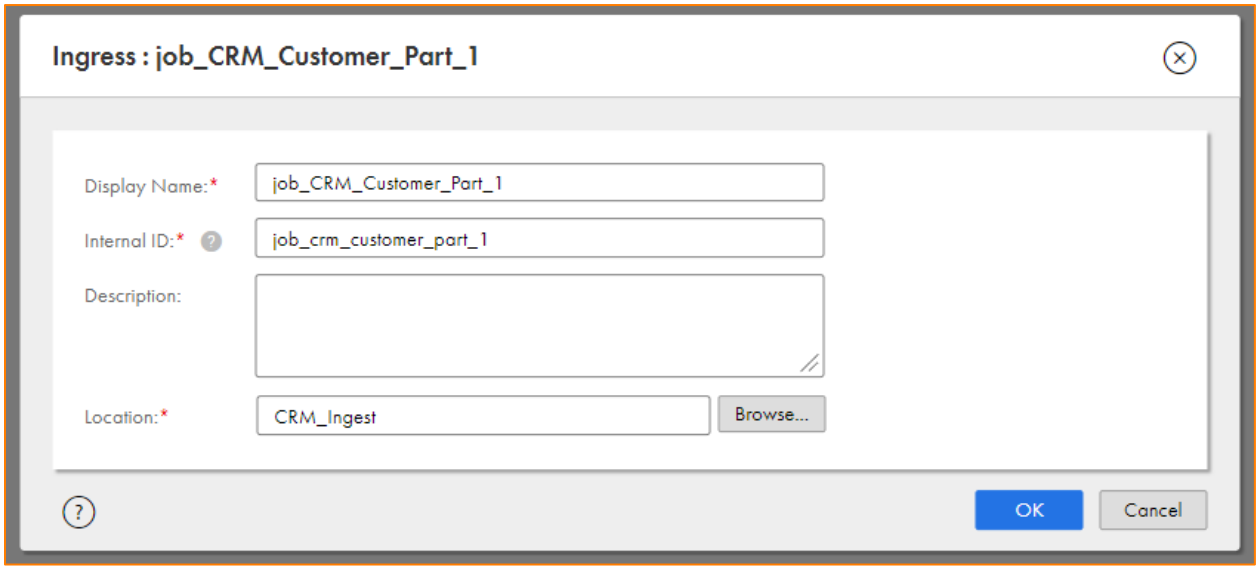
© Informatica.

Instructions/Steps – Create Business 360 job and test:

1. In the Business 360 Console, click on New and open the Jobs tab. The various job types are listed here:

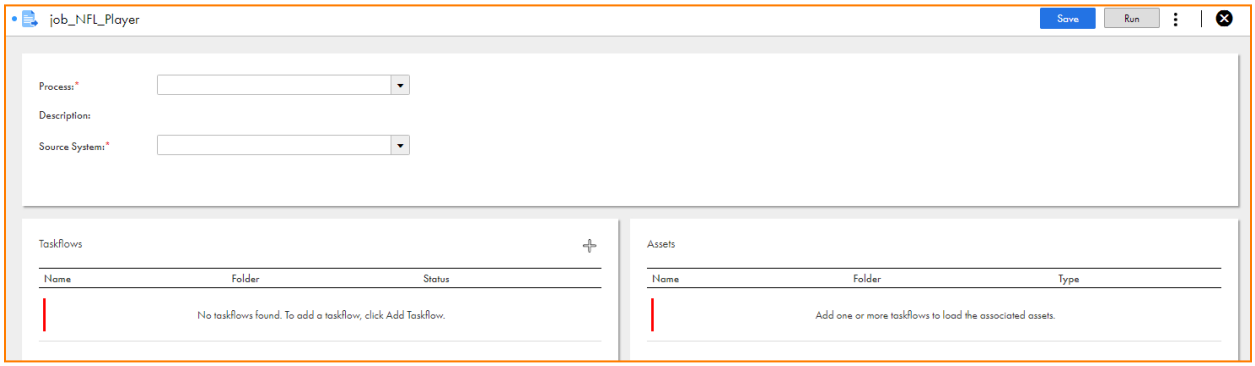


2. Highlight Ingress and click on Create.
3. Give it a name and ensure the location is your NFL_customization folder:

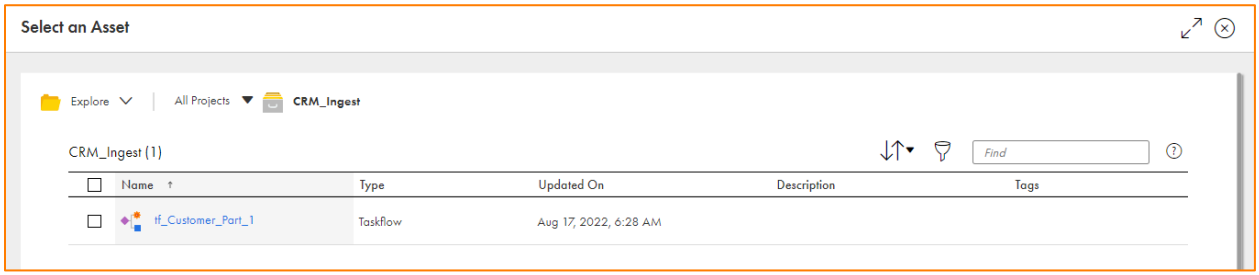


© Informatica.

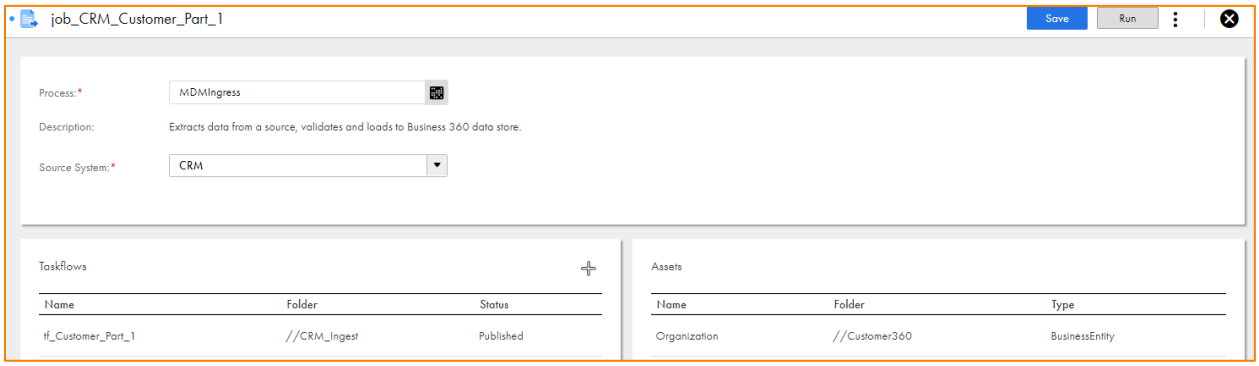
4. Click Ok. The job specification page appears:



5. The Process is MDMIngress; there is no need to change this
6. Select the Source System as CRM
7. In the Taskflows pane, click the "+" sign to add a taskflow. The select screen appears:



8. Select the taskflow you created earlier and click Select. You see the taskflow added, and in the Assets pane you see the entity affected:



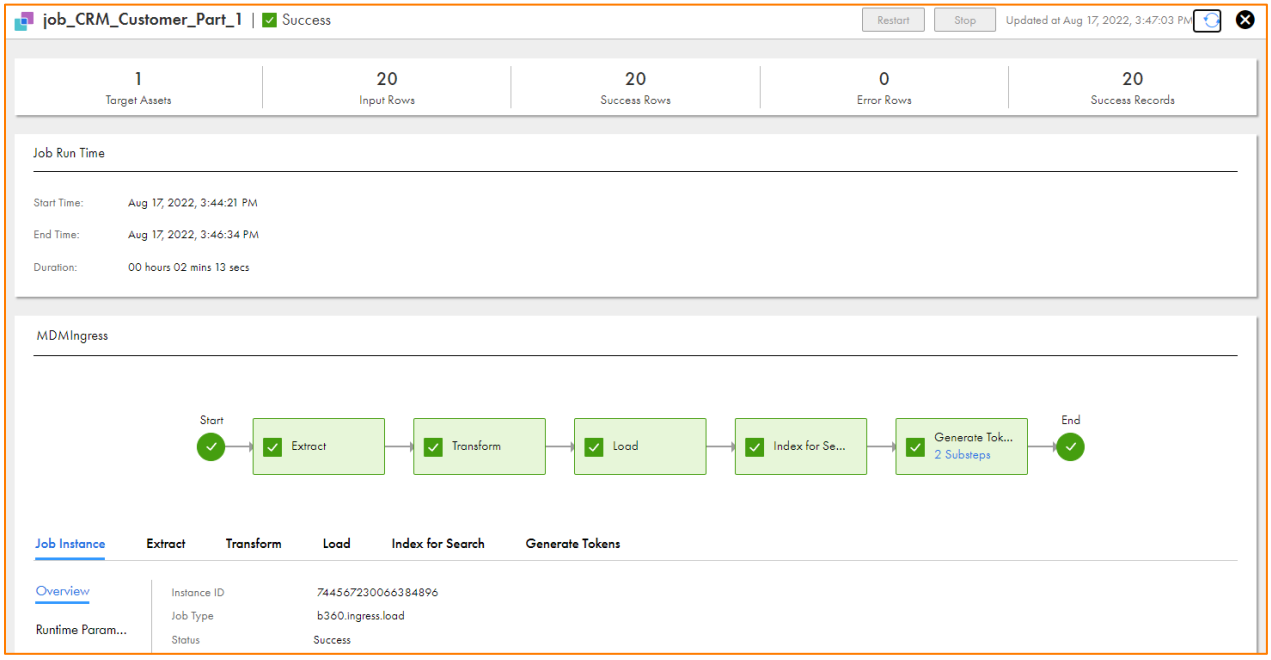
9. Click Save

And now you can run the job!

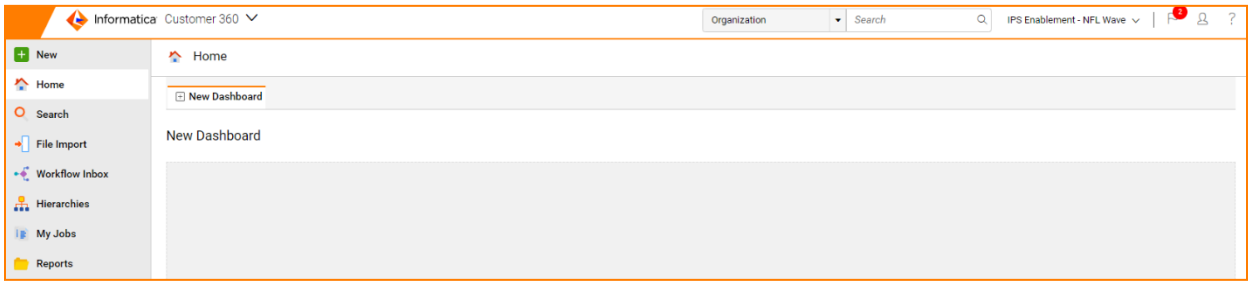
10. Click Run.

© Informatica.

11. You will get a popup telling you the job has started. Click on the link in the popup or go to My Jobs to monitor the progress
12. When the job is finished, you should see a screen like this showing the statistics:



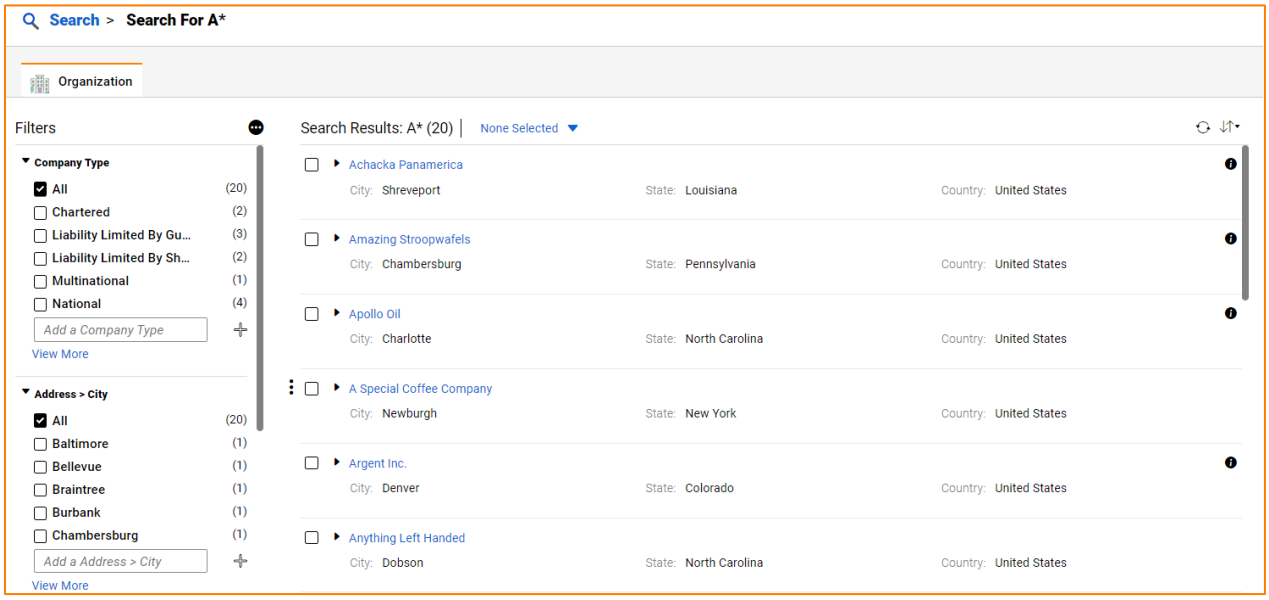
13. You can see the metrics per step by clicking on the steps. Since all records are successful, let's check the loaded records.
14. Click on the Transform step and you will see three errors. This is the built-in validation of the DUNS number stored in the Alternative Identifier. The data will still be loaded.
15. Go to Customer 360. The Home screen appears:



16. In the search box at the top of the screen, search for Organization. All organizations loaded in this exercise start with an A, so you can search for A*

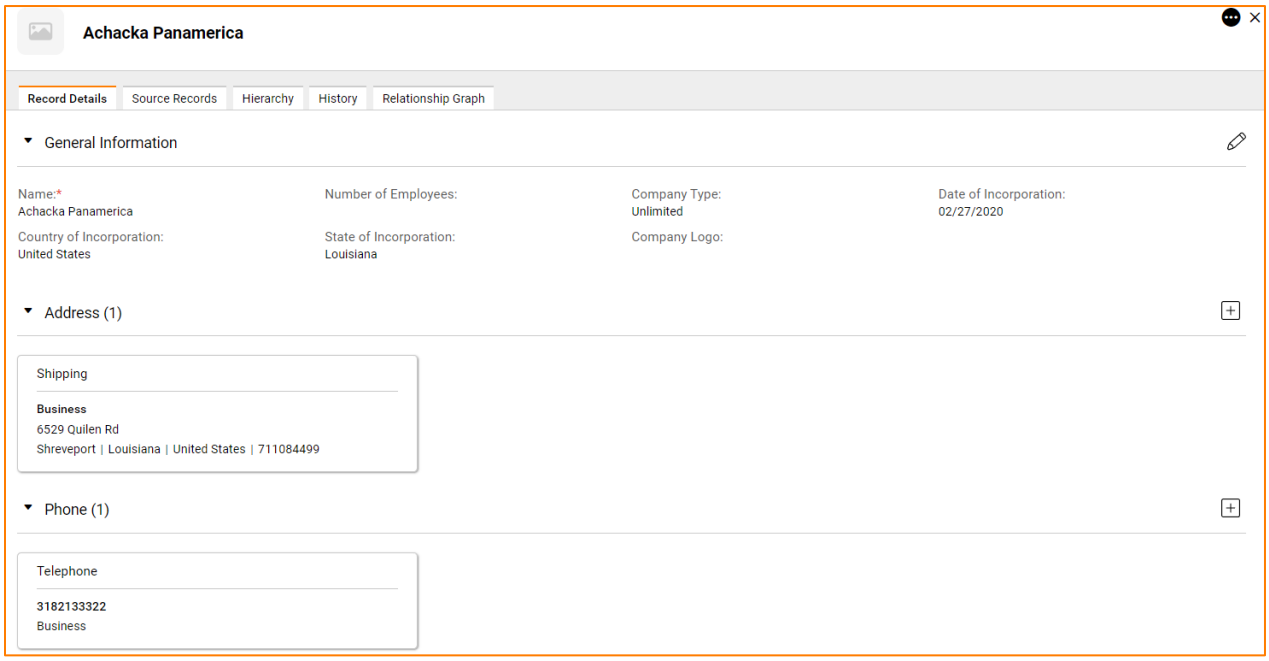
© Informatica.

17. The search results screen appears:



You see the facets on the left-hand side to refine the search results.

18. Click on one of the Organizations found to view the details. You will see the Address, Phone, and Identifier child elements populated:

19. 

This concludes the first lab, loading Organization with child elements and out-of-the-box error handling.

© Informatica.

LAB Part 2: Multiple child elements

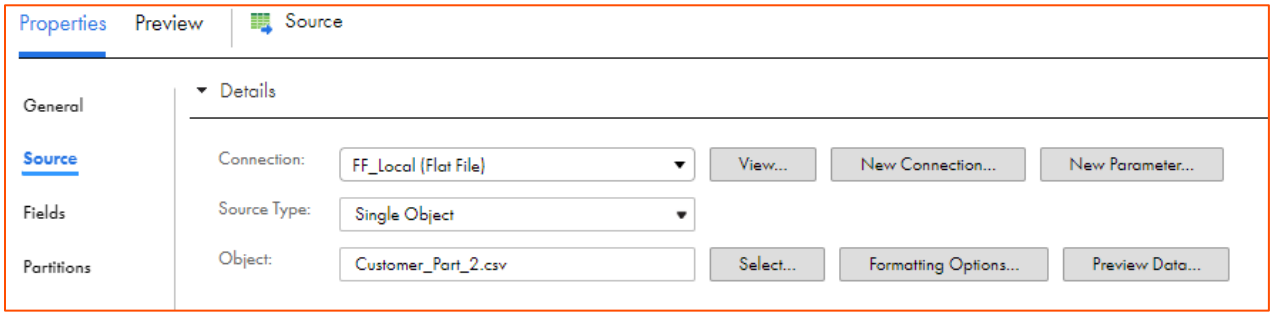
DURATION: 45 minutes

Purpose:

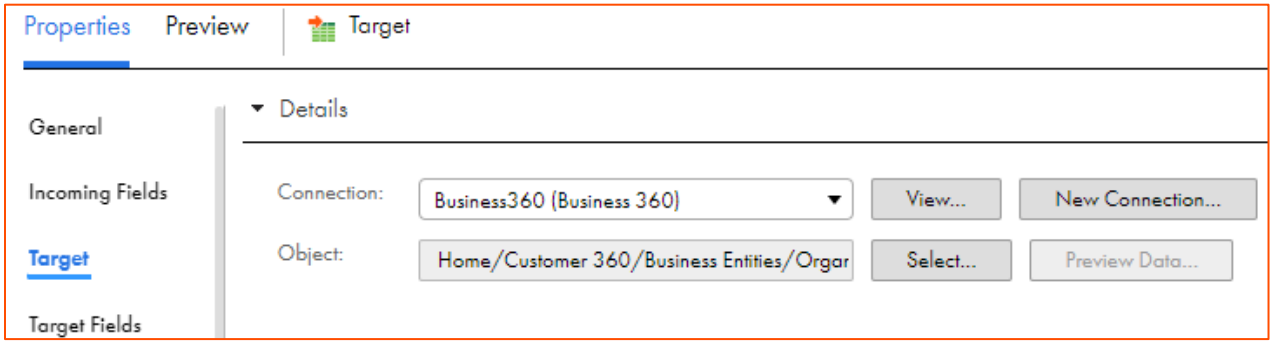
The purpose of this lab is to load customers with two addresses and phone numbers in one process. The mapping you will build contains multiple target objects to load the specific elements to the Business360 landing area. Using the same key for the root object will ensure all elements are placed into one Organization document.

Instructions/Steps – Create Mapping, Task, and Taskflow:

1. Log into IDMC and go to Data Integration.
2. Create a new mapping m_Customer_Part_2 and save it in the CRM_Ingest project
3. Select the file Customer_Part_2.csv as source:



4. Select Organization as Target, as done in the previous mapping:



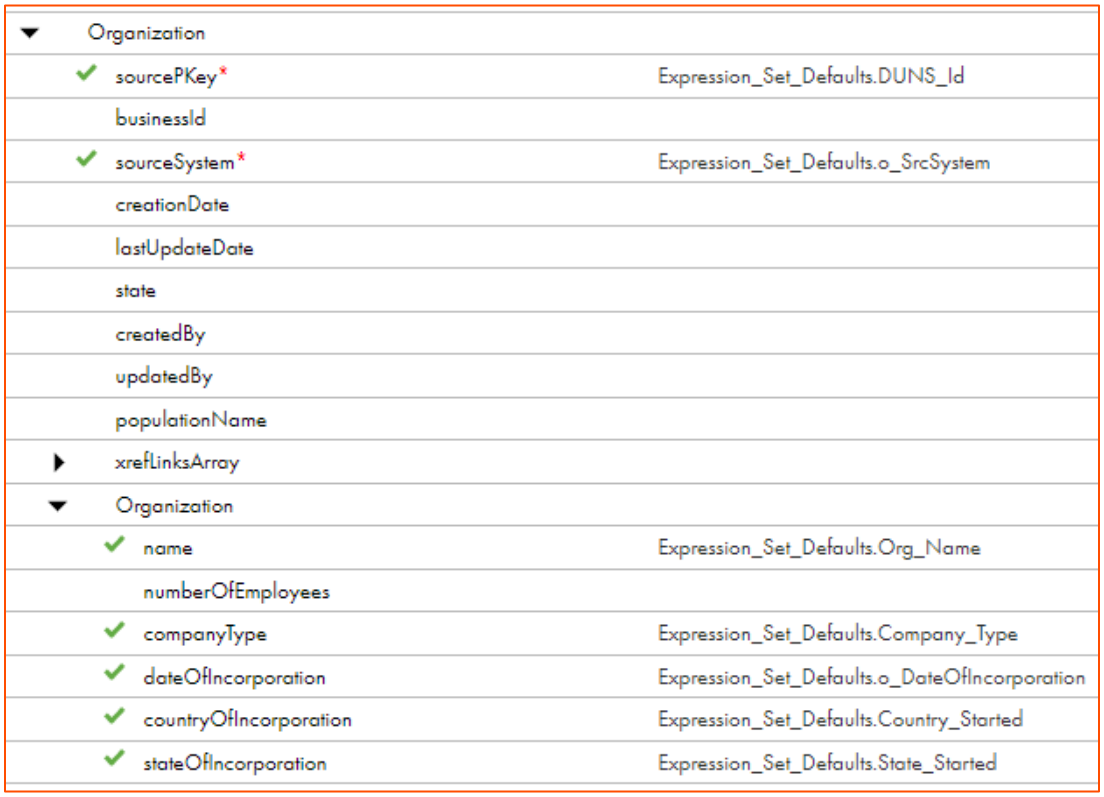
5. Name the target Target_Organization_root

© Informatica.

6. Create an Expression, Expression_Set_Defaults, and link it to the source. Set the following defaults:

Field Name	Expression
o_SrcSystem	'CRM'
o_AltIdentifierType	'DUNS'
o_DateOfIncorporation	To_Date(Date_Started, 'YYYYMMDD')

7. Connect the expression to the target and map the fields for the root:



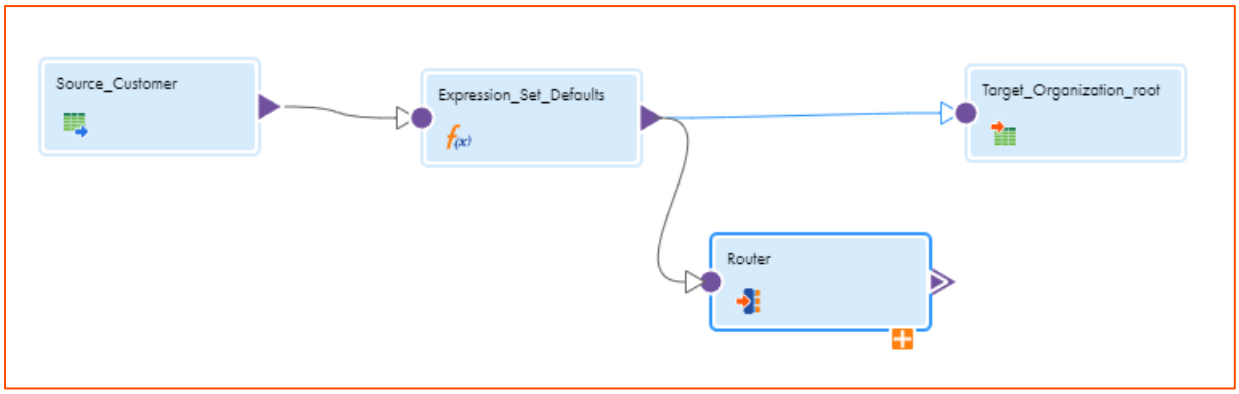
© Informatica.

8. When a child element has only one occurrence, like AlternateIdentifier, you can add this to the mapping for the root element. The best practice is to avoid having too many targets in the mapping. You should only create separate targets when a child element occurs more than once.

   Open the AlternateIdentifierArray, open AlternateIdentifier and map the fields as shown here:

AlternateIdentifierArray	
AlternateIdentifier	Expression_Set_Defaults
sourcePKey	Expression_Set_Defaults.DUNS_Id
parentId	Expression_Set_Defaults.DUNS_Id
fieldGroup_State	
idType	Expression_Set_Defaults.o_AltIdentifierType
altIdValue	Expression_Set_Defaults.DUNS_Id
issuingAuthority	
IdentifierStatus	
effectiveStartDate	
effectiveEndDate	

9. Save the mapping. It should be valid.
10. Drag a Router to the mapping canvas from the Transformation toolbar on the left.
11. Connect it to the expression. The mapping will look like this:



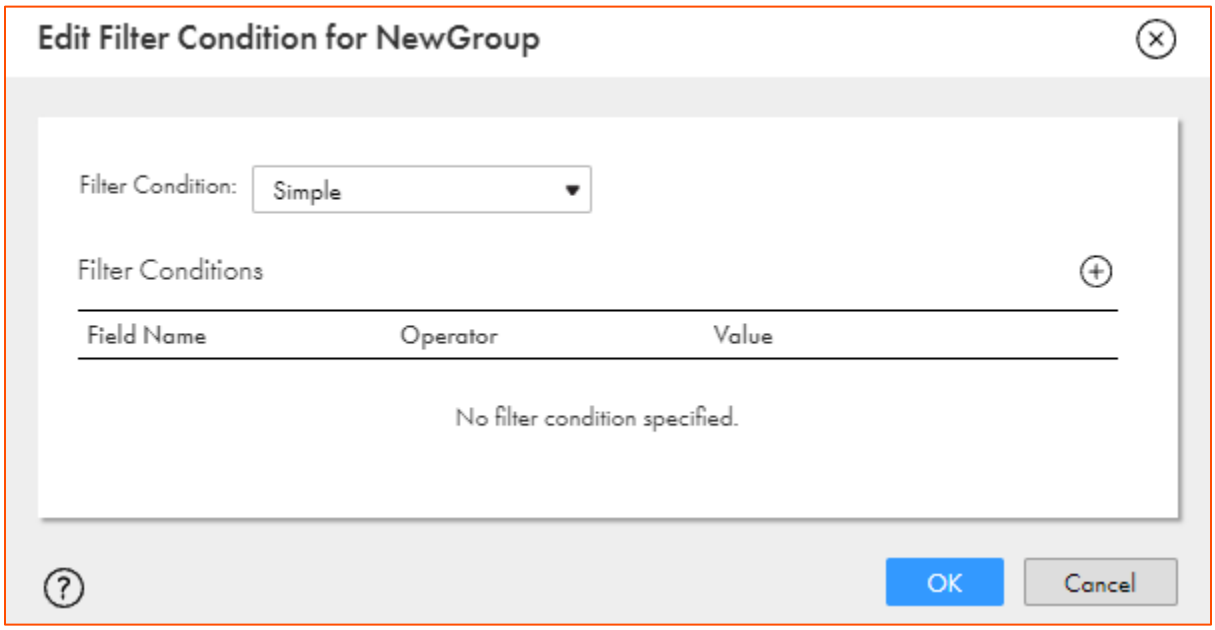
12. In the Router properties, click on Output Groups. You will see a default1 group here for all input records that do not fulfill any of the conditions.

© Informatica.

13. Click on the (+) sign to add an output group. The screen will change like this:

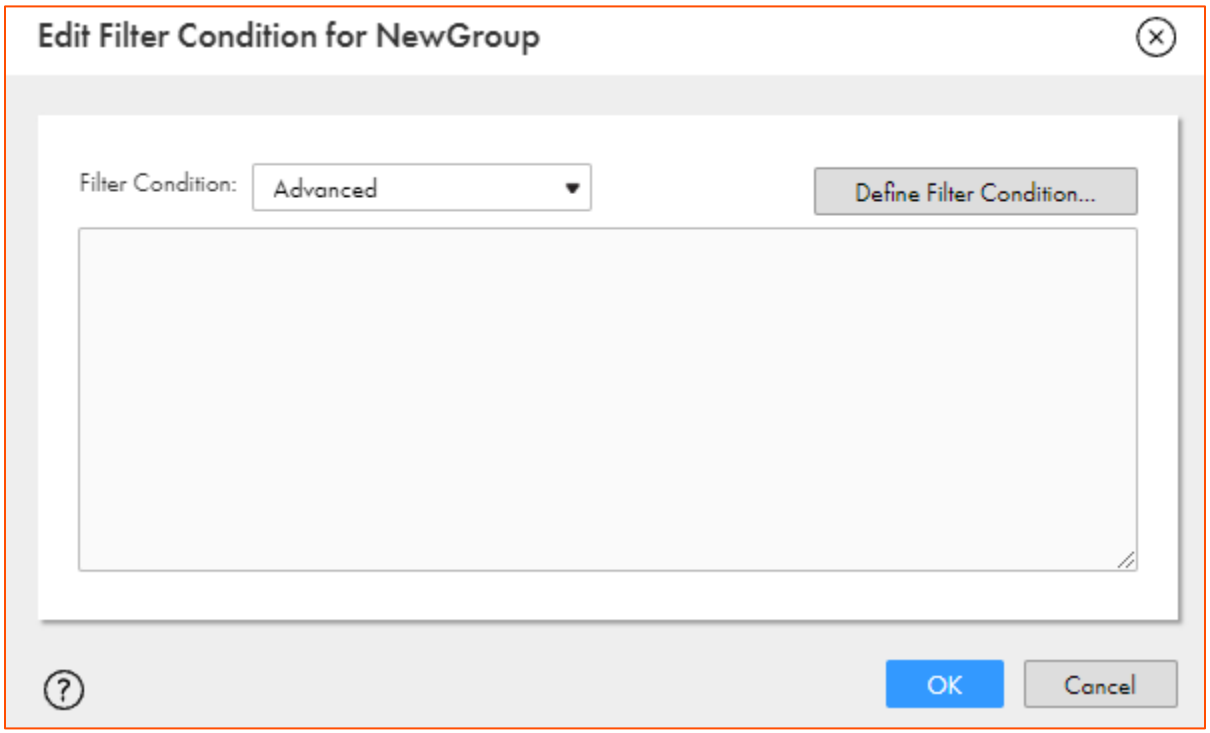


14. Click on Configure to add the condition. The following screen appears:

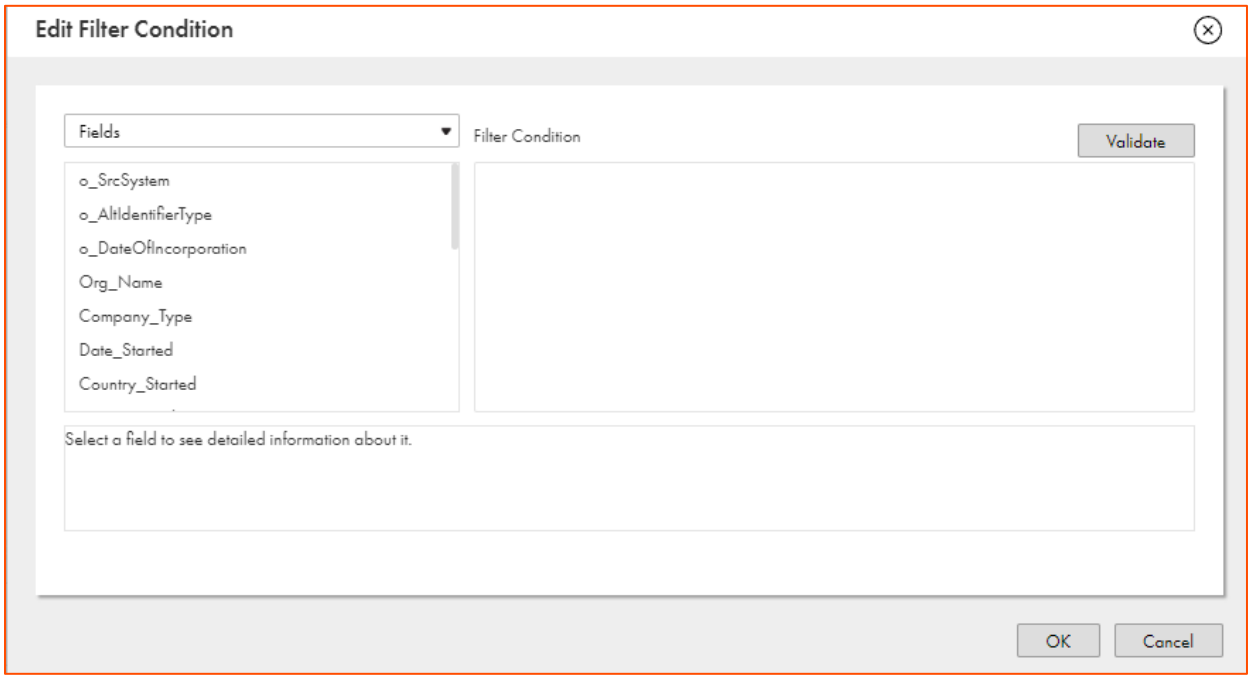


© Informatica.

15. Change the Filter Condition type to Advanced. The following screen appears:



16. Click on Define Filter Condition. The following screen appears, similar to the expression editor:



© Informatica.

17. Enter the condition "`Not IsNull(Ship_Street)` ". You can either type it in or use the Fields and Functions dropdown.
18. Give the output group a logical name, like ShippingAddress. The router will now look like this:



19. Add three more output groups, for Billing Address, Phone, and Fax, all with the Not IsNull condition. The router will look like this:

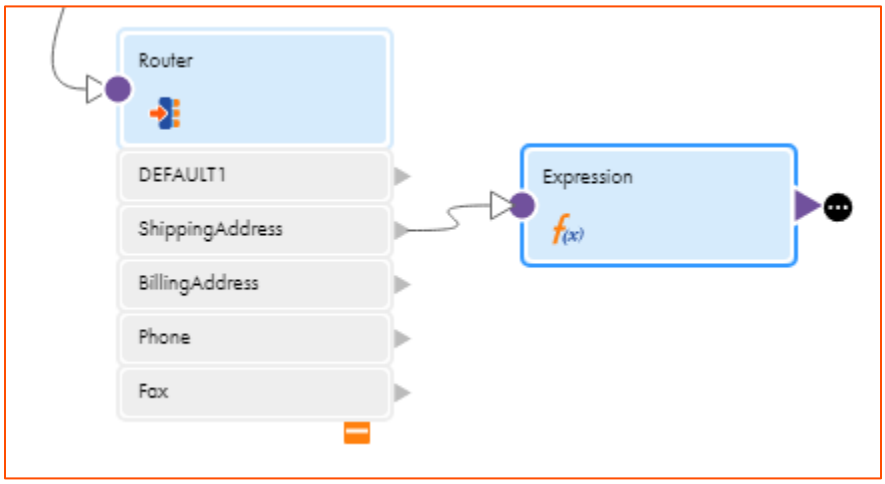


20. In the mapping canvas, click on the orange + sign you see below the router. The output groups appear:



© Informatica.

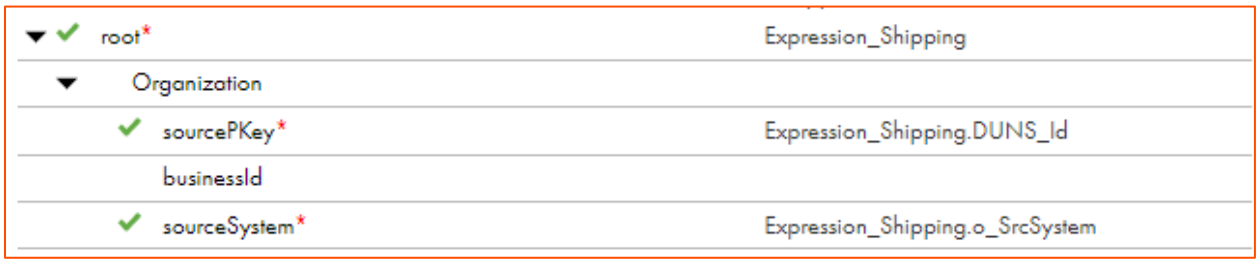
21. Add an expression and link it to the ShippinAddress output group:



22. Rename the expression to Expression_Shipping and configure it to populate the following fields:

Field Name	Expression
o_sourcePKey_Shipping	DUNS_Id || '-Ship'
o_AddressType	'Shipping'
o_AddressUsageType	'Business

23. Add a new Target and name it Target_ShippingAddress.
24. Select the Business360 connector and select the Organization business entity.
25. Connect it to the Expression_Shipping
26. In Target_Shipping_Address, go to the field mapping.
27. In the root, map the sourcePKey and the source System from DUNS-Id and o_SecSystem:



28. Expand Organization, PostalAddressArray, and PostalAddress.

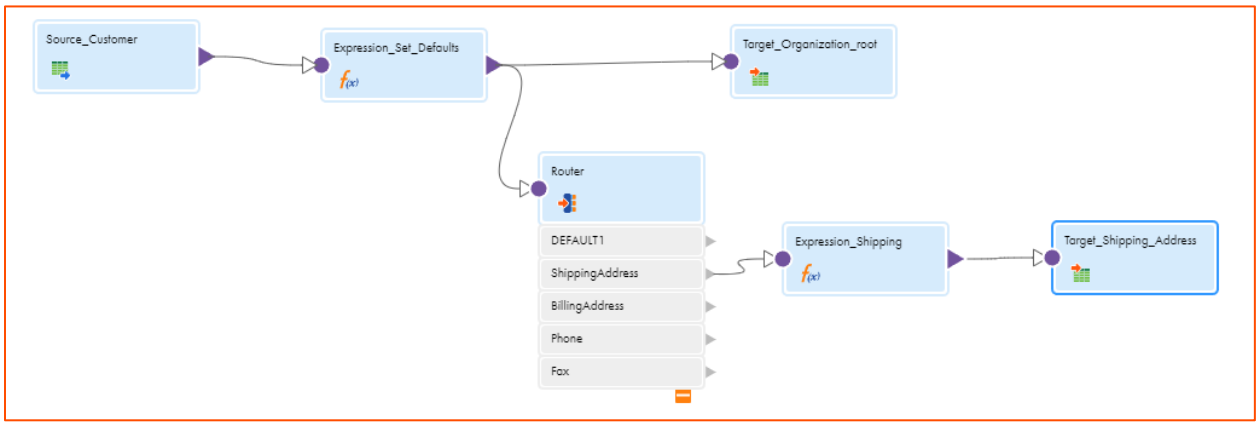
© Informatica.

29. Map the fields as shown below:

PostalAddressArray	
PostalAddress	Expression_Shipping
sourcePKey	Expression_Shipping.o_sourcePKey_Shipping
parentId	Expression_Shipping.DUNS_Id
fieldGroup_State	
defaultIndicator	
addressType	Expression_Shipping.o_AddressType
usageType	Expression_Shipping.o_AddressUsageType
addressStatus	
addressLine1	Expression_Shipping.Ship_Street
addressLine2	
addressLine3	
country	Expression_Shipping.Ship_Country
state	Expression_Shipping.Ship_State
city	Expression_Shipping.Ship_City
county	
postalCode	Expression_Shipping.Ship_Zip

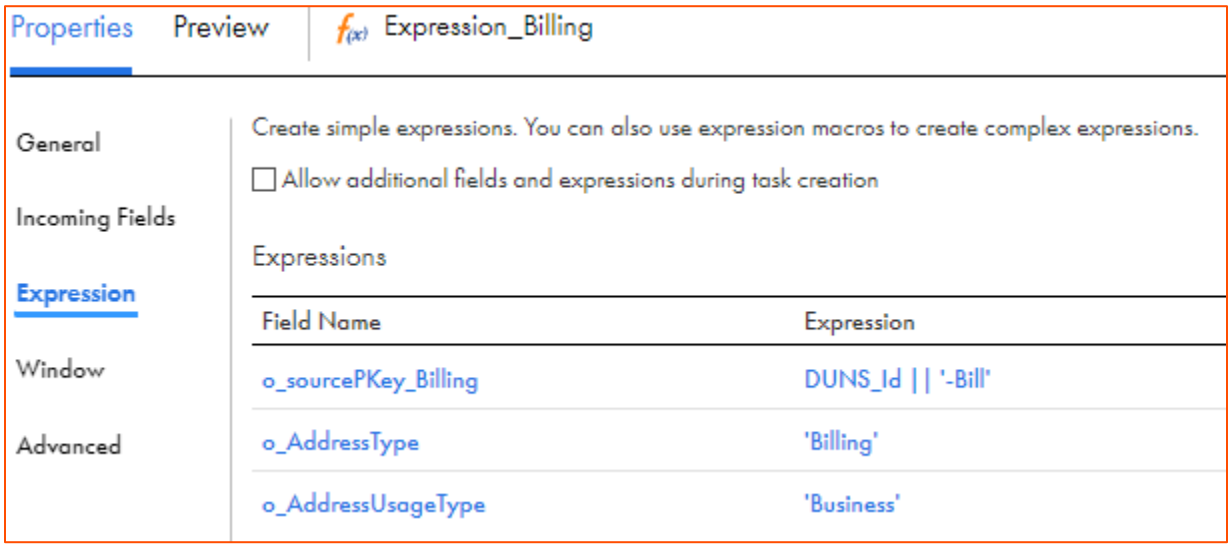
30. Because you have mapped the root element with the same sourcePKey as the target Target_Organization_root, it will both be loaded to the same Organization document in the MDM SaaS data store.

31. The mapping now looks as follows:



© Informatica.

32. Repeat these steps for BillingAddress. The expression should look like this:

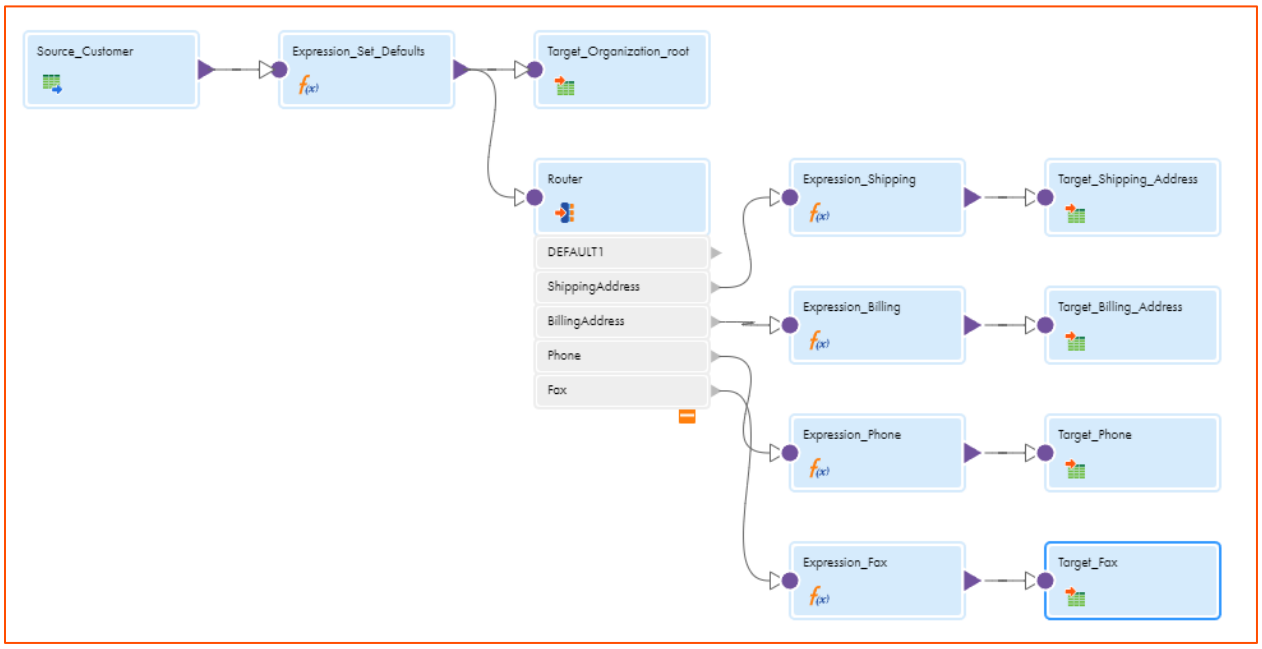


33. Add another target and call in Target_Billing_Address.
34. Select Organization again as target and map the same as for shipping, except now use the Billing fields.
35. Optionally, do the same for Phone and Fax, both with UsageType 'Business'. For phone, the PhoneType is 'Telephone', and for fax it is 'Fax'. Map it to the child element PhoneCommunication. This is the field mapping for Phone:



© Informatica.

36. The completed mapping looks like this:

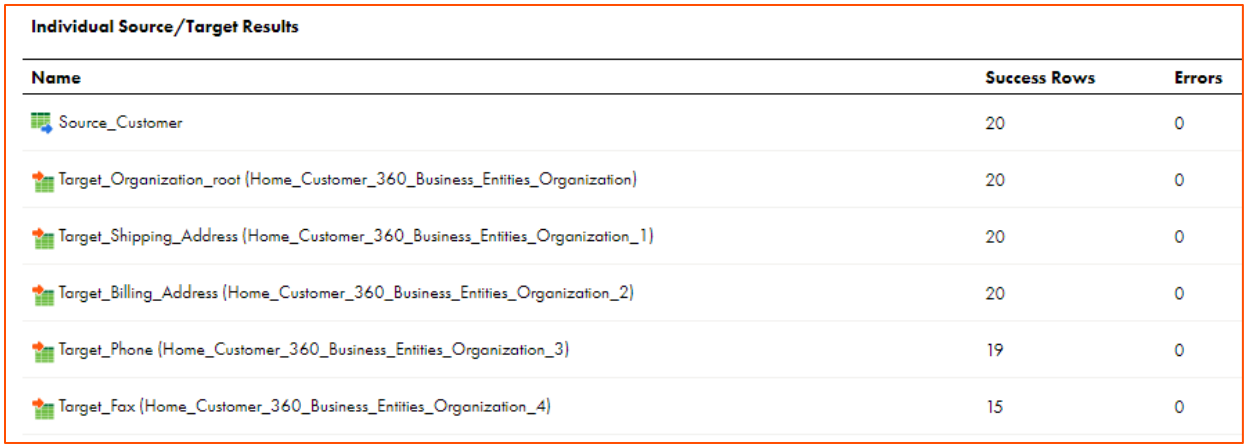


37. Don't forget to add the mapping parameter jobInstanceId, like in the first mapping. Without this parameter, the process will not run.
38. Next, create the mapping task and taskflow as in the first part.
39. Publish the taskflow.

Instructions/Steps – Business360 job and test:

40. Go to the Business360 Console and create the job.
41. Run the job
42. To view the details of the loaded targets in CDI, go to Data Integration and click on My Jobs. You see the job here. Click on View Subtasks
43. Click on the Instance Name, which is the mapping task name.
44. You can see detailed load statistics here. A few input records don't have Fax or Phone, so these statistics are as expected:

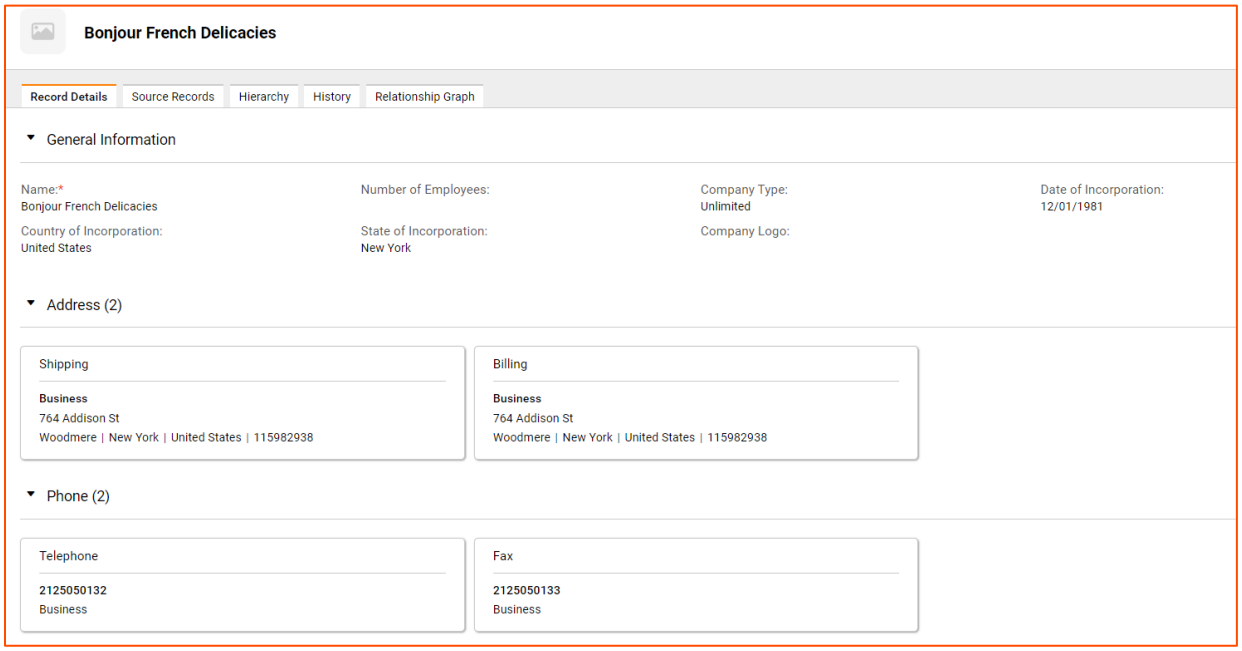
© Informatica.



Individual Source/Target Results

Name	Success Rows	Errors
Source_Customer	20	0
Target_Organization_root (Home_Customer_360_Business_Entities_Organization)	20	0
Target_Shipping_Address (Home_Customer_360_Business_Entities_Organization_1)	20	0
Target_Billing_Address (Home_Customer_360_Business_Entities_Organization_2)	20	0
Target_Phone (Home_Customer_360_Business_Entities_Organization_3)	19	0
Target_Fax (Home_Customer_360_Business_Entities_Organization_4)	15	0

45. When the job is finished in the Business360 Console, go to Customer360 to check the results. All loaded Organizations in this exercise start with a B, so search for B*.
46. Open one of the Organizations, and you see two addresses and phones:



This concludes Lab Part 2.

© Informatica.

LAB Part 3: Custom error handling

DURATION: 45 minutes

Purpose:

In the previous exercises, you have noticed that data validations as designed in the data model are executed in the Transform phase. However, if records contain an error, it will still be loaded. This is the default behavior as of the August 2022 release.

Data can also be rejected in the Load phase, mainly when values in dependent picklists cannot be found, for example, a state code does not exist in a country.

This exercise will show how to catch errors in a CDI mapping, to avoid loading invalid data in the MDM SaaS data store.

Instructions/Steps – Create Crosswalk:

The input file Customer_Part_3.csv contains phone types, to indicate which type of phone number is stored. These types are different from used in the reference data in the MDM SaaS data store.

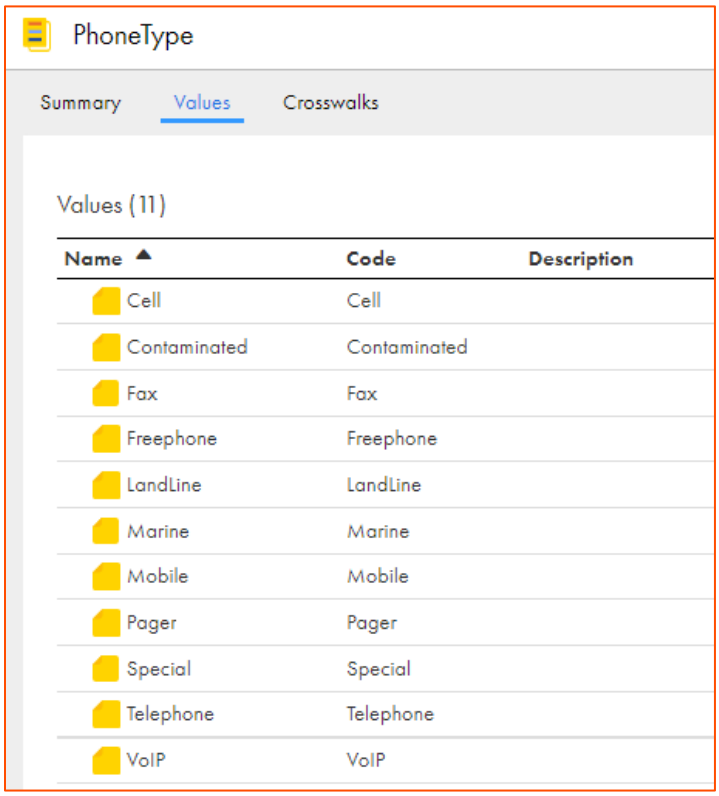
Using a crosswalk will let you translate the source codes to the MDM reference data in the transform phase. However, records with errors will still be loaded with invalid data nullified.

First, you will build the crosswalk, and then the crosswalk will be used as a dictionary for validation in the CDI mapping.

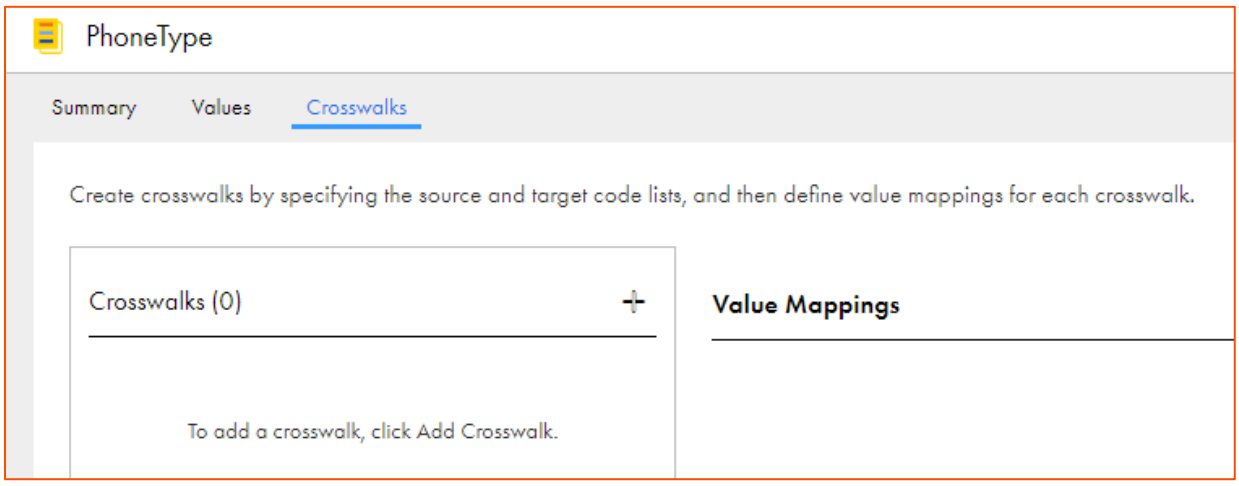
1. Open the file Customer_Part_3.csv and inspect the Phone Types. You will see the following:
   - Phone
   - Mob and Mobile
   - Fax and Faxx
   - Direct
   - AltPhone
   - Cell
2. Go to the Business 360 Console and go to Explore.
3. Open project Reference30 and open PhoneType

© Informatica.

4. Click on the Values tab; you see the values here:



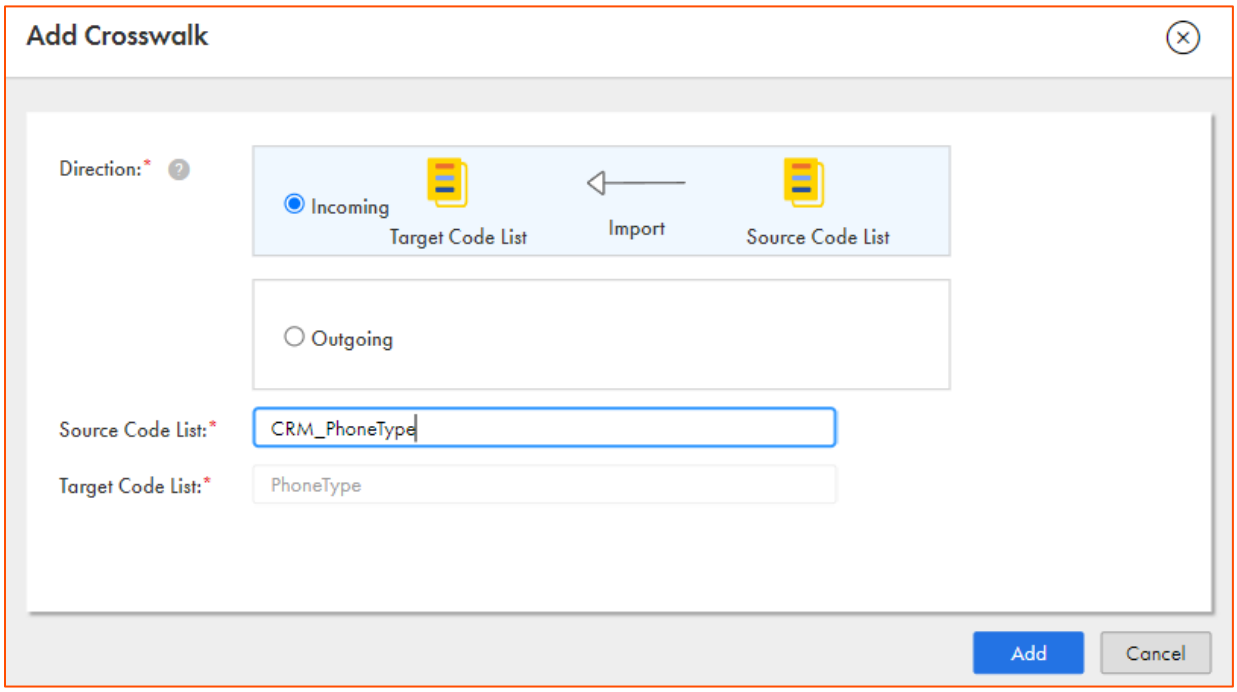
5. Click on Crosswalks. The following screen appears:



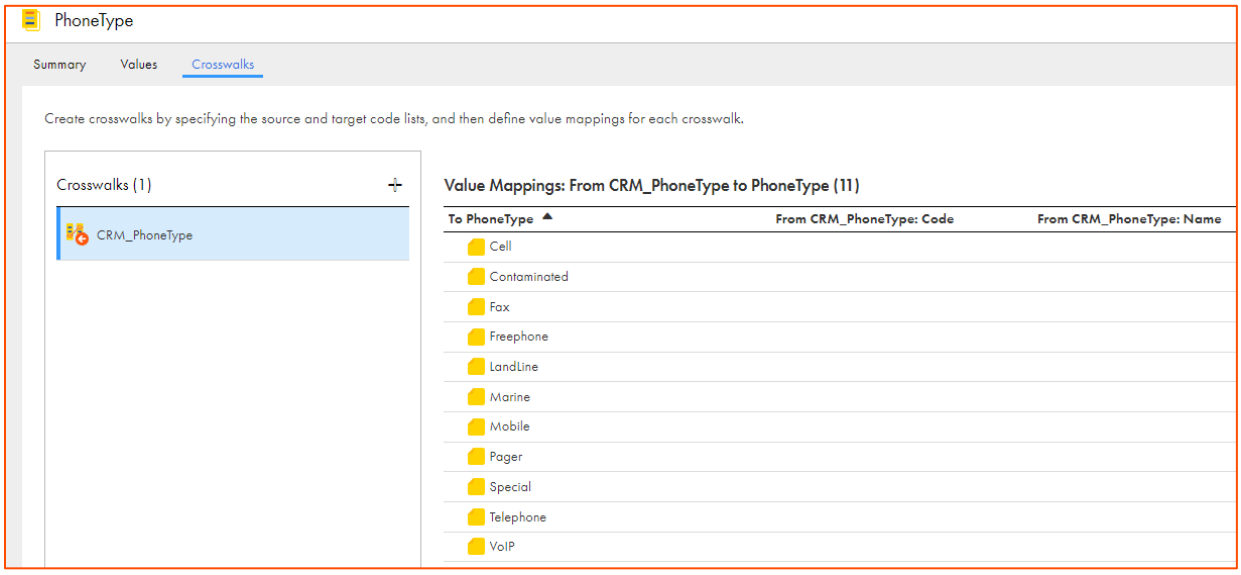
6. Click on the + sign to add a crosswalk.

© Informatica.

7. On the next screen, give it the name CRM_PhoneType and keep the direction as Incoming:

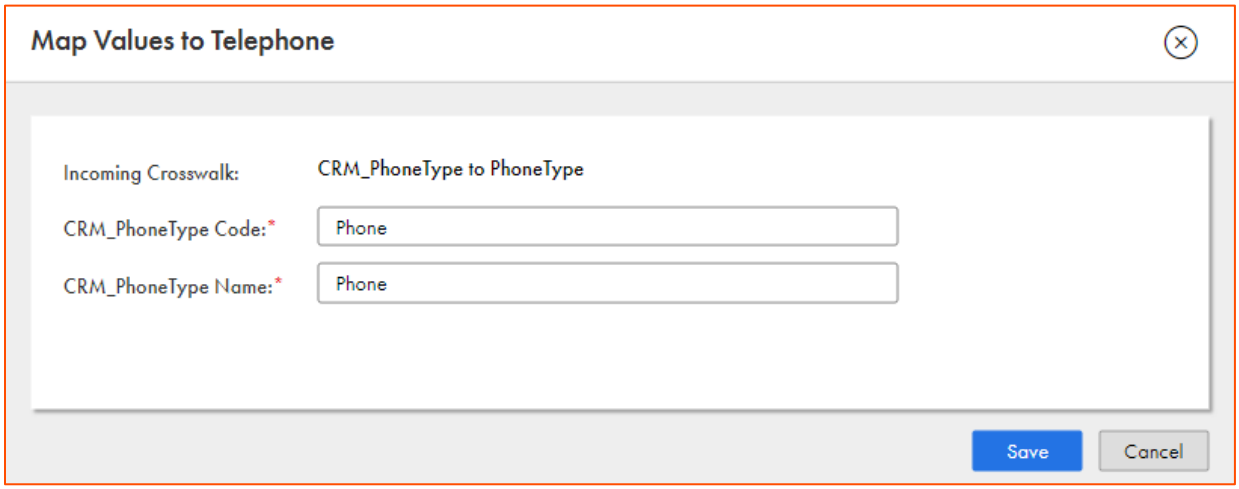


8. Click Add. You se the crosswalk created and the target values populated from PhoneType:



© Informatica.

9. Highlight Telephone and click on the pencil button that appears on the right. Map the value Phone to Telephone on the next screen:



10. In the same way, map the values as follows:

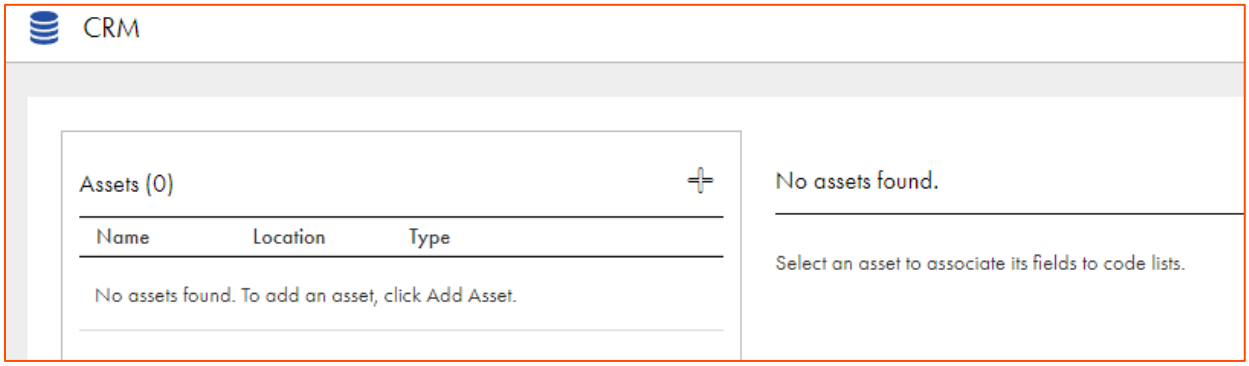
Value Mappings: From CRM_PhoneType to PhoneType (11)

To PhoneType ▲	From CRM_PhoneType: Code	From CRM_PhoneType: Name
Cell	Cell	Cell
Contaminated		
Fax	Fax	Fax
Freephone		
LandLine	Direct	Direct
Marine		
Mobile	Mobile	Mobile
Pager		
Special	AltPhone	Alternative Phone Number
Telephone	Phone	Phone
VoIP		

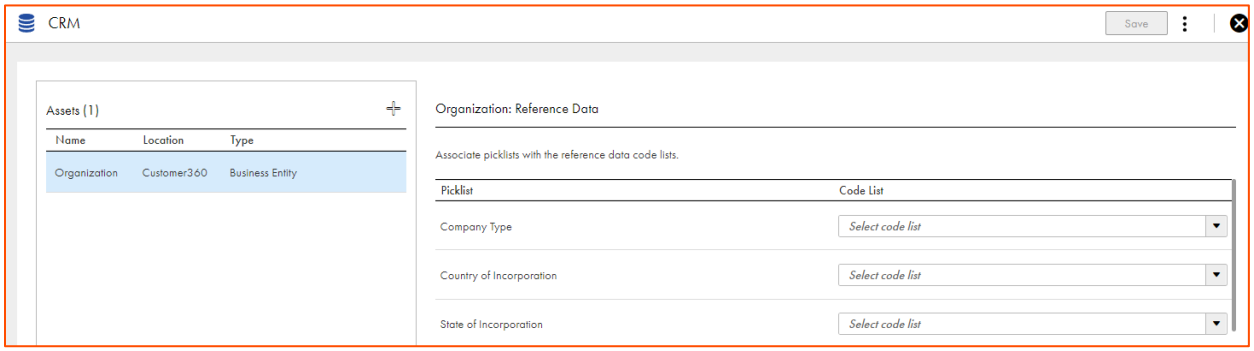
11. Note that you can only map one source value to one target in this interface. To map multiple source values, you must use the Reference360 Service. Reference360 also allows importing values from excel; this is useful for larger code sets.

© Informatica.

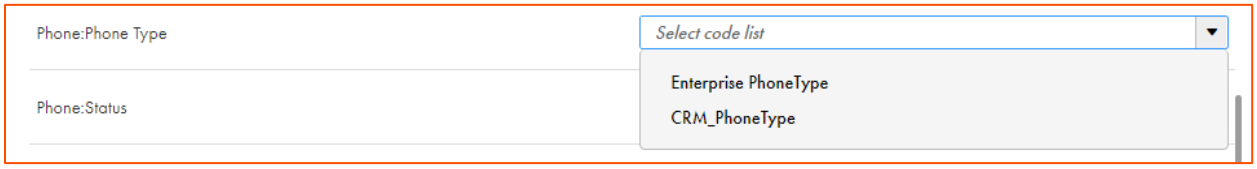
12. Next is to register this crosswalk to be used for code translation in the business application. To do this, go to your CRM_Ingest project and open the source system CRM. The following appears:



13. Click on the + sign to add an Asset, and select Organization. The screen now looks like this, with all Picklists mentioned:



14. Scroll down to Phone Type and select the code list:



15. Select the code list CRM_Phone type. This tells the application that the input values for the CRM source should be found in the code list CRM_PhoneType, and the crosswalk must be used to translate the codes to the Enterprise PhoneType. This translation is done in the Transform phase of the data load.
16. Save the source system.

© Informatica.

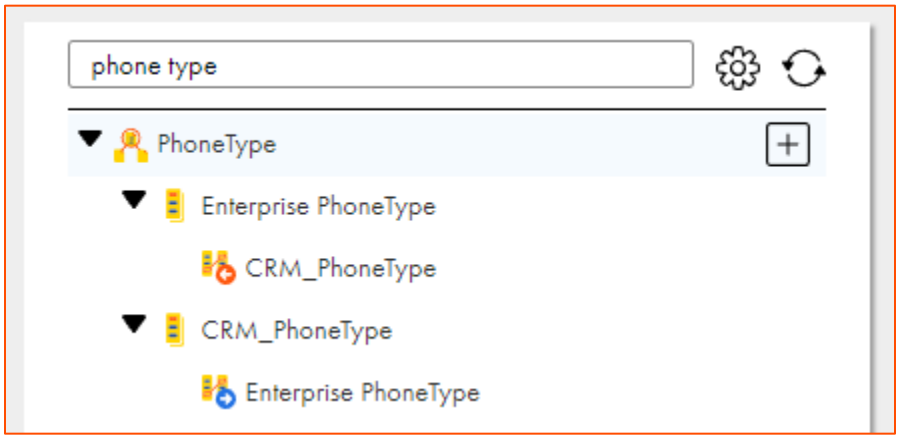
Instructions/Steps – Create CDQ validation:

When you would run an ingest process using this setup, any source code not found in the crosswalk will be flagged in the Transform phase, but the data is still loaded, and the code will be nullified.

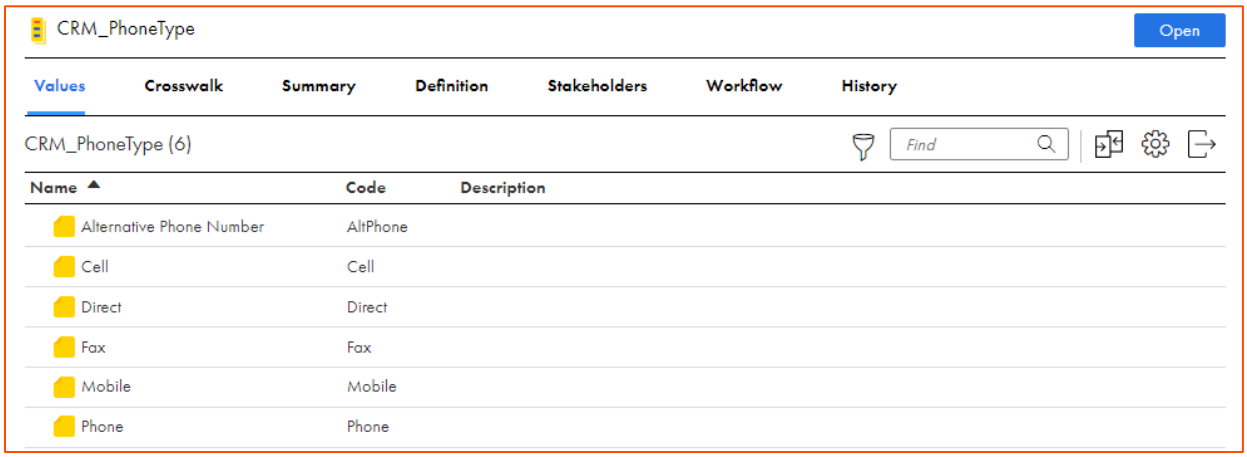
This may be acceptable for some customer projects, but there will be many cases where the data must be rejected. As of the August 2022 release, there is no option to reject these errors automatically, so it must be done in the CDI mapping.

The following steps will show how to use a CDQ dictionary derived from the crosswalk, and a cleanse function to do the validation. This will then be incorporated in the CDI mapping.

1. Go to Reference 360 and go to Explore
2. Search for phone type. The following result comes up:

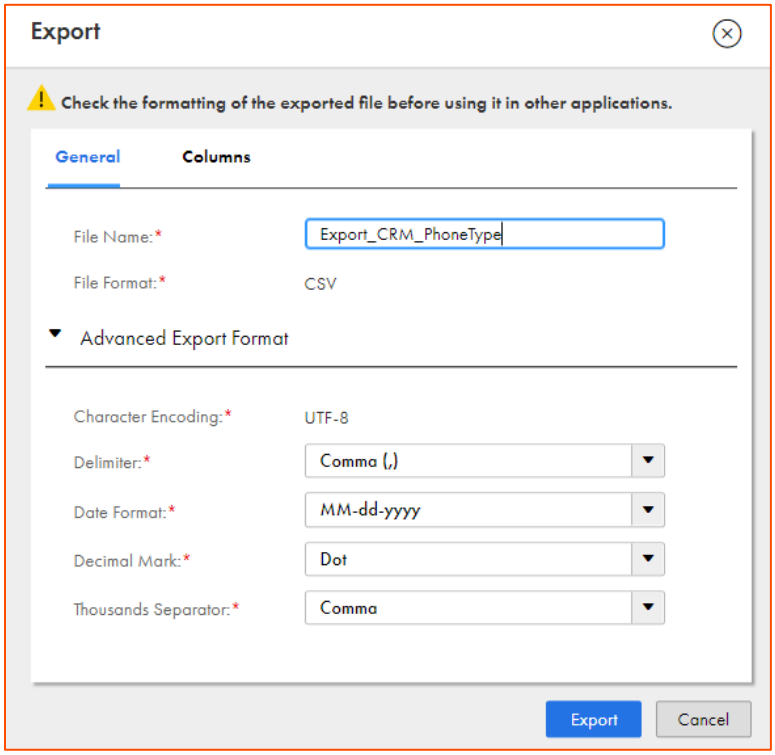


3. Click on CRM_Phone_Type to see the values. These are the values you have entered in creating the crosswalk in the last part:



© Informatica.

4. Click on the Export icon to export the values. The CDQ function will check against the values, but will not translate. Therefore, you can use the values from the reference data instead of the crosswalk. If you want to do the translation in CDI, you need to use the crosswalk.
5. The following screen appears. Give the file a meaningful name and click Export:



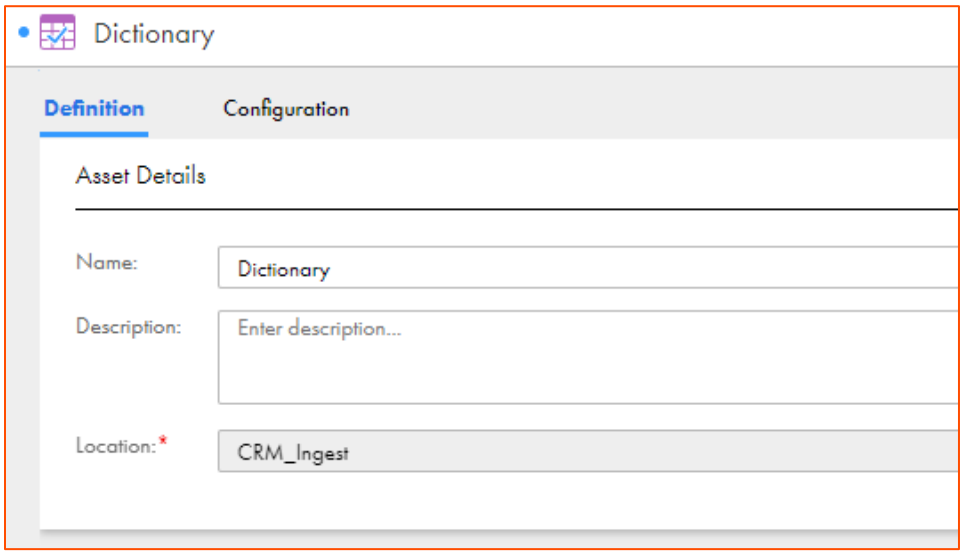
6. The export file, opened in Excel looks like this. You don't have excel on the image, but you can open the file in notepad to see it as text:

	A	B	C	D	E
1	Name	Code	Description	status.key	effectiveDate
2	Phone	Phone			
3	Mobile	Mobile			
4	Alternative Phone Number	AltPhone			
5	Fax	Fax			
6	Direct	Direct			
7	Cell	Cell			
8					

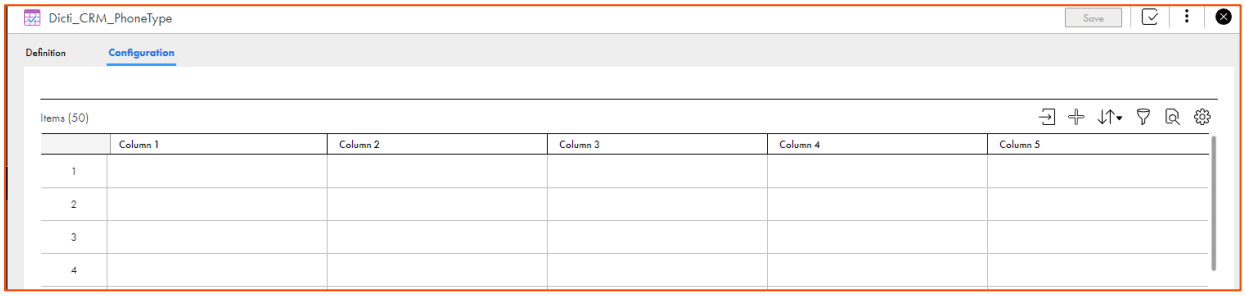
7. Go to Data Quality.

© Informatica.

8. Click on New and click on Dictionary. The following screen comes up:

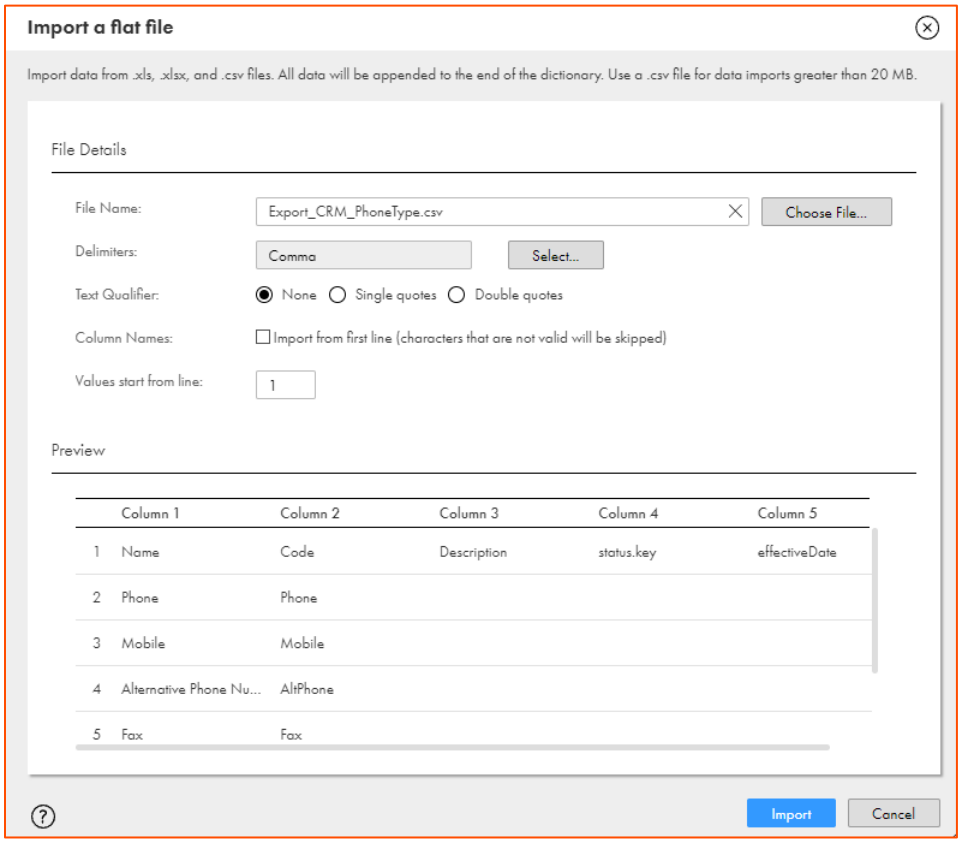


9. Name the dictionary Dict_CRM_PhoneType and ensure the location is set to CRM_Ingest to keep all assets in the same project.
10. Click Save.
11. Click on Configuration. You see a screen appear where you can enter the values:

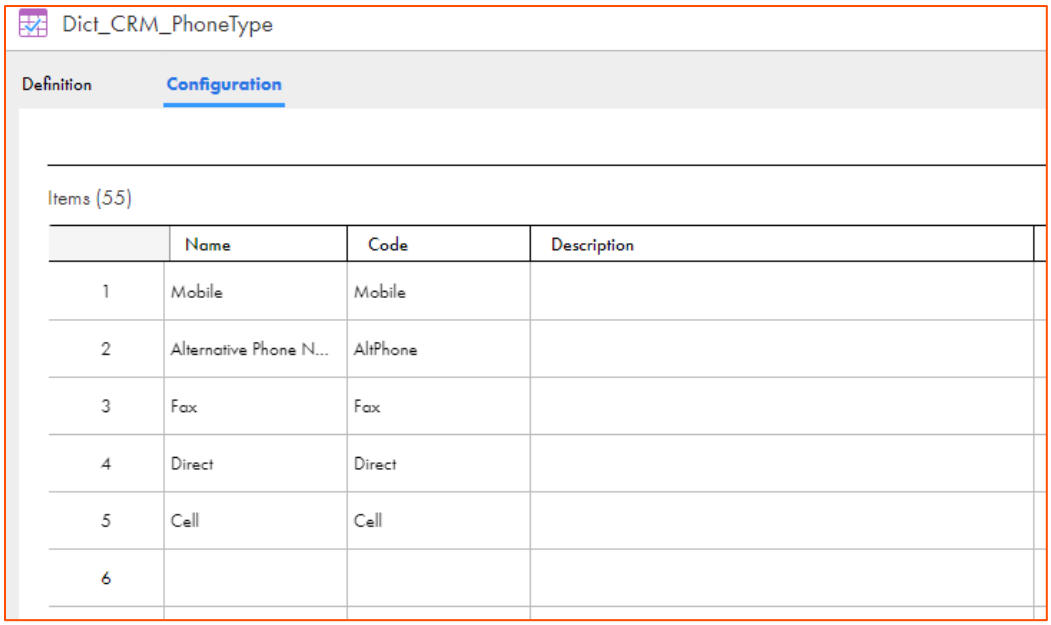


© Informatica.

12. Click on the Import icon. On the next screen, select your export file from Reference 360. The following screen appears:

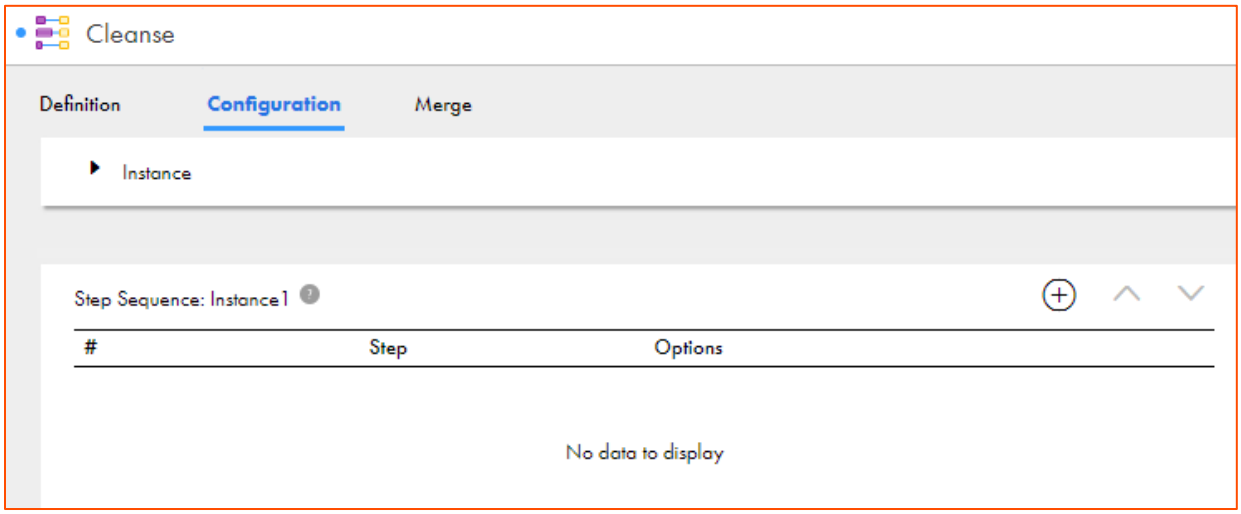


13. Check the checkbox to import the column names and set “Values start from line:” to 2.
14. Click Import. The data will be imported and should look like this:

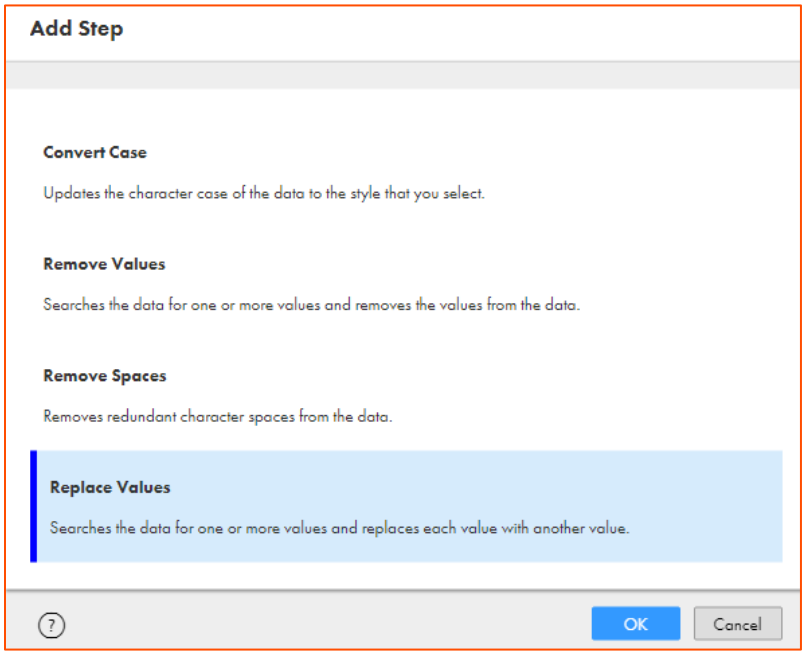


© Informatica.

15. Save the Dictionary.
16. Click on New and click on Cleanse to create a validation function using this Dictionary.
17. Name it Cleanse_CRM_PhoneType and ensure the location is CRM_Ingest.
18. Go to Configuration. The following screen appears, showing no cleanse steps yet created:

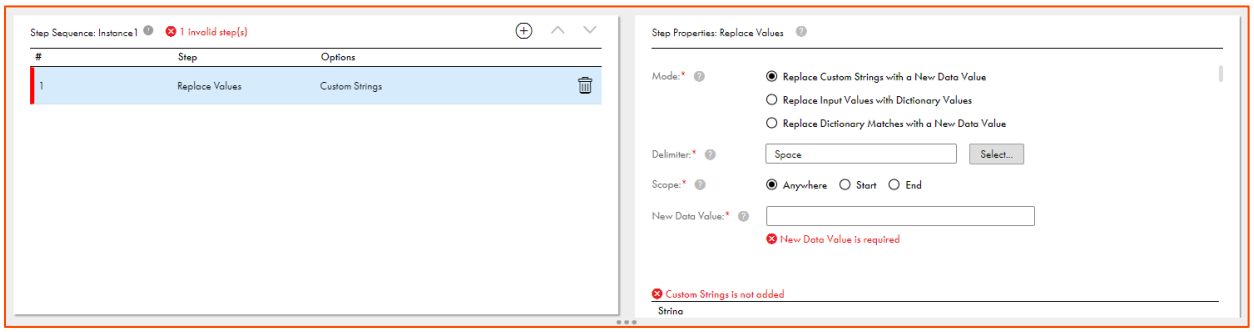


19. Click on the "+" sign to add a step. The following screen appears, where you select Replace Values:

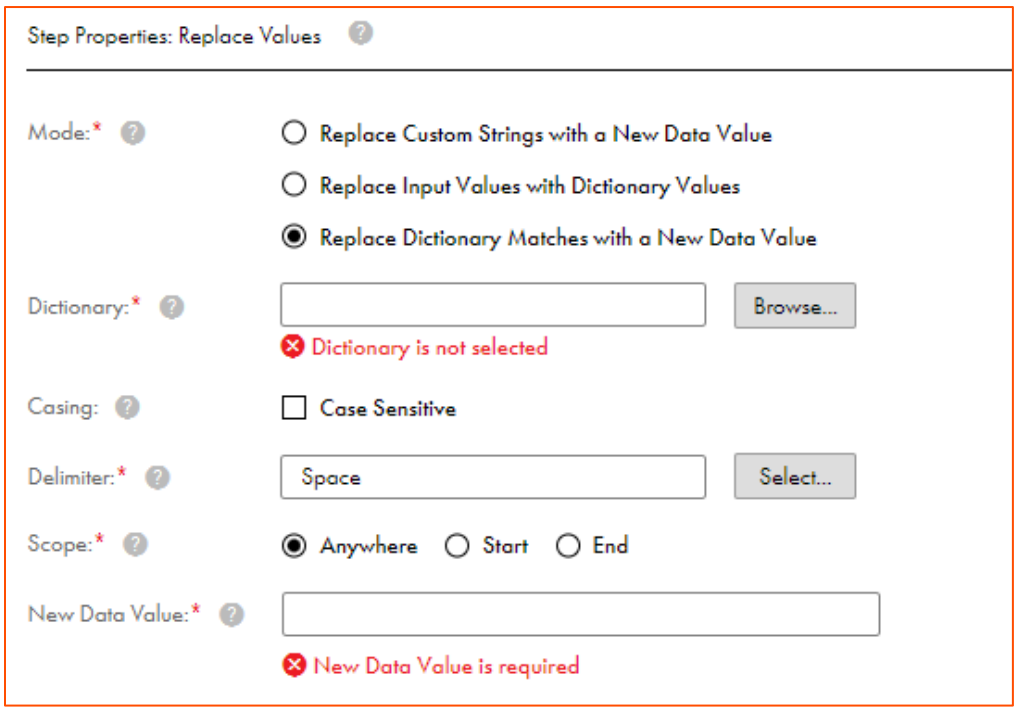


© Informatica.

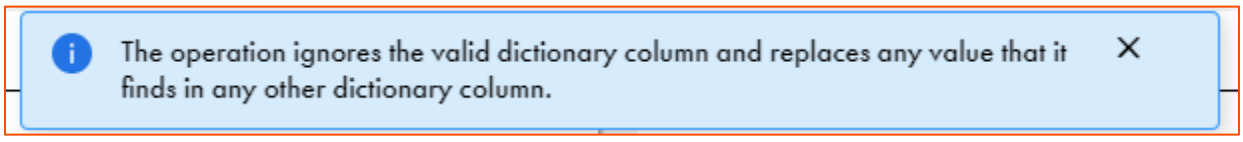
20. The screen shows now that you have to configure the step properties:



21. Select the third option, “Replace Dictionary Matches with a New Data Value”. It will now ask for a dictionary:



22. Select the Dictionary you just created. You see an important popup appear, which may be confusing at first:

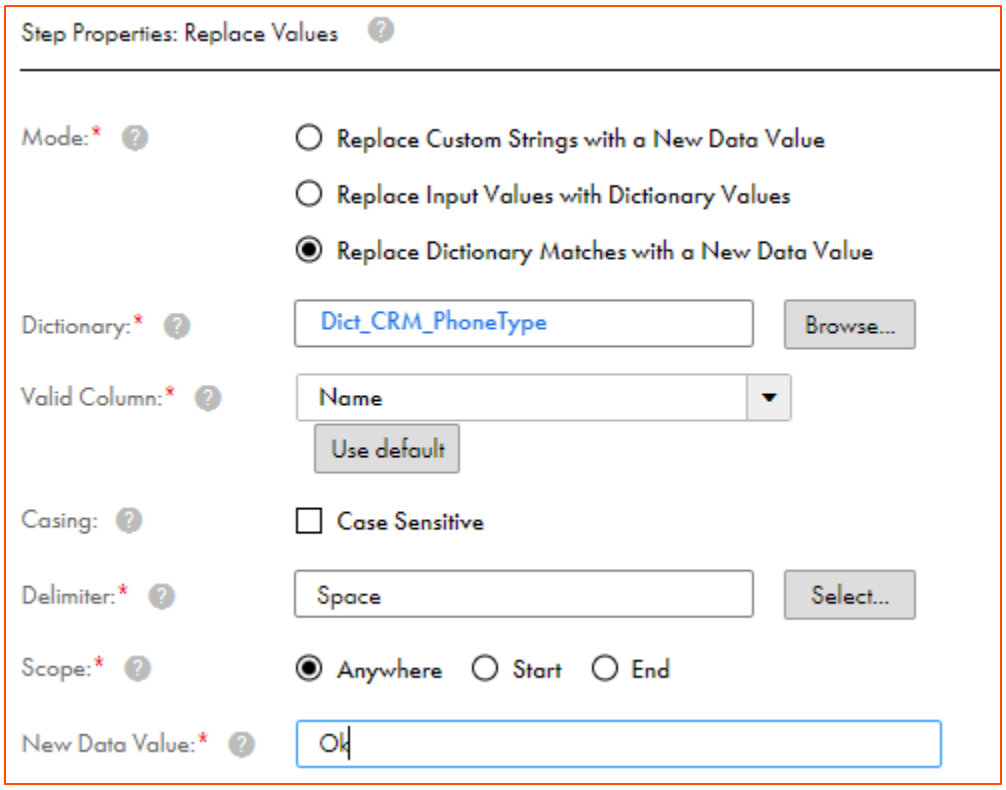


The function will parse all columns in the Dictionary, except the one you specify as Valid. This will be the output column when you select to replace input values with dictionary values, which you would do in standardization solutions. The function you are building now only does

© Informatica.

validation and gives an “Ok” output when the data is found. When you scroll down in the properties pane, you see the data just imported, and the codes used in the crosswalk are stored in the field Code. If you select the field Code as Valid Column, the values in there will be ignored.

23. Keep the field Name as valid column, so you are checking the input file against the Code, and enter Ok as New Data Value:



24. Save the cleanse function.
25. Test the function by using the bottom pane. Enter a few input values, select the runtime environment as your secure agent and click the Test button. An example is shown here:



26. You see the valid input data translated to Ok and the invalid data is kept as is.
27. You can save the cleanse function again if you want to keep the test data.

© Informatica.

Instructions/Steps – Create CDI mapping with validation:

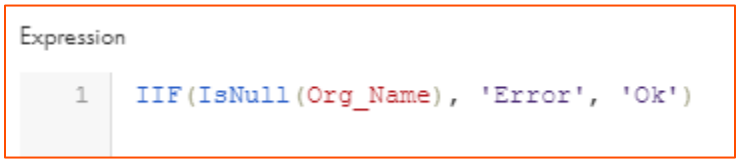
The final step in this exercise is to create the mapping that includes the cleanse function for validation.

In this mapping, you will do three validations:

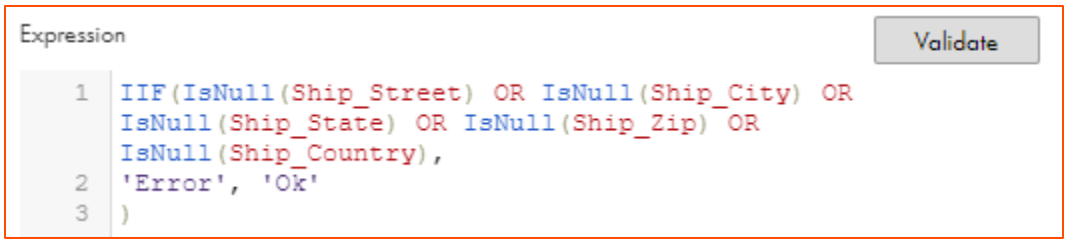
- Customer Name must not be empty
- Address must be complete, i.e. all address fields are populated
- Phone type must be valid

When an error occurs, the data must be reported in an error file and should not be loaded to MDM SaaS.

1. Go to Data Integration and create a new mapping
2. Name the mapping m_Customer_Part_3
3. Use the file Customer_Part_3.csv as source.
4. Use the business entity Organization as target and name the target instance Target_Organization_Root.
5. Add an expression and connect it to the source. Name it Expression_Validations
6. Add an output field and name it o_Name_Valid.
7. Add the following expression:



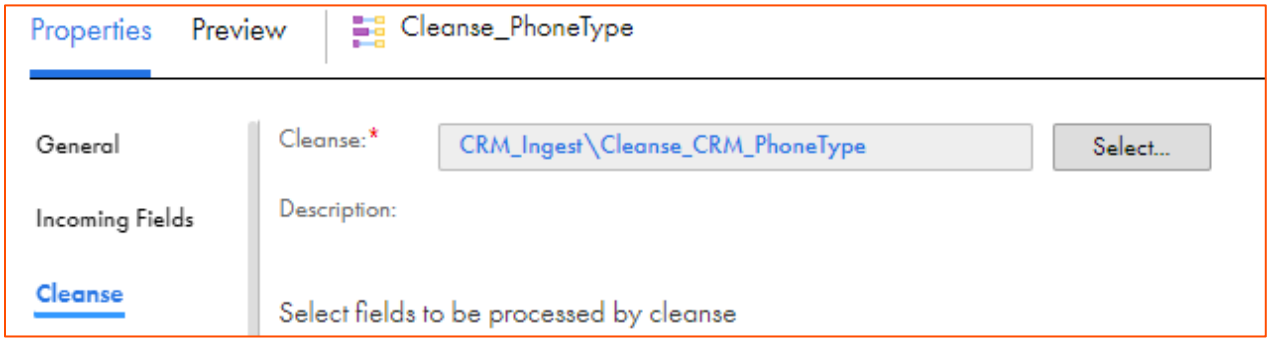
8. Add a second output field and call it o_Address_Valid, and add the following expression:



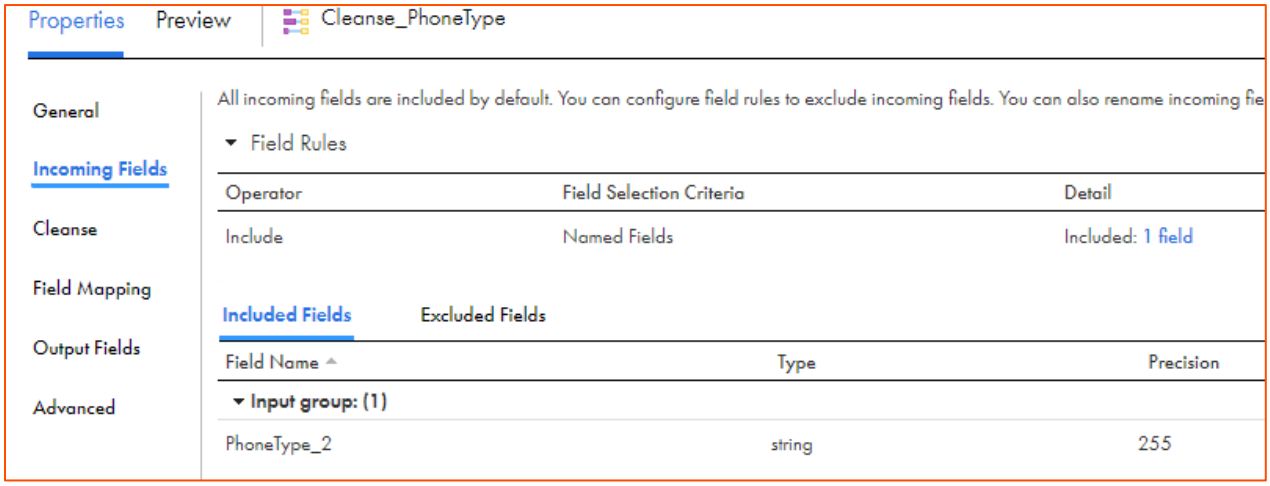
9. The expresion "`IIF(IsNull(Ship_Street) OR IsNull(Ship_City) OR IsNull(Ship_State) OR IsNull(Ship_Zip) OR IsNull(Ship_Country), 'Error', 'Ok' )`" checks if any field of address is null. If yes, the address is incomplete, so the output field will be 'Error'.
10. Add a Cleanse transformation to the mapping, and connect it to the source. Give it the name Cleanse_PhoneType.

© Informatica.

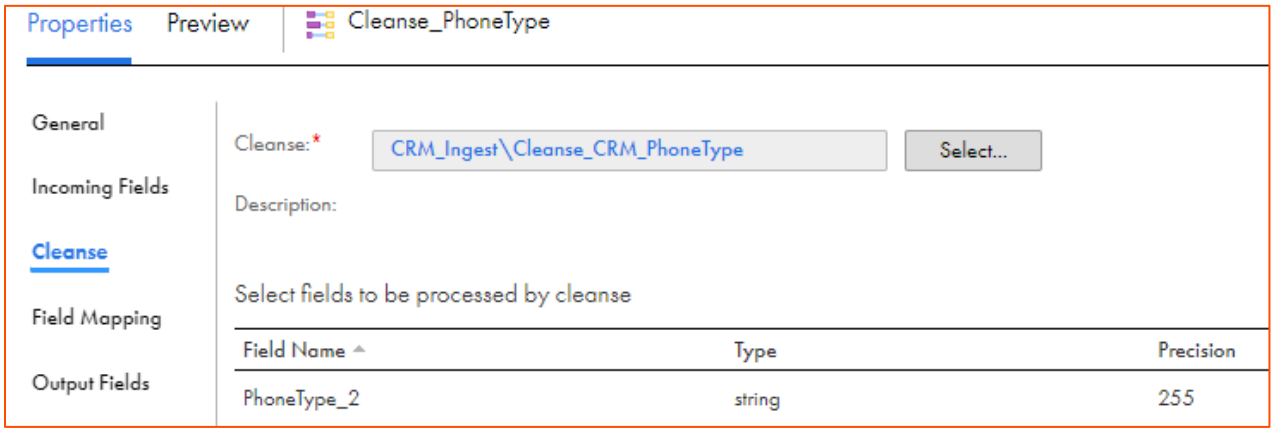
11. In the Properties pane, go to Cleanse and select the Cleanse function Cleanse_CRM_PhoneType you created earlier:



12. Go to Incoming Fields and change the field Rules to select Name Fields. Configure the transformation to have PhoneType_2 as input:

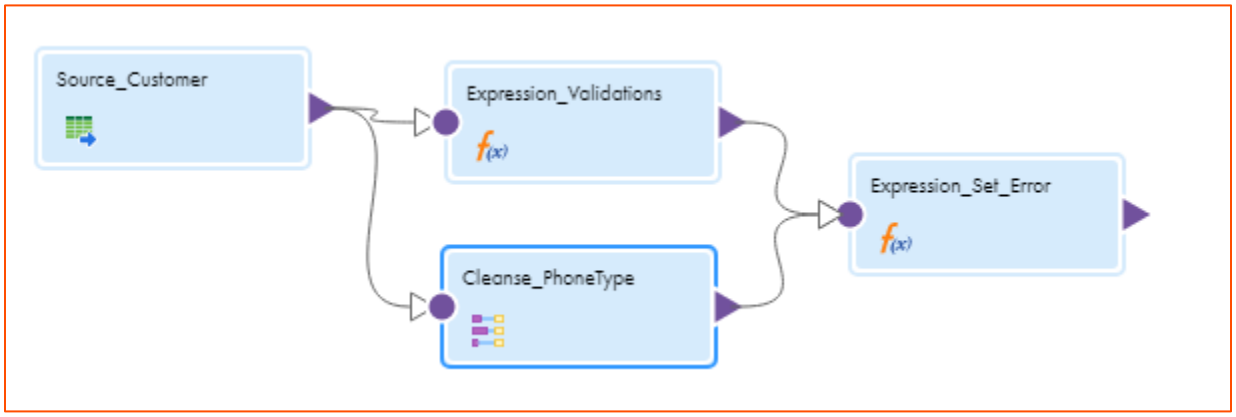


13. In this lab, you will only validate PhoneType_2. The field PhoneType_1 contains only the type “Phone”, which is valid.
14. Go back to Cleanse and select the fields to be processed:

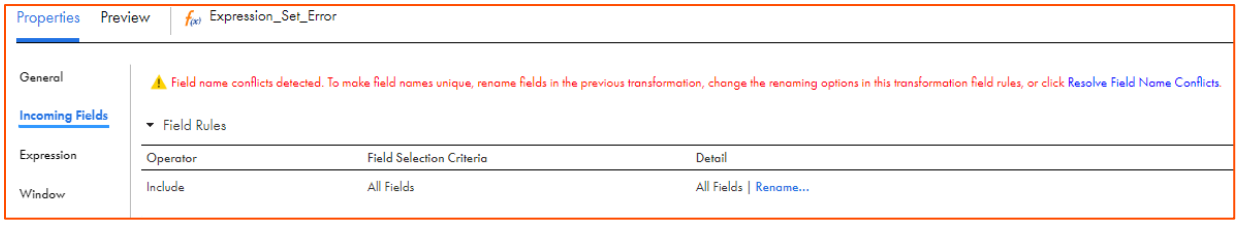


© Informatica.

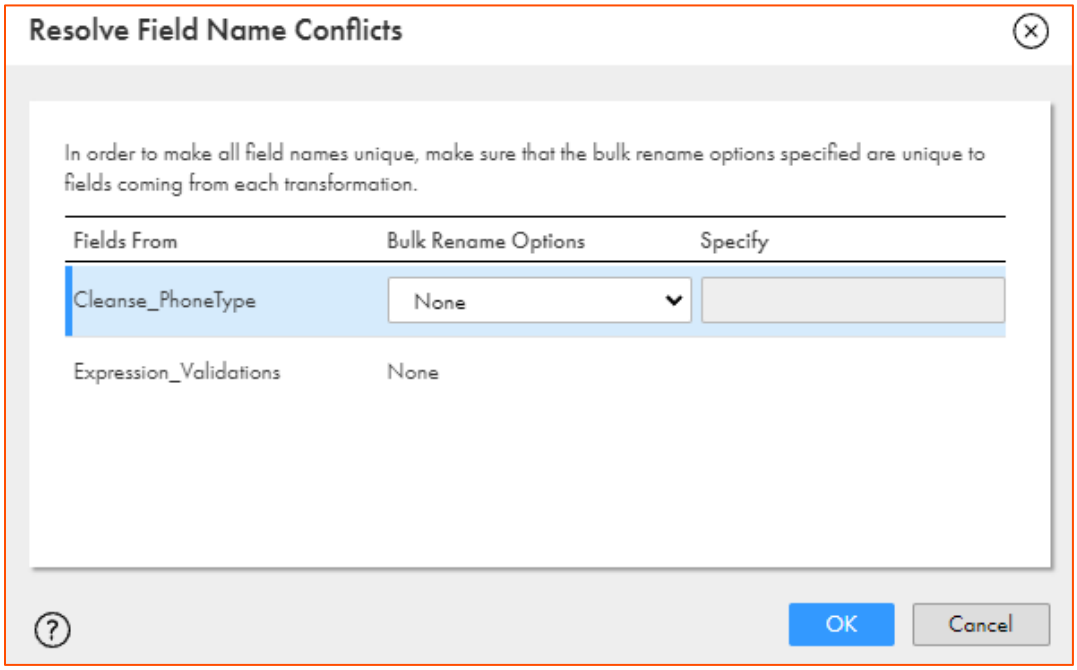
15. Add an Expression to the mapping, call it Expression_Set_Error, and connect it to Expression_Validations and to Cleanse_PhoneType:



16. Click on the expression and go to Incoming Fields. You see an error message about field name conflicts:

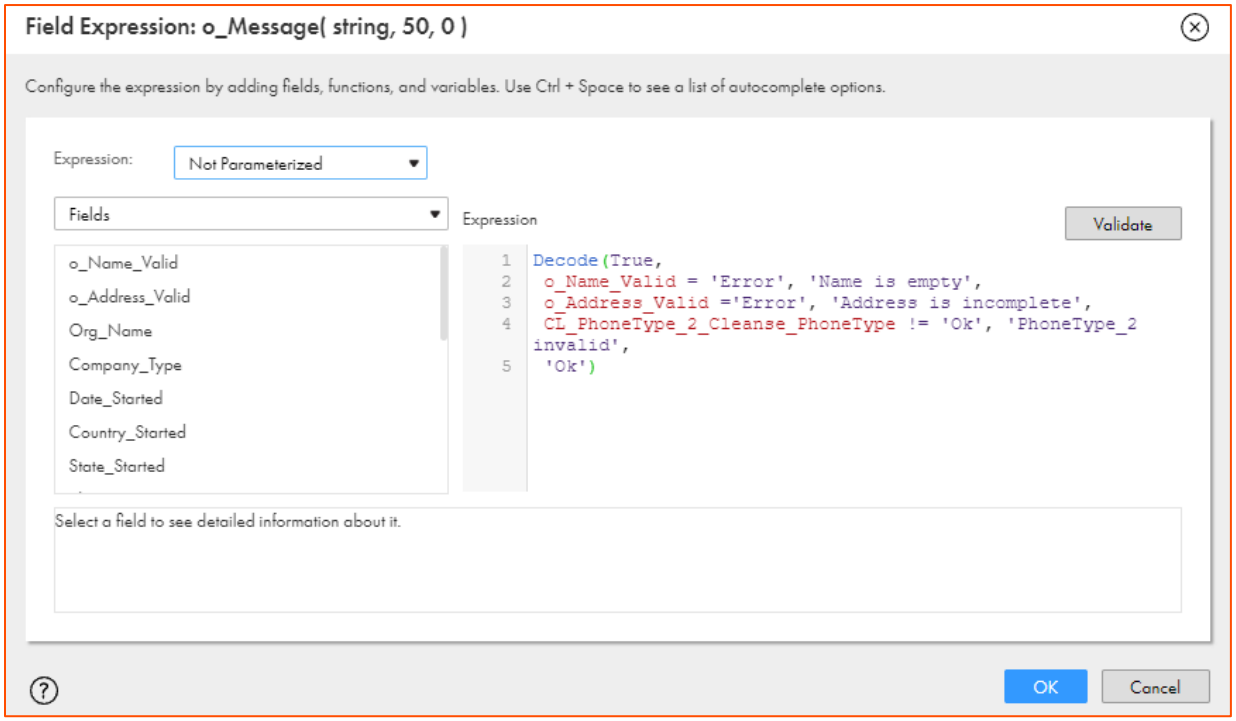


17. Click on the blue text “Resolve Field Name Conflicts.” The following screen comes up:



© Informatica.

18. For Cleanse_PhoneType, set the Bulk Rename Options to Prefix and specify CL_
19. Click ok, and you see the conflicts are resolved.
20. Click on expression and add an output field o_Message with precision 50
21. Click on Configure to add the function. This time you will use a Decode to do multiple checks. You can copy the text from below, or enter it via the expression editor



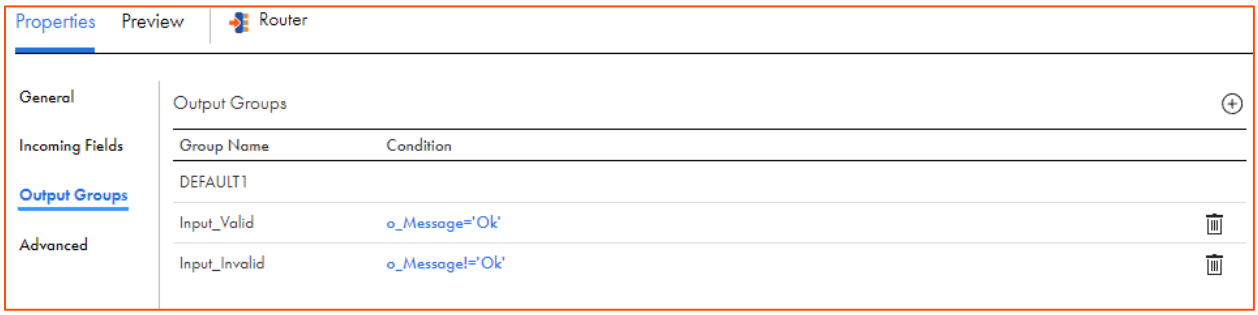
22. You can copy the expression from here:

```
Decode(True,
 o_Name_Valid = 'Error', 'Name is empty',
 o_Address_Valid ='Error', 'Address is incomplete',
 CL_PhoneType_2_Cleanse_PhoneType != 'Ok', 'PhoneType_2 invalid',
 'Ok')
```

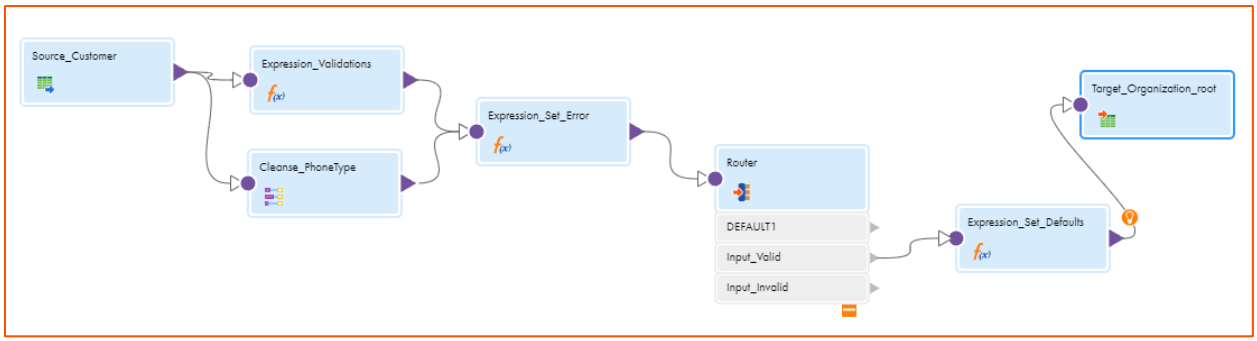
23. Next, add a Router and connect it to the Expression_Set_Error
24. Add two output groups. Call the first output group Input_Valid and set the condition to “o_Message = ‘Ok’ “.
25. Name the second output group Input_Invalid and set the condition to “o_Message != ‘Ok’ “.

© Informatica.

26. The Output groups should now look like this. It is best practice not to use the Default group.



27. Add an expression Expression_Set_Defaults and connect it to the Input_Valid output group. Connect the Target_Organization_root to the expression. The mapping will now look like this:

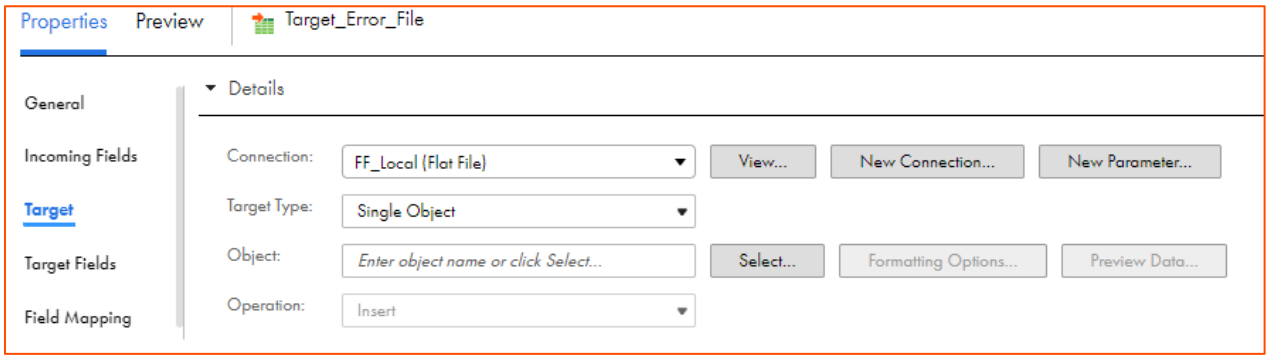


28. Add output fields as shown in the below table:

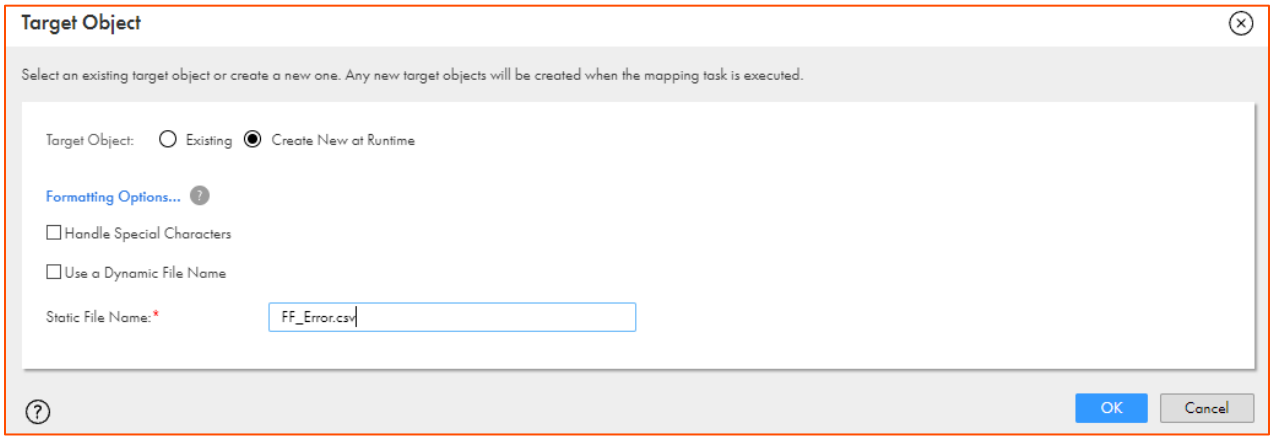
Field Name	Expression
o_SrcSystem	'CRM'
o_AltIdentifierType	'DUNS'
o_DateOfIncorporation	To_Date(Date_Started, 'YYYYMMDD')
o_AddressType	'Shipping'
o_UsageType	'Business'
o_sourcePKey_Address	DUNS_id || '-Ship'
o_sourcePKey_Phone_1	DUNS_Id || '-Phone'
o_sourcePKey_Phone_2	DUNS_Id || PhoneType_2

© Informatica.

29. Go to the target and complete the field mapping. Map the root object, the Address, the Phone from PhoneNumber_1, and the Identifier. Refer to the previous exercise for the detailed field mapping.
30. Optionally, create a second target, similar to the previous exercise, to store PhoneNumber_2. Map the source system and id, and map the Phone element
31. Add an expression Expression_Write_Error and connect it to the Input_Invalid output group of the router.
32. Go to Incoming Fields in the properties and select Named Fields.
33. Only select the source fields and the error message field.
34. Add a target and connect it to Expression_Write_Error. Name it Target_Error_File
35. Select the Flat File connection and click on Select next to the object:



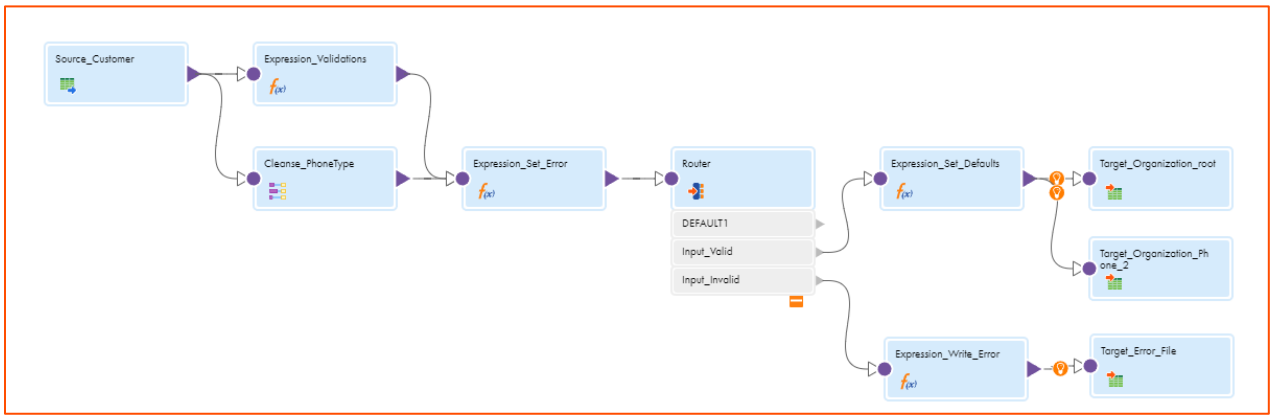
36. Click “Create New at Runtime” and specify the file name:



37. And finally, add the InOut mapping parameter jobInstanceId

© Informatica.

38. Save the mapping. It should look like this:



39. The next steps are to create the mapping task, taskflow, and job definition as in the previous exercises. For detailed instructions, refer to the steps in these.

40. Run the job and wait for it to finish. It will take something like five minutes. The job statistics should look like this:



41. Go to the Files directory and open the file FF_Error.csv. For ease of reading, copy it to your laptop and open it in excel. You will see six error messages there:

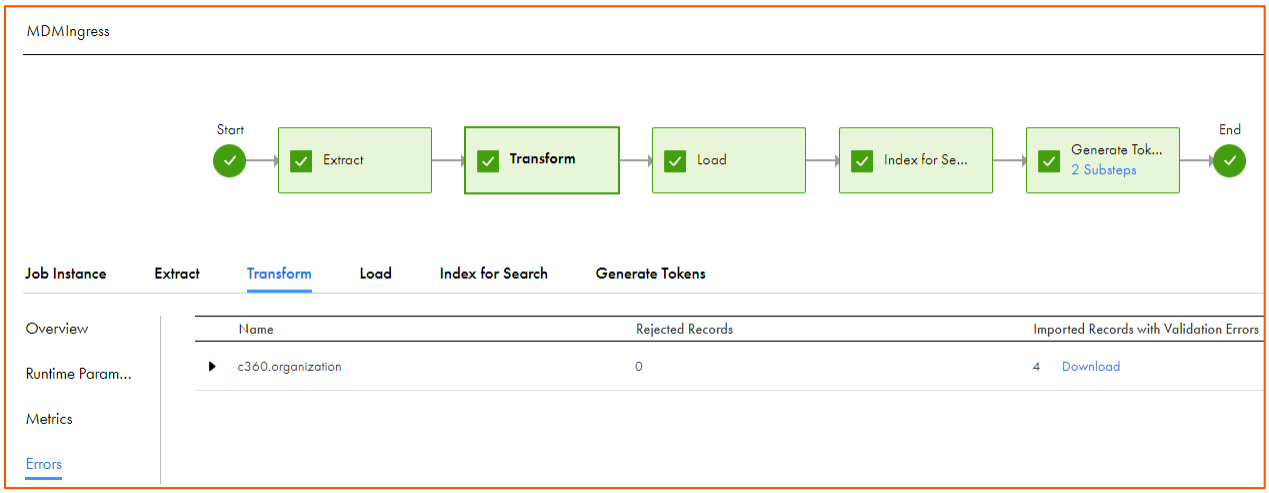
A	F	G	H	I	J	M	N	P
Org_Name	Ship_Street	Ship_City	Ship_State	Ship_Zip	Ship_Country	PhoneType_2	PhoneNumber_2	o_Message
Custom Bikes by Mark	700 S 4th St Ste A	Hamburg	PA	195269212	US	Mob	4025634516	PhoneType_2 invalid
Cravings Cakes	720 S Magnolia Ave	Melbourne	FL	329356441	US	Faxx	9198824050	PhoneType_2 invalid
Cadburee Chocolate	722 Industry Rd S	Cocoa		329265827	US	Cell	5125318999	Address is incomplete
	7720 W 98th St	Oak Lawn	IL	604572300	US	Mobile	2123795399	Name is empty
CrossWalk Supplies	1290 Ave of The Americas		NY	101040101	US	Fax	9107966889	Address is incomplete
Chew Chew Pet Food		Englewood	CO	380287909	US	Fax	3212548444	Address is incomplete

(for ease of viewing, the irrelevant columns are hidden).

This is a basic error handling design; based on requirements, you can implement an error severity so that records with less severe errors will be reported and passed through to the MDM SaaS data store.

© Informatica.

42. Next, go to the transform phase of the job and click on Errors:



43. You see four errors here. When you download and open the error file, you will see that these all refer to the invalid format of the DUNS number. This is an out-of-the-box validation in the data model. If you want to catch these errors in the CDI mapping too, you can incorporate the DUNS validation rule into your mapping.
44. Go to Customer360 and check whether the data loaded successfully. All customers in this input file begin with a C, so you can search on “C*”.

This concludes this exercise.

© Informatica.

LAB Part 4: Customer relationships

DURATION: 30 minutes

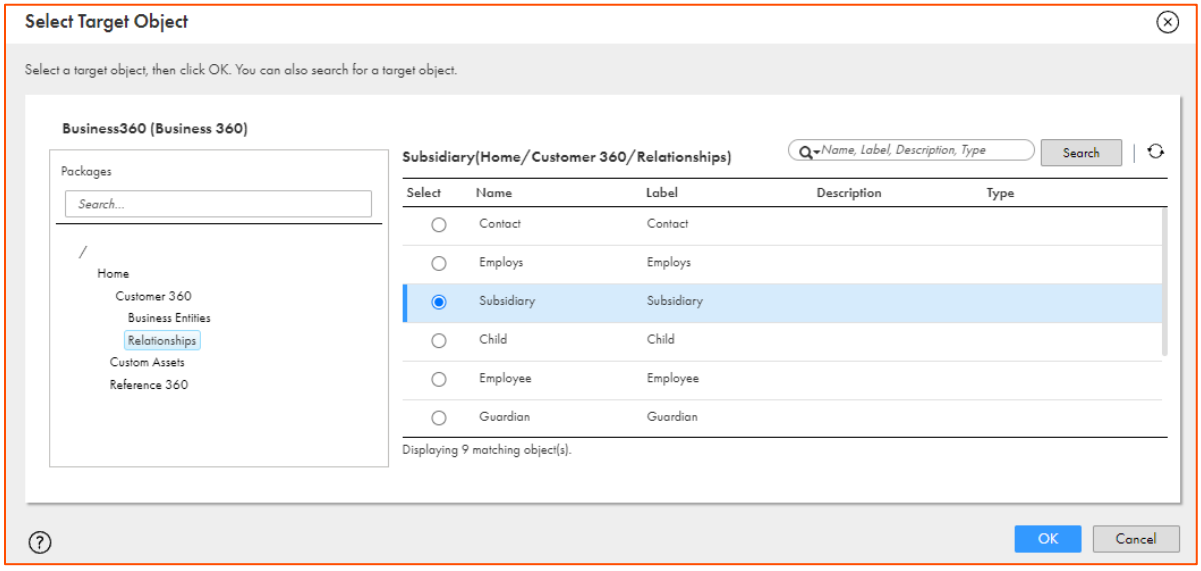
Purpose:

Customers can be related to each other, for example, parent companies and subsidiaries or employee or household relationships. This exercise aims to create relationships between the organizations we have loaded before.

Instructions/Steps – Create CDI mapping

The CDI mapping is relatively easy, reading a source file, setting default values, and loading into the MDM SaaS data store. You will also create the mapping task, taskflow, and job definition.

1. Go to Data Integration and click New, Mapping. Name it m_CRM_Customer_Relationships.
2. Click on the source and name it Source_Relationships
3. Click on the source properties and select the flat file connection
4. Select Customer_Relationships.csv as the source file.
5. Click on the target and name it Target_Subsidiary
6. Click on the target properties and select the Business 360 connection
7. Click on Select object and navigate to Customer 360 / Relationships to find the relationship.
8. Select Subsidiary and click Ok:



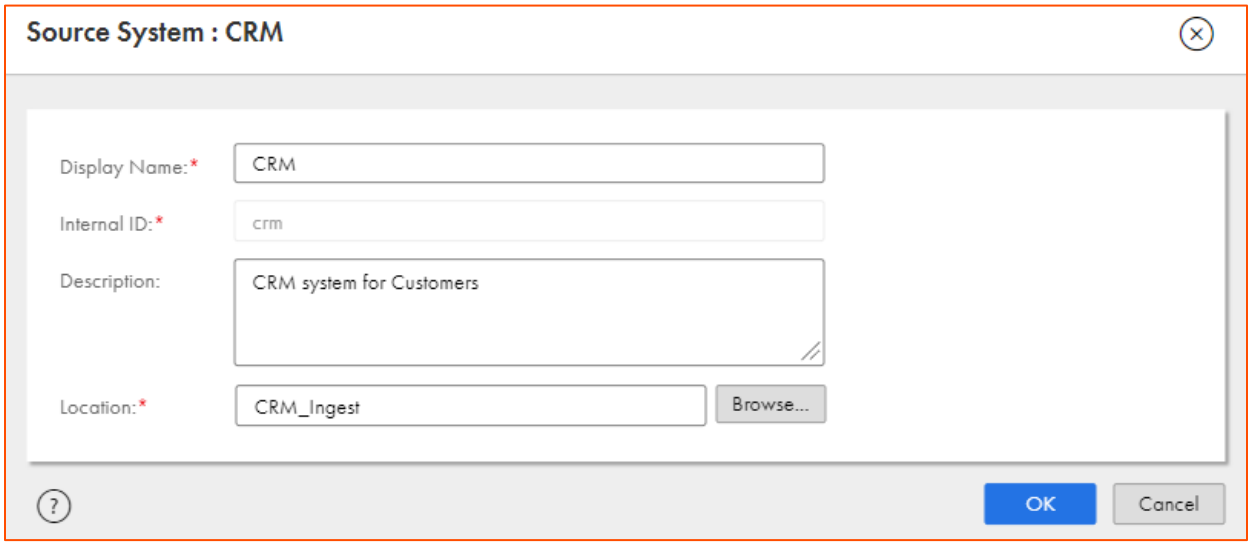
9. Add an expression and connect it to the source. Name it Expression_Defaults.

© Informatica.

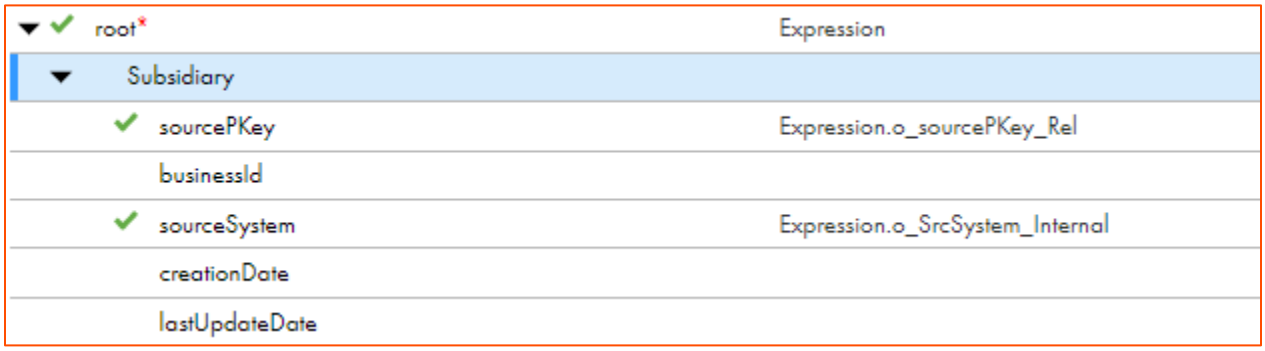
10. Set the following output values:

Field Name	Precision	Expression
o_SrcSystem_Internal	10	'crm' (All lowercase)
o_sourcePkey_Rel	50	parent_DUNS || '-' || child_DUNS

Relationships must be created with the Internal source system name, which is always lowercase. You can find this by opening the source system.



11. Connect the expression to the target.
12. Go to the target and go to field mapping.
13. Expand Subsidiary and map the fields as follows:



© Informatica.

14. Expand the Subsidiary below these fields, and expand the _from and _to containers. In both, expand the id. Then map as follows:

Subsidiary	
_from	
businessEntity	
id	
✓ sourcePKey	Expression.parent_DUNS
✓ sourceSystem	Expression.o_SrcSystem_Internal
businessId	
_to	
businessEntity	
id	
✓ sourcePKey	Expression.child_DUNS
✓ sourceSystem	Expression.o_SrcSystem_Internal

15. Add the In-Out parameter jobInstanceId. Don't forget to set the precision to 100.
16. Save the mapping. It should look like this:

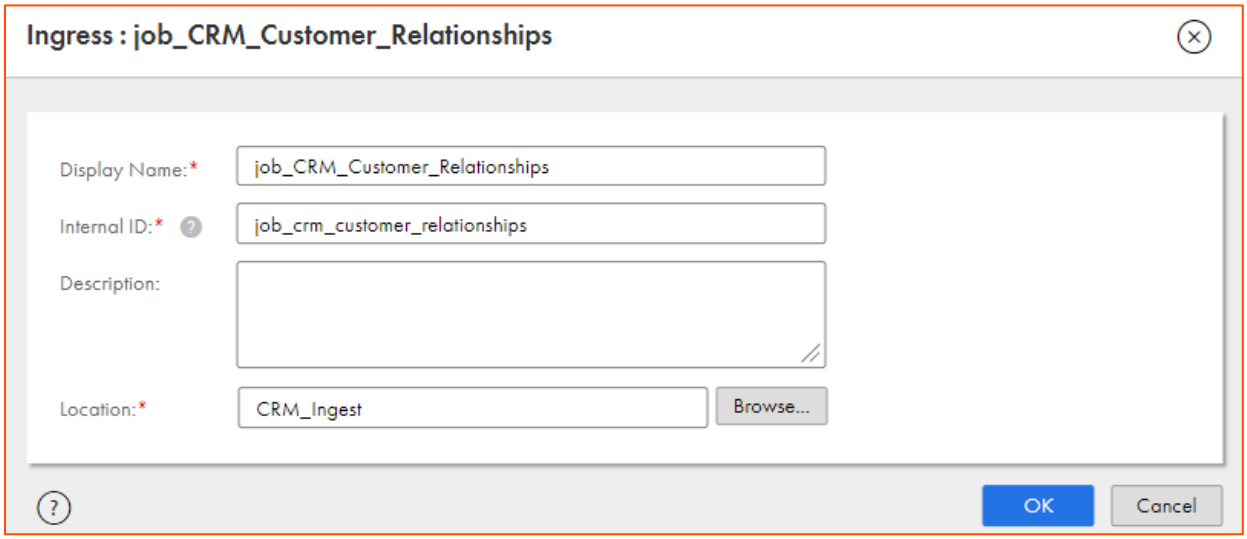


17. Create the mapping task and taskflow as you have done before.
18. Publish the taskflow. This step is often forgotten.

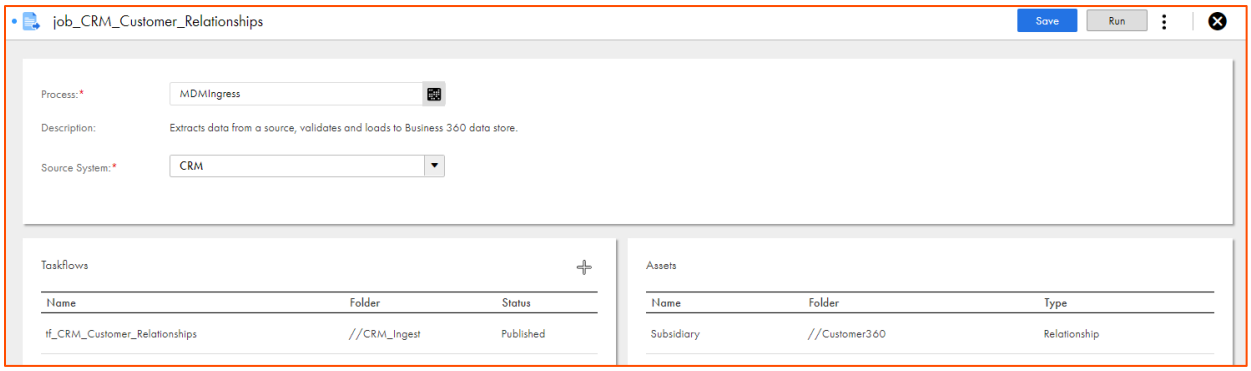
© Informatica.

Instructions/Steps – Create job definition and run

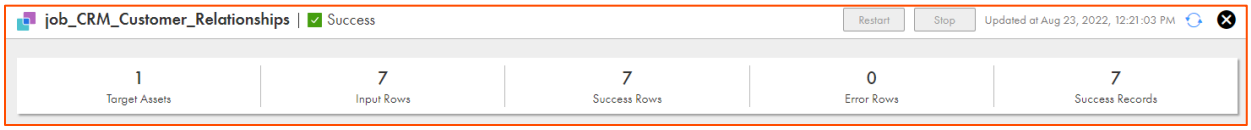
19. Go to Business 360 and create the job definition. Make sure you set the location to your CRM_Ingest project:



20. Set the source system to CRM and add your taskflow. In the Assets pane on the right-hand side, you will see the relationship:

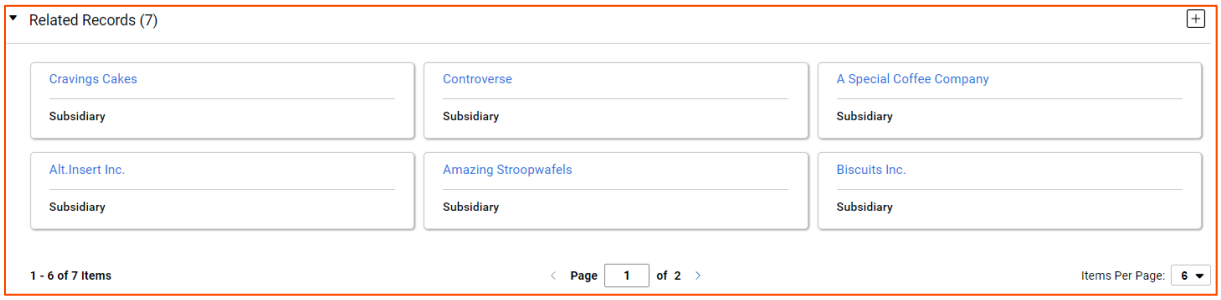


21. Save and run the job.
22. The job should load 7 records, as shown in the statistics:

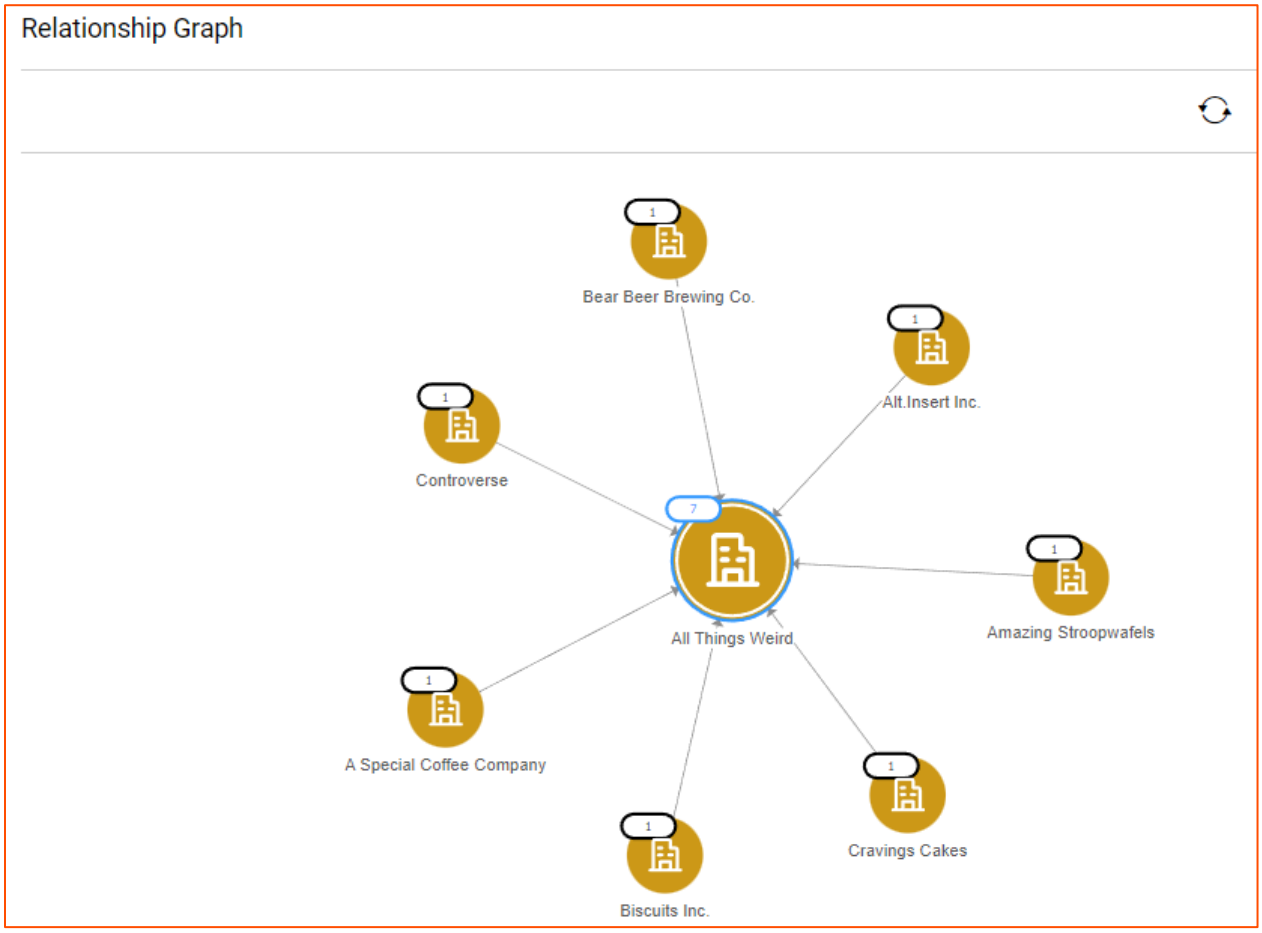


© Informatica.

23. If you have fewer, the common cause is the sourcePKey field is defined as only 10 characters, resulting in duplicate keys.
24. Go to Customer360 and search for the organization “All Things Weird”.
25. Open the organization and scroll to the bottom. You can see the relationships here:



26. You can also go the the Relationship Graph to view the relationships:



This concludes the exercise.

© Informatica.